



Cactus

Chemometrics and Analytical Chemistry Tools

User Guide

A user-friendly software for exploring, preprocessing and modelling analytical data
through chemometric and machine-learning workflows
without writing code

Version 0.1.0

Developed by

DataBloom and the University of Turin



UNIVERSITÀ
DI TORINO



This guide describes the application from the perspective of the end user. It does not document internal source code, algorithms, or implementation details.

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Chapter 1

Introduction

1.1 What Cactus Is

Cactus is a software application for the analysis of tabular analytical data using chemometric and machine-learning methods. It is designed primarily for numeric and high-dimensional datasets, such as spectroscopic data containing many predictor variables, but it can also be used with more general classification and regression datasets.

The application provides a guided workflow covering the main stages of data analysis:

1. **Import** data from CSV, TXT or Excel files.
2. **Edit** the dataset by selecting samples and variables, assigning variable roles, managing missing values and defining Training and Test sets.
3. **Explore** the data using graphical tools and statistical tests.
4. **Preprocess** predictor and outcome variables using common chemometric transformations.
5. **Model** the data using unsupervised, classification, one-class and regression methods.
6. **Predict** new samples using a previously saved model.

Cactus is built with R and Shiny, but no knowledge of R or programming is required to use the application.

1.2 Who This Guide Is For

This guide is intended for anyone who wants to explore, analyse and model their data without writing code.

It describes the main application workflow, the available controls and the practical meaning of the most important settings. Statistical and chemometric concepts are introduced only where they are necessary for selecting and using a method correctly.

1.3 Supported Data Structure

Cactus expects the dataset to be organised as a table in which each row represents a sample or observation and each column represents a variable.

Sample labels	class	13C/12C honey	13C/12C protein fraction	% dextrose	% fructose	NDP2 moisture (g/100g)	HMF (mg/kg)	dextrose	color
honeyview oak 1	honeyview	-26.22	-26.03	28.8	17.62	7.02	15.52	1.7	23.6
honeyview oak 2	honeyview	-26.2	-26.01	28.75	17.66	6.98	15.36	1.76	23.66
honeyview oak 3	honeyview	-26.18	-26.07	28.91	17.55	7.04	14.95	1.65	0.961805556
honeyview oak 4	honeyview	-26.18	-26.03	28.84	17.73	7.02	15.08	1.72	24.14
honeyview oak 5	honeyview	-26.13	-26.02	28.78	17.68	7	15.43	1.78	23.29
honeyview oak 6	honeyview	-26.16	-26.03	28.94	17.6	7.06	15.27	1.65	23.45
honeyview oak 7	honeyview	-26.2	-26.06	28.74	17.65	7	15.34	1.63	24.28
honeyview 1	honeyview	-25.87	-25.81	25.1	30.5	8.8	15.08	1.81	25.03
honeyview 2	honeyview	-26.02	-25.86	25.33	30.51	8.94	14.89	1.77	24.67
honeyview oak 8	honeyview	-25.88	-26	25.45	30.38	9.06	15.38	1.74	23.89
honeyview 4	honeyview	-25.93	-25.79	25.64	30.72	8.8	15.22	1.86	24.67
muscalyptus 1	muscalyptus	-25.8	-25.6	30.8	41.2	6.44	15.7	1.76	26.03
muscalyptus 2	muscalyptus	-25.85	-26.48	31.11	40.7	6.2	15.44	2.83	25.83
muscalyptus 3	muscalyptus	-25.78	-26.57	30.82	42.23	6.4	15.11	2.82	26.26
muscalyptus 4	muscalyptus	-25.83	-26.61	30.75	42.22	6.35	15.53	2.95	25.95
muscalyptus 5	muscalyptus	-25.77	-26.66	30.78	42.16	6.43	16.02	2.73	26.11
muscalyptus 6	muscalyptus	-25.8	-26.6	30.85	42.16	6.38	15.87	1.88	26.43
muscalyptus 7	muscalyptus	-25.85	-26.53	30.81	42.12	6.35	15.45	2.83	25.91
muscalyptus 8	muscalyptus	-25.85	-26.48	30.9	40.7	6.2	15.76	2.65	26.05
muscalyptus 9	muscalyptus	-25.96	-26.33	31.07	40.55	6.27	15.38	2.83	25.78
muscalyptus 10	muscalyptus	-25.68	-26.72	30.95	41.08	6.11	15.92	2.869	26.22
muscalyptus 11	muscalyptus	-25.94	-26.39	31.06	40.94	6.19	16.03	2.9	25.87
muscalyptus 12	muscalyptus	-25.71	-26.75	30.88	40.86	6.52	15.83	2.82	25.91
Shesnat Monti Cima 1	chestnut	-25.74	-25.98	25.91	40.22	7.36	17.52	1.73	24.53
Shesnat Monti Cima 2	chestnut	-25.87	-25.9	25.9	40.2	8.11	17.26	2.06	25.15
Shesnat Monti Cima 3	chestnut	-25.75	-25.96	25.87	40.18	7.73	18.03	1.86	23.38
Shesnat Monti Cima 4	chestnut	-25.73	-25.95	26.03	40.34	7.93	16.42	1.91	24.56
Shesnat Monti Cima 5	chestnut	-25.76	-26.04	25.96	40.79	7.85	17.15	1.88	24.91

(a) Example of Excel dataset

```

sample:4284,677,4287,744,6111,771,4225,428,4219,485,4223,342,4227,199,4283,856,6176,471,4218,77,4242,426,4246,483,
treat_01:1,1816565;-1,455374;-1,454477;-1,451232;-1,451896;-1,451880;-1,468853;-1,467867;-1,469731;-1,468436;-1,469543
treat_02:1,1817889;-1,4515396;-1,454477;-1,451272;-1,451776;-1,469718;-1,468186;-1,466887;-1,469423;-1,465085;-1,46909
treat_03:1,4952454;-0,993772;-1,492776;-0,992423;-0,990788;-0,998883;-0,998789;-0,998132;-0,998212;-0,99216;-0,99216;-0,992
treat_04:1,4682823;-1,4681815;-1,4680864;-0,995941;-0,995256;-0,995684;-0,995527;-0,993784;-0,992149;-0,991585;-0,99
treat_05:1,4699845;-1,4697899;-1,469608;-1,468472;-1,468377;-1,468181;-1,468222;-0,998278;-0,995584;-0,99582;-0,99683
treat_06:1,4699888;-0,998111;-0,998087;-0,997784;-0,998043;-0,998221;-0,998111;-0,998181;-0,998091;-0,998427;-0,99802
treat_07:1,1813132;-1,181285;-1,469881;-1,469132;-1,469278;-1,469616;-1,469479;-1,469484;-1,469121;-1,469145;-1,469
treat_08:1,122822;-1,126734;-1,119444;-1,118242;-1,118771;-1,11915;-1,115281;-1,119944;-1,112429;-1,118891;-1,118
treat_09:1,181897;-1,181825;-1,188813;-1,469846;-1,469772;-1,469698;-1,469177;-1,469133;-1,469121;-1,469068;-1,469
treat_10:1,469227;-1,469559;-1,469428;-1,469657;-1,469448;-1,469442;-1,469243;-1,469143;-1,469115;-1,469081;-1,469
treat_11:1,468471;-1,468422;-1,468409;-1,468444;-1,468423;-1,467949;-1,467778;-1,467823;-1,467843;-1,467841;-1,467
treat_12:1,182679;-1,182586;-1,188764;-1,46988;-1,469841;-1,469727;-1,469615;-1,469517;-1,469193;-1,469282;-1,469
treat_13:1,181837;-1,181725;-1,188802;-1,469585;-1,469541;-1,469541;-1,469541;-1,469541;-1,469541;-1,469541;-1,469
treat_14:1,467789;-1,467726;-1,467623;-1,467625;-1,467188;-1,467188;-1,467188;-1,467188;-1,467188;-1,467188;-1,467
treat_15:1,469523;-1,468763;-1,468234;-1,468294;-1,468483;-1,468285;-1,468189;-1,467914;-1,467791;-1,467588;-1,4679
treat_16:1,467659;-1,467629;-1,467629;-1,467629;-1,467629;-1,467629;-1,467629;-1,467629;-1,467629;-1,467629;-1,467
treat_17:1,467187;-1,467248;-1,467118;-1,468097;-1,468482;-1,467625;-1,468586;-1,468189;-1,468368;-1,468368;-1,468
treat_18:1,4681543;-1,4679146;-1,467813;-1,467813;-1,467732;-1,467531;-1,467813;-1,467387;-1,467234;-1,467187;-1,467
treat_19:1,469294;-1,469125;-1,467899;-1,468821;-1,468482;-1,468232;-1,468124;-1,468338;-1,467847;-1,467588;-1,468
treat_20:1,469648;-1,468449;-1,467122;-1,465431;-1,465293;-1,465293;-1,465112;-1,464934;-1,464819;-1,464781;-1,464
treat_21:1,467624;-1,466939;-1,466122;-1,466382;-1,466285;-1,466181;-1,465933;-1,465788;-1,465581;-1,465278;-1,46568
treat_22:1,469246;-1,468183;-1,468183;-1,468994;-1,467732;-1,465748;-1,465481;-1,465233;-1,464885;-1,464885;-1,464
treat_23:1,4682254;-1,4688034;-1,467855;-1,467795;-1,467622;-1,467582;-1,467285;-1,467143;-1,467077;-1,466876;-1,467
treat_24:1,469634;-1,4697256;-1,469612;-1,469256;-1,469081;-1,469185;-1,469045;-1,468992;-1,468788;-1,468594;-1,468
treat_25:1,469731;-1,469582;-1,469486;-1,469489;-1,469214;-1,469483;-1,469454;-1,469427;-1,469417;-1,469211;-1,46928

```

(b) Example of csv/txt structured dataset

Figure 1.1: Examples of different types of datasets that can be imported in Cactus

During the Edit step, columns can be assigned one of the following roles:

- **Predictor (X)** - Numeric variables used as model inputs, such as spectral intensities, chemical measurements or instrumental signals.
- **Classification outcome (Y)** - Categorical variables containing class or group labels.
- **Regression outcome (Y)** - Numeric variables used as response values in regression models.
- **Date/time** - Variables used to order samples, create time-based splits or produce temporal visualizations.

Note

A dataset may contain multiple classification and regression outcomes. The variable to be analysed is selected within the relevant modelling module.

1.4 Application Workflow

The main sections of Cactus follow the order shown below:

Stage	Main functions
Import	Load the original dataset
Edit	Modify data, assign roles, manage missing values and split samples
Explore	Visualizations and statistical analyses
Preprocess	Transform predictor and outcome variables
Model	Unsupervised, classification, one-class and regression methods
Predict	Apply a saved model to new data

The workflow is sequential. Sections that require processed or approved data remain unavailable until the necessary previous steps have been completed.

Tip

If a section is hidden or disabled, check that the preceding stage has been completed and confirmed. This behaviour prevents analyses from being run on incomplete or unapproved data.

Chapter 2

Installation and Launching

Cactus is distributed as a self-contained installer for Windows, macOS, and Ubuntu/Debian-based Linux. R and all required R packages are bundled with the application - you do not need to install R, or any R package, separately.

2.1 Downloading Cactus

Go to the Cactus Releases page and download the installer for your operating system:

- **Windows** - CactusSetup.exe
- **macOS** - CactusSetup.dmg
- **Ubuntu / Debian** - CactusSetup.deb

Download

Get the latest version from the [Releases](#) page.

Platform	How to install
Windows	Download .exe and run the installer, then follow the setup wizard
macOS	Download .dmg , drag Cactus into Applications. On first launch: right-click the app → Open (needed because the app isn't yet signed with an Apple Developer ID)
Linux (Ubuntu)	Download .deb and install from a terminal with <code>sudo apt install ./CactusSetup.deb</code> (rather than double-clicking, so dependencies are resolved automatically)

System requirements

- Windows 10/11
- macOS on Apple Silicon
- Ubuntu 22.04 LTS or later

Figure 2.1: Cactus Releases page on GitHub, with the three installers listed

i Note

Always download the installer from the official Cactus releases page. Do not run an installer you received from an untrusted source.

2.2 Installing on Windows

1. Double-click `CactusSetup.exe`.
2. Follow the setup wizard: choose whether to create a desktop icon, then click **Install**.
3. The installer downloads the bundled R runtime as part of the installation - this step needs an internet connection and can take a few minutes depending on your connection speed. A progress bar shows the download.
4. Once installation completes, click **Finish**. Cactus starts automatically (this can be unchecked on the last wizard page if you prefer to start it later).

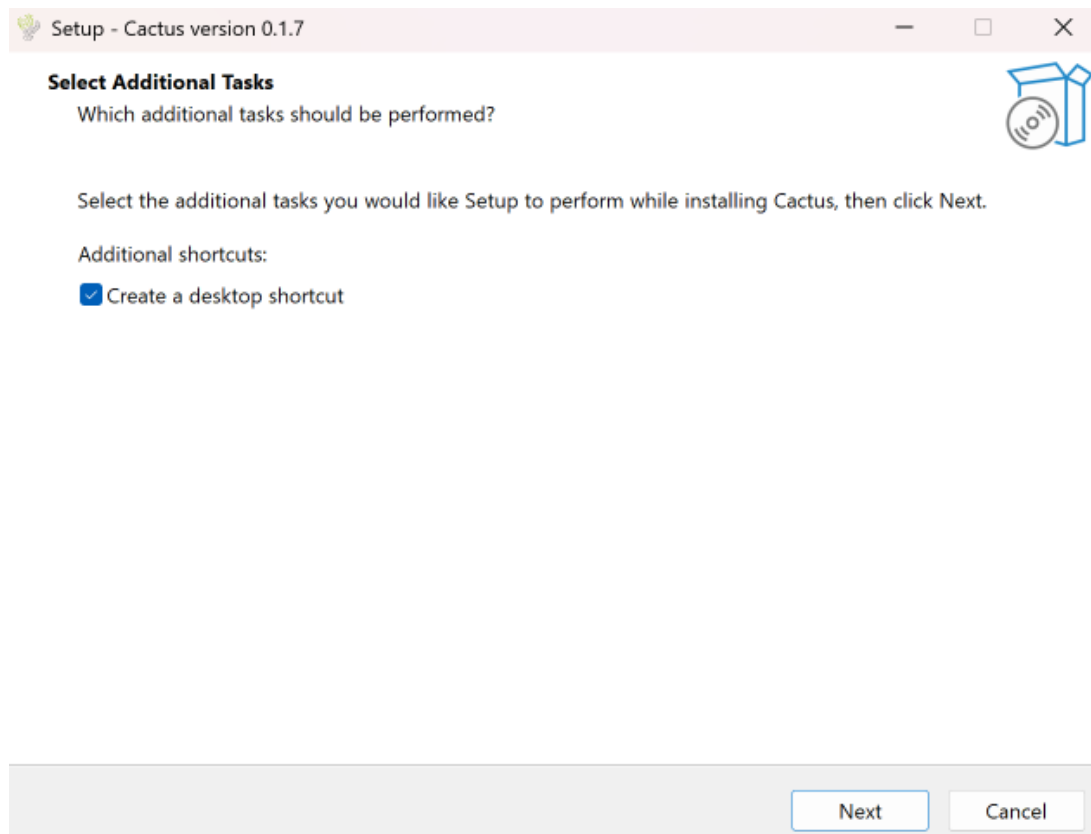


Figure 2.2: Windows installer wizard (welcome / install progress / finish pages)

Note

No administrator rights are required - Cactus installs into your personal user folder, not Program Files.

Tip

Windows may show an **”Unknown publisher”** warning the first time you run the installer, because it is not (yet) signed with a paid code-signing certificate. This is expected - click **More info** then **Run anyway** to proceed.

2.3 Installing on macOS

1. Open **CactusSetup.dmg**. A temporary disk image opens showing the **Cactus** icon and a shortcut to the **Applications** folder.
2. Drag **Cactus** onto **Applications** and wait for the copy to complete.
3. Eject **CactusSetup**, then open the **Applications** folder and locate Cactus there.
4. The first time you launch Cactus, **right-click** (or Control-click) the app and choose **Open**. When macOS displays the security warning, click **Open** again.
5. If macOS still blocks the application, open **System Settings > Privacy & Security**, scroll down to the security section and click **Open Anyway** next to the message referring to Cactus. Confirm the action when prompted.

This procedure is normally required only for the first launch because Cactus is not signed with an Apple Developer ID. After it has been authorised, the application can be opened normally.

Warning

Do not run Cactus directly from the mounted **.dmg** installer. Always copy it to **Applications** first. The disk image is read-only, so automatic updates cannot be installed correctly when Cactus is launched from there.

If updates do not work, verify that Cactus is running from **/Applications/Cactus.app**. In Finder, right-click the running Cactus icon in the Dock and choose **Options > Show in Finder**.

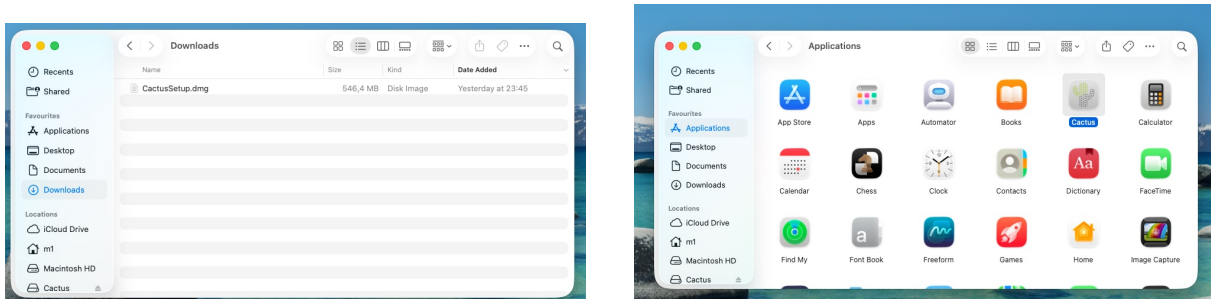


Figure 2.3: macOS .dmg window with the Cactus icon and Applications shortcut

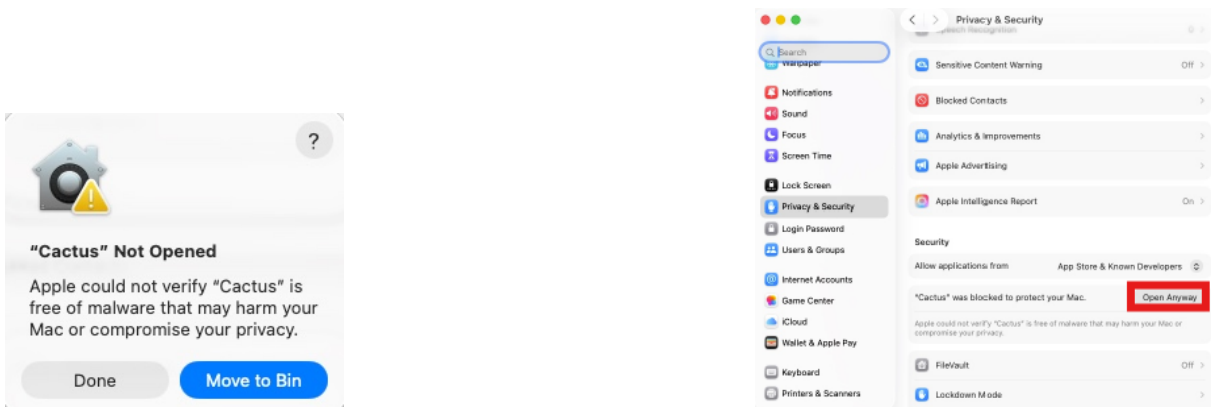


Figure 2.4: macOS security warning and Privacy & Security options used to allow Cactus to open

2.4 Installing on Ubuntu / Debian

The .deb package should be installed from a terminal with `apt`, rather than by double-clicking it, so that its dependencies are resolved automatically:

1. Open a terminal in the folder where you downloaded the file (or `cd` into it).
2. Run:

```
sudo apt install ./CactusSetup.deb
```
3. Enter your password when prompted. `apt` downloads any missing system dependencies and installs Cactus.
4. Once installation completes, launch Cactus from your applications menu, or by running `cactus` from a terminal.

2.5 First launch

The first time you start Cactus (on any platform), a splash screen with the Cactus logo appears for a few seconds while the application starts up in the background.



Loading...

Figure 2.5: Cactus splash screen shown on startup

After the splash screen, the **login / sign-up** window opens.

Note

If a newer version of Cactus is available, an update pop-up may appear at this point instead of (or shortly before) the login screen - see Section 2.9.

2.6 Creating an account

If you do not have an account yet, click **Sign Up!** on the login screen and fill in the registration form:

- **First name, Last name, Email, Country** - required.
- **Affiliation** (university or company) and **Role** (Student, PhD Student, Researcher, Professor, Industry, Other) - optional. Selecting **Other** for Role reveals a text field to specify it.
- **Password / Confirm Password** - click the eye icon inside either field to show or hide what you have typed.

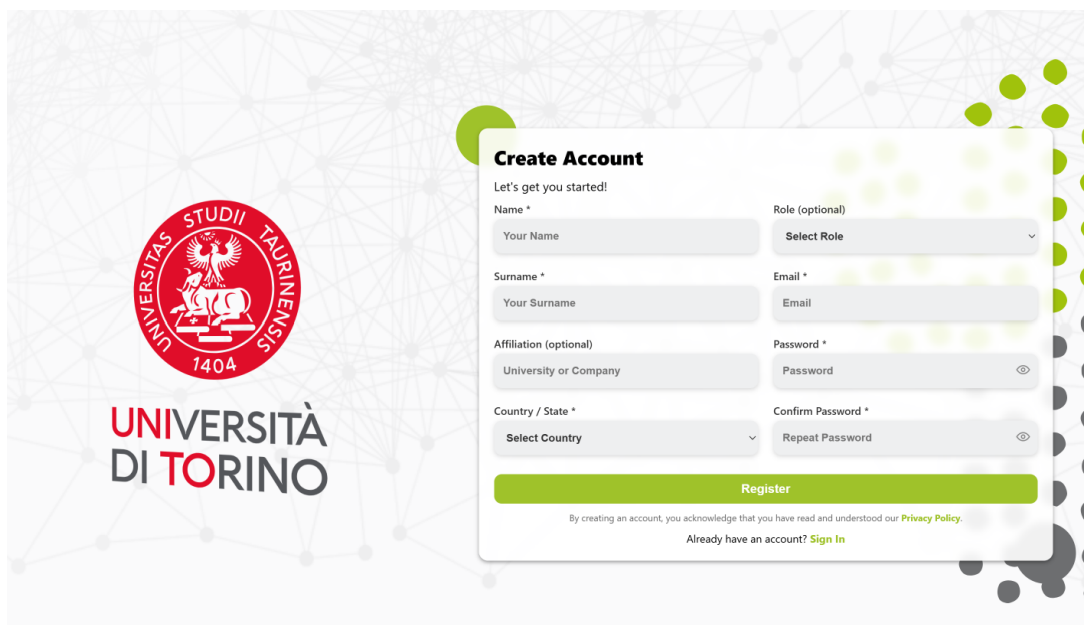
The image shows a web registration form titled "Create Account" overlaid on a background featuring the University of Turin logo and a network diagram. The form includes fields for Name, Surname, Affiliation, Country/State, Role, Email, Password, and Confirm Password. A green "Register" button is at the bottom, with a link to the Privacy Policy and a "Sign In" link for existing users.

Figure 2.6: Sign-up form with all fields visible

By clicking **Register** you acknowledge the Cactus Privacy Policy, linked below the button. Cactus then sends a confirmation code to your email address - enter it on the next screen to complete your registration.

The image shows the OTP confirmation screen. It features the University of Turin logo and a network diagram background. A form titled "Enter the confirmation code sent via email" contains a single input field for the "Confirmation code (OTP)". Below the input field is a green "Confirm Registration" button and a link labeled "Back to Sign Up".

Figure 2.7: OTP confirmation code screen

2.7 Logging in

Enter your email and password on the login screen and click **Sign In**. Check **Remember Me** to stay logged in on this computer for future launches. If you forgot your password, use the **Forgot Password?** link to receive a reset code by email.

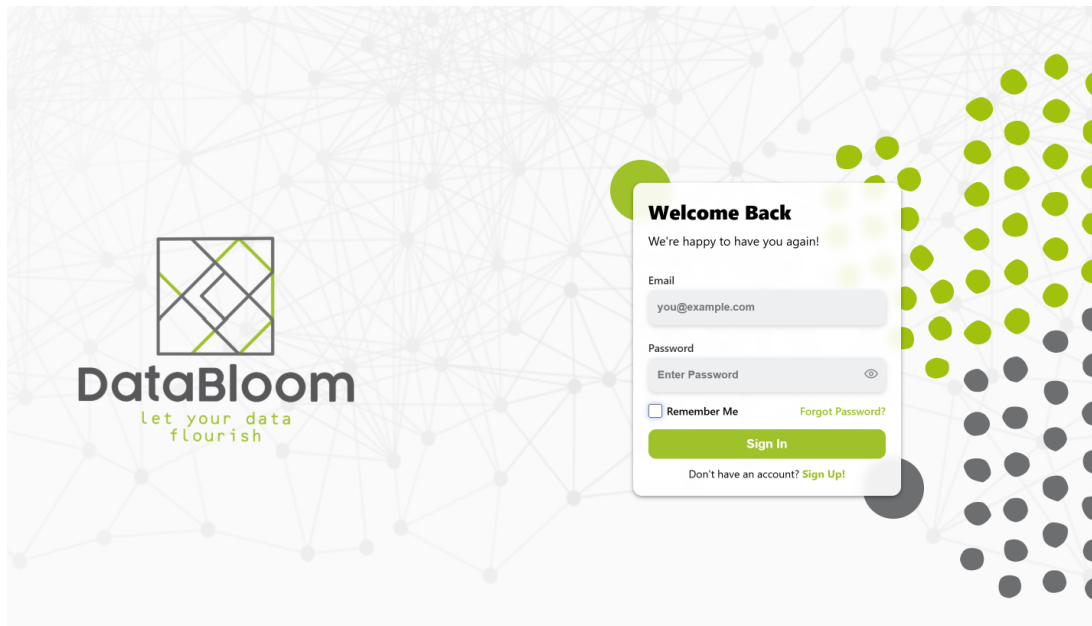


Figure 2.8: Login form with Remember Me and Forgot Password

Once you are logged in, Cactus finishes loading (this can take a little longer the very first time) and the main application opens, described in Chapter 3.

2.8 Resetting your password

If you forgot your password, click **Forgot Password?** on the login screen:

1. Enter your account email and click **Send Reset Code**. Cactus sends a confirmation code to that address.
2. On the next screen, enter the **confirmation code (OTP)** from the email, then your **new password** and **confirm** it (use the eye icon on either field to check what you typed).
3. Click **Confirm Reset**. You are returned to the login screen and can sign in with your new password.



Figure 2.9: Request reset-code screen (email field, Send Reset Code button)

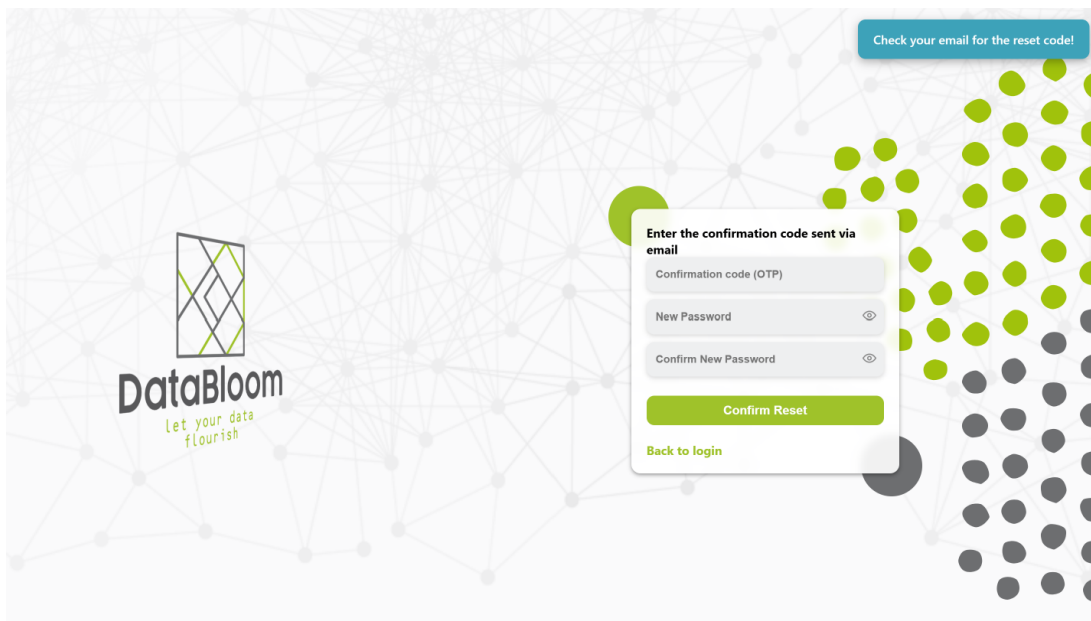


Figure 2.10: Reset-password screen (OTP code, new password, confirm password)

2.9 Automatic updates

Cactus checks for a newer version automatically every time it starts. If one is available, a small pop-up window appears with a summary of what changed.

- Click **Download** to install the update - Cactus downloads it, applies it, and restarts itself automatically. This can take a moment depending on your connection.
- Click **Not now** to continue using the current version; you will be asked again the next time you start Cactus.

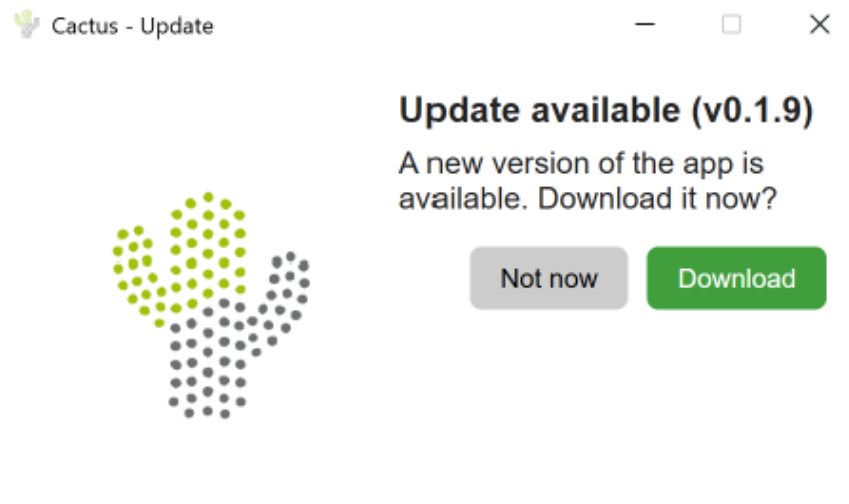


Figure 2.11: Update available pop-up with Download / Not now buttons

2.10 Closing the application

Simply close the Cactus window. This stops the application and the background R process automatically - there is no separate console window to close.

Chapter 3

Application Overview

3.1 Main layout

Cactus uses a classic dashboard layout with three regions:

- **Header (top bar)** - shows the Cactus logo, a **theme toggle** button (sun/moon icon) that switches between light and dark appearance, and a **gear icon** that opens the right-hand settings panel.
- **Sidebar (left)** - the main navigation menu. Each entry corresponds to one stage of the analysis pipeline described in Chapter 1.
- **Body (centre)** - the content of the currently selected sidebar tab: forms, tables and plots.

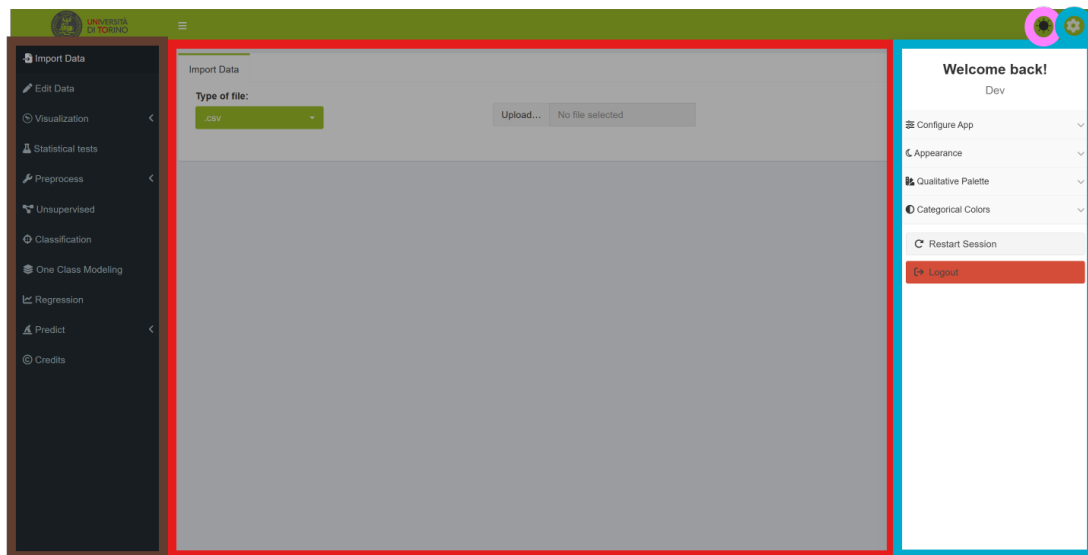


Figure 3.1: Main application layout (header, sidebar, body)

3.2 Sidebar navigation

The left sidebar lists, from top to bottom:

1. **Import Data**
2. **Edit Data**
3. **Visualization** (One Variable / Two Variables / Multiple Variables)
4. **Statistical Tests**
5. **Preprocess** (Predictors / Outcome / Data Preprocessed)
6. **Unsupervised**
7. **Classification**
8. **One Class Modeling**
9. **Regression**
10. **Predict** (Upload Model / Prediction)
11. **Credits**

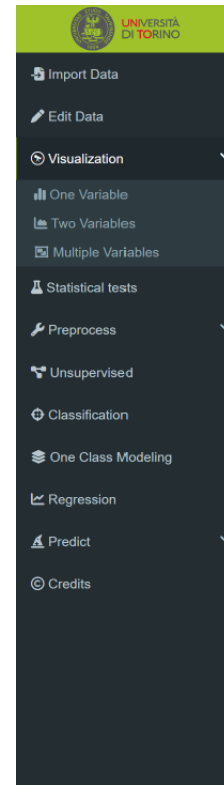


Figure 3.2: Left sidebar navigation with all sections unlocked.

Warning

Several sidebar entries are **gated**, meaning that they only appear or become available after the required previous steps have been completed.

- The **Preprocess** submenu only appears after clicking **Approve Dataset** in the **Edited Data** tab.
- **Unsupervised**, **Classification**, **One Class Modeling**, and **Regression** become available only after preprocessing has been completed.

If a section is empty or displays a message asking you to complete a previous step, return to the required module and finish that step first.

3.3 Right-hand settings panel

Clicking the gear icon in the header opens a panel on the right with several collapsible sections:

- **Qualitative Palette** - the default colour palette used for continuous/gradient plots (choose any ColorBrewer palette, a saved custom palette, or build a new one via "Create your Palette").
- **Categorical Colors** - eight default colours used for categorical/factor variables across all plots; click **Edit Colors** to change them.
- **Appearance** - a **Dark mode** switch, saved as your default for future sessions.

- **Configure App** - lets you show or hide individual optional tabs and options throughout the app (e.g. which chart types appear under Visualization, which cross-validation methods are offered, which metrics are shown in classification/ regression/SIMCA results, and the PLS-DA classification rule). Hiding a tab here does not delete any data; it only declutters the interface.
- **Restart Session / Logout** - at the bottom of the panel.

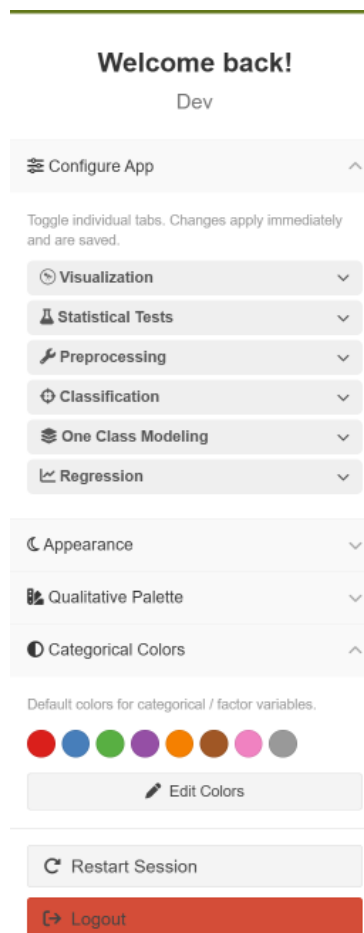


Figure 3.3: Right-hand settings panel (Appearance and Configure App sections open).

Tip

If an expected chart type, metric, or cross-validation method seems to be missing from a module, check **Configure App** in this panel - it may simply be switched off.

3.4 Common UI elements

These elements recur throughout the application:

- **Boxes** - most content is grouped into titled boxes. A box with a green header is a "success"-styled box, typically used for confirmed/completed steps.
- **Maximize button** (top-right corner of a box, expand icon) - enlarges that box (and any plot/table inside it) to fill the screen; click again to restore.

- **Info button** (circled "i" icon) - opens a short explanation or a References modal for that box's content.
- **Settings / gear dropdown** - above many tables and plots; controls how many rows to display, plot colours, labels, and similar cosmetic options.
- **Download / Export buttons** - appear next to tables (CSV/Excel export) and plots (image export); see Chapter 8 and Chapter 18.
- **Confirmation pop-ups** - destructive or pipeline-changing actions (removing columns, resetting a pipeline, approving a dataset) always ask for confirmation before proceeding.

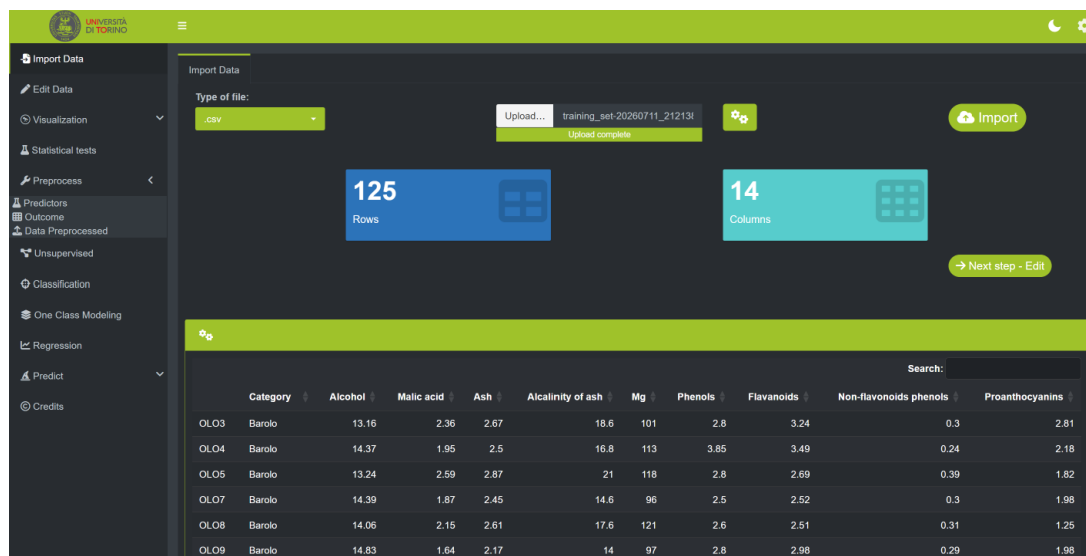


Figure 3.4: Dark theme

3.5 Moving from import to analysis to export

The intended path through the application is linear:

1. Import your dataset (Chapter 5).
2. Clean and configure it in Edit Data, then click **Approve Dataset** (Chapter 6).
3. Optionally explore it visually and statistically (Chapters 7 and 9).
4. Preprocess the predictor and/or outcome variables (Chapter 10).
5. Run one or more modelling techniques appropriate to your goal: unsupervised exploration (Chapter 11), classification (Chapter 13), one-class modelling (Chapter 14) or regression (Chapter 15).
6. Save a trained model, then later reuse it on new data via Chapter 16.

Any change to the Edit step (re-approving the dataset, changing which variables are predictors) automatically resets downstream models, so that results are never left inconsistent with the data used to produce them.

Chapter 4

Import and Data Management - Overview

This chapter gives a high-level view of how data enters Cactus and how it is prepared for analysis. Full, step-by-step instructions for each screen are given in Chapter 5 (Import module) and Chapter 6 (Edit module).

4.1 Getting data into the application

Data is loaded on the **Import Data** tab by uploading a single file in one of three formats: CSV, TXT (delimited text) or Excel (.xls/.xlsx). The file should contain one row per sample and one column per variable, with column headers on the first row (unless you explicitly indicate otherwise). Advanced options let you control the delimiter, decimal symbol, Excel sheet number, whether the first column holds row labels, and whether to automatically drop empty or constant columns.

4.2 Expected data format

- **One table, one dataset:** all variables (predictors, class labels, regression targets, dates) must be columns of the same file.
- **Row labels:** if the first column contains unique sample identifiers, enable "Use first column as row labels" so it is not treated as a data variable. Duplicate values in that column will block the import.
- **Column types:** Cactus automatically detects numeric, categorical and date/time columns after import. Numeric columns with text mixed in (e.g. stray units) will not be recognised as numeric.

4.3 Class and response columns

Cactus does not require you to designate class or response columns at import time. Instead, once the data is imported you assign the *role* of every column during the Edit Pipeline step:

- **Classification variables** - categorical columns used as class labels (Classification/Discrimination and One Class Modeling).
- **Regression variables** - numeric columns used as a modelling target (Regression).
- **Date variables** - optional date/time columns.

- **Predictor (X) variables** - everything else that should be used as model input; by default all remaining numeric columns.

Any column left unassigned at the end of the Edit Pipeline must be explicitly deleted before you can proceed - Cactus will not silently guess its role.

4.4 Approving the dataset

Once columns are typed and assigned, and any missing values are resolved, the **Edited Data** tab shows the final preview and an **Approve Dataset** button. Approving the dataset is the point of no return that unlocks Visualization, Statistical tests, Preprocessing and all modelling tabs; changing your mind afterwards means editing again and re-approving, which resets any models you had already trained.

4.5 Common import errors

- **File type mismatch** - the uploaded file's extension does not match the selected "Type of file"; choose the matching type or file.
- **Duplicate row labels** - occurs when "Use first column as row labels" is enabled but that column has repeated values; either uncheck the option or make the identifiers unique before importing.
- **Date parsing warnings** - shown when a column looks date-like but could not be parsed reliably; check the source formatting of that column.

See Chapter 5 for the complete, screen-by-screen reference, and Chapter 18 for a consolidated list of error messages and fixes.

Chapter 5

Import

The Import module (sidebar entry **Import Data**) is the first screen you see: it loads your data file into Cactus so it can be cleaned, explored and modelled in the next steps. Follow the steps below in order.

5.1 Step 1 - Choose the file type

Open the **Type of file** dropdown at the top of the page and select the format of your dataset: **.csv**, **.txt**, or **.xls/.xlsx**. Pick the option that matches your file's real extension - Cactus uses it to decide how to read the file.

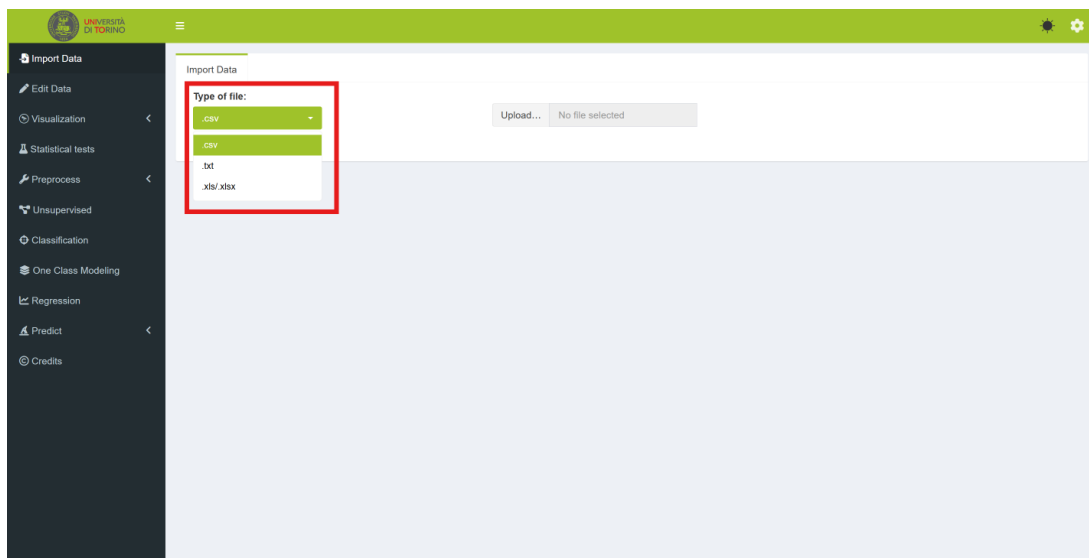


Figure 5.1: Type of file dropdown

5.2 Step 2 - Upload your file

Click **Upload...** The file browser that opens will only let you pick files matching the type you selected in Step 1 - if your file does not appear in the list, go back and check that you chose the correct extension.

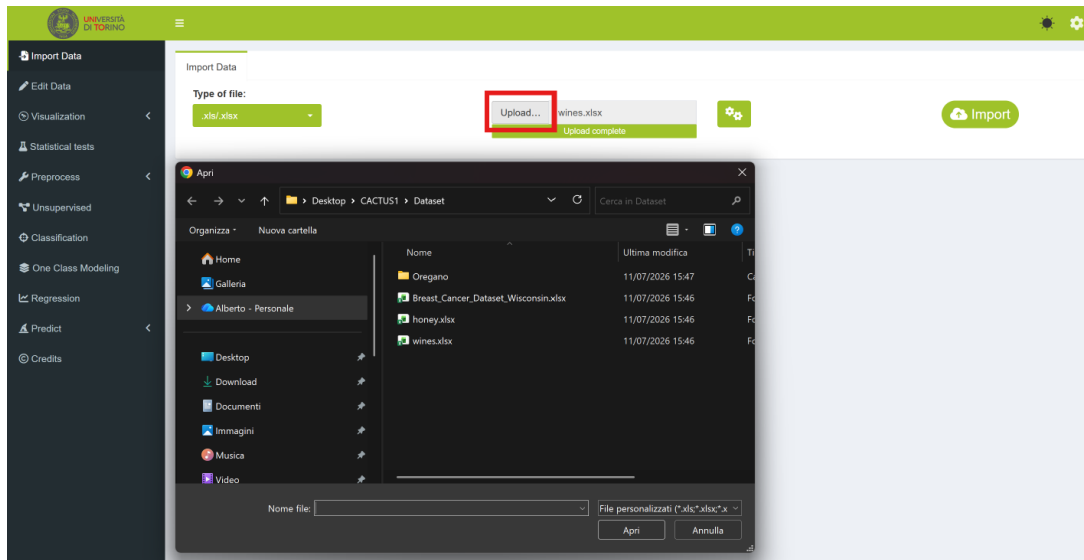


Figure 5.2: Upload button and file browser filtered to the chosen file type

5.3 Step 3 - Adjust advanced options (optional)

If your file needs special handling, click the gear icon to open **Advanced options**:

- **Sheet number** - which worksheet to import (Excel files only, default: 1).
- **Delimiter** - the column separator (CSV/TXT only; default comma for CSV, tab for TXT; leave blank to auto-detect).
- **Decimal symbol** - the character used for decimals (default: . - change this to a comma if your file was exported with European settings).
- **Use first column as row labels** - turn this on if your first column holds sample names/IDs rather than data (default: on).
- **Remove constant columns** - drops columns where every row has the same value (default: off).
- **First row is data (no header)** - turn this on if your file has no column-name row; columns are then auto-named `var_1`, `var_2`, ... (default: off).
- **Remove empty columns** - drops columns that are entirely empty (default: on).

Most files need no changes here - only open this panel if Step 4 fails or the preview looks wrong.

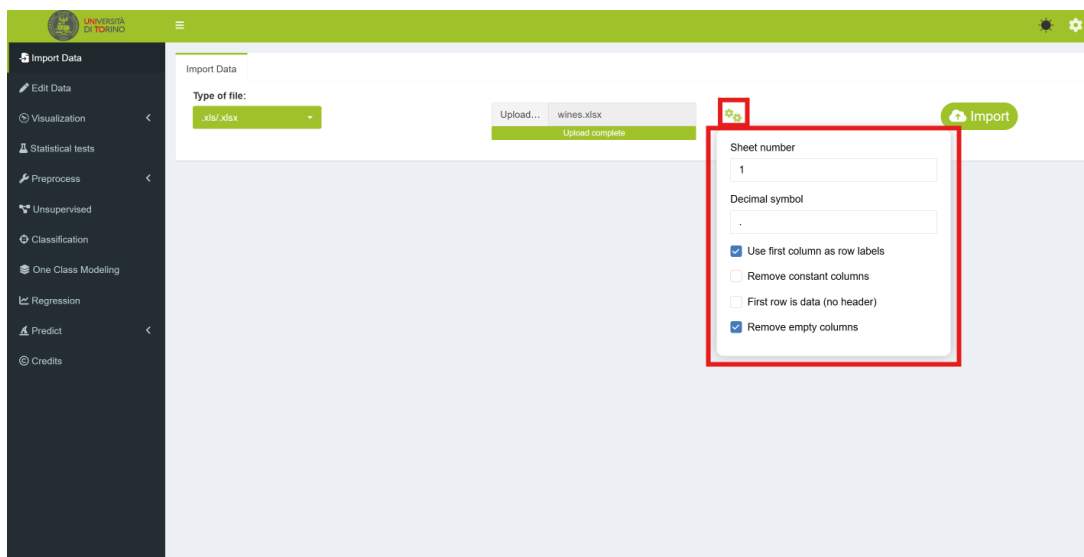


Figure 5.3: Advanced options dropdown open

5.4 Step 4 - Click Import

Click the green **Import** button. A pop-up confirms how many rows and columns were imported, or explains what went wrong (see Common mistakes below).

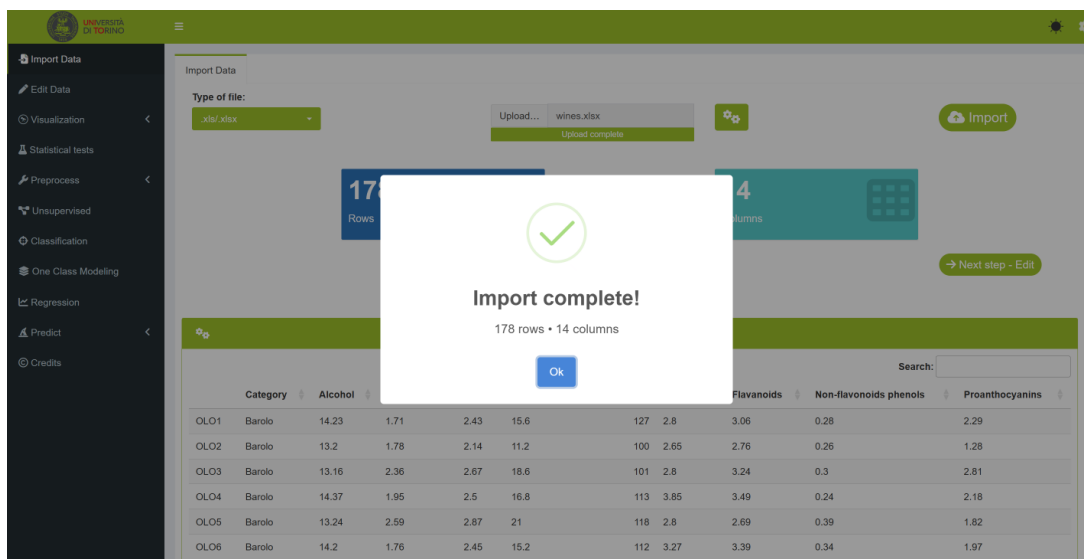


Figure 5.4: Import success pop-up with row/column counts

5.5 Step 5 - Check the preview table

Scroll through the preview table to confirm the data looks right: column names, numeric values, and dates should all appear as expected. Use the gear-icon settings dropdown above the table to show more rows/columns or to download the imported data as CSV.

Warning

Displaying the entire dataset on screen by selecting the **Show entire dataset** checkbox may significantly slow down the application. It is therefore recommended to avoid this option. This table is intended only as a preview to verify that the data import was successful.

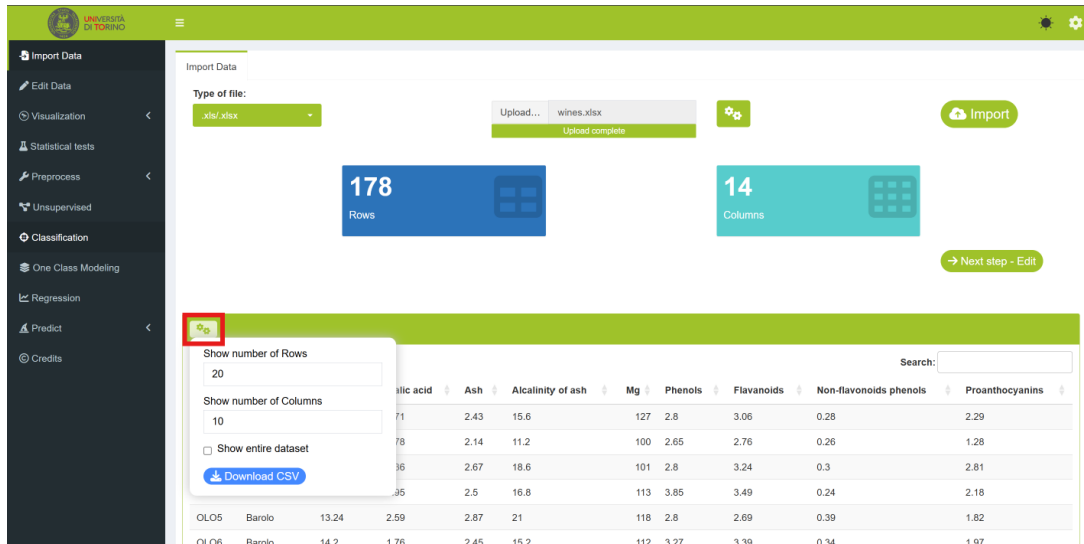


Figure 5.5: Data preview table with row/column count boxes

Tip

Cactus automatically tries to detect date/time columns. A successful import can generally be confirmed when all numeric columns are correctly recognized and their values are displayed properly in the preview table. Note that Cactus uses a dot as the decimal separator. If a column you expected to be numeric is imported as text, check the source file for stray characters or an incorrect decimal separator, then re-import the dataset.

5.6 Step 6 - Move to the Edit module

Once the preview looks correct, click **Next step - Edit** to continue.

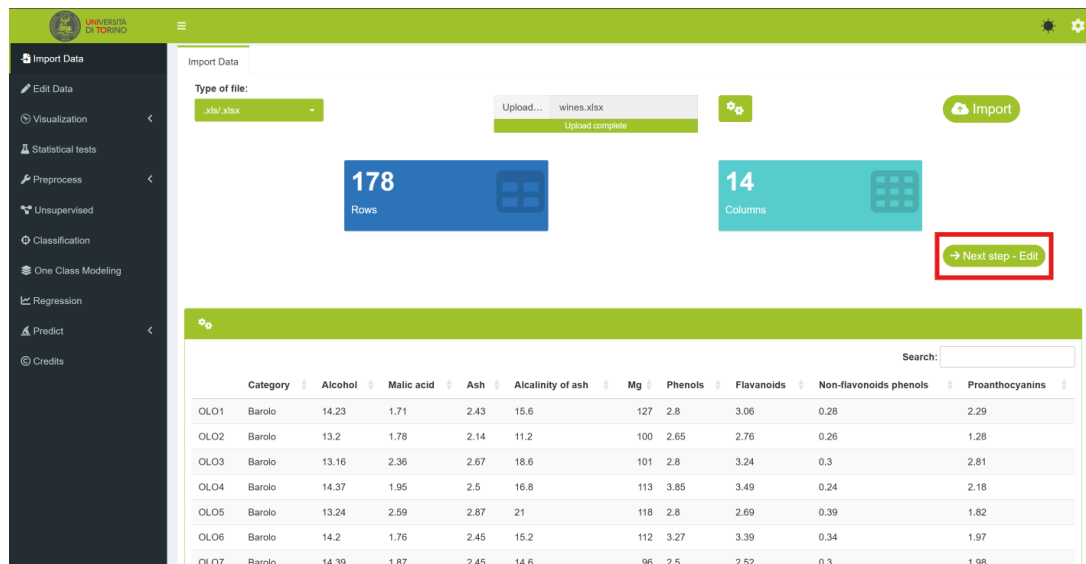


Figure 5.6: Next step - Edit button

5.7 Common mistakes

- Selecting the wrong **Type of file** for the uploaded extension - blocked with a "File type mismatch" pop-up.
- Leaving **Use first column as row labels** checked when that column contains duplicate values - import is blocked with an error asking you to uncheck the option.
- Enabling **Show entire dataset** in the preview settings on a very large dataset - you will be asked to confirm, since this can slow down the browser.
- Forgetting to adjust the **Delimiter/Decimal symbol** for files exported from software using European conventions (semicolon delimiter, comma decimal).

Chapter 6

Edit

The Edit module (sidebar entry **Edit Data**) cleans your imported data and prepares it for analysis. It contains four tabs, which must be completed in order: **Edit**, **Split Train/Test**, **Missing Values**, and **Edited Data**. Each tab becomes available only after the previous step has been completed. The six summary panels shown inside the green box in Figure 6.1 are updated automatically after each subsequent step.

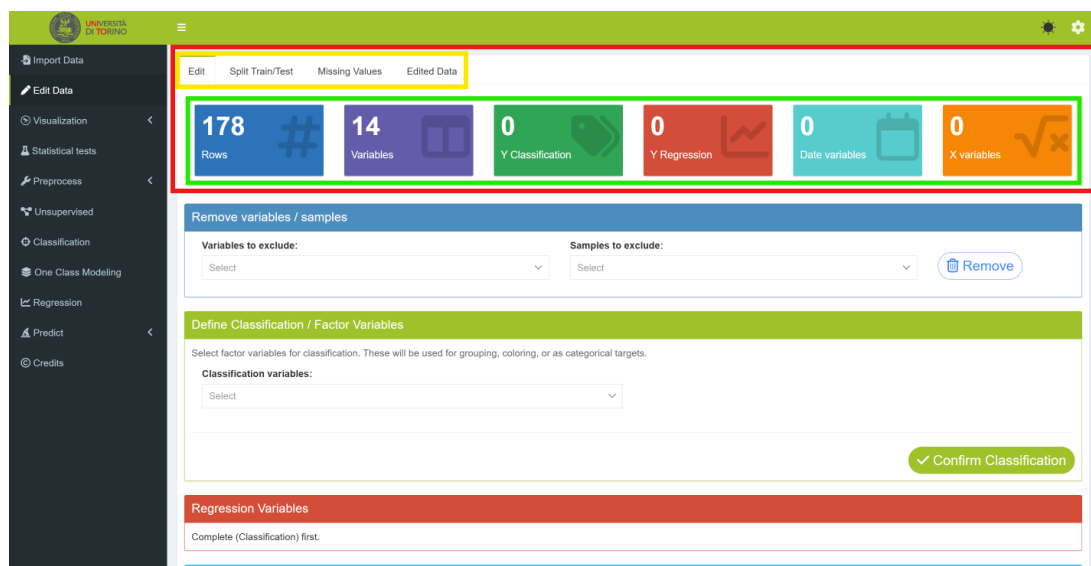


Figure 6.1: Edit Data Steps

6.1 Step 1 - Remove variables or samples (optional)

If some columns or rows should not be included in the analysis, select them and click **Remove**. This step is entirely optional: if there are no variables or samples to remove, proceed without making any changes.

The screenshot shows a panel titled "Remove variables / samples" with a blue header. It contains two dropdown menus: "Variables to exclude:" and "Samples to exclude:", both with a "Select" button. To the right is a blue "Remove" button with a trash icon. Below these is a yellow box labeled "Removed items:" containing two sections: "Variables:" with a list of "Alcohol" and "Malic acid", and "Rows:" with a list of "OL01" and "OL02".

Figure 6.2: Remove variables / samples step

6.2 Step 2 - Confirm Classification variables

Select any columns that contain categorical information (e.g. class labels), choose which groups to keep, and then click **Confirm Classification**. If your dataset does not contain any categorical variables, simply click **Confirm Classification** without selecting any columns.

The screenshot shows a panel titled "Define Classification / Factor Variables" with a green header. It contains a text input field for "Classification variables:" with a dropdown menu showing "Category" and a close button "x". Below this is a yellow box labeled "Levels for Category:" containing three items: "Barberis" (selected with a yellow circle), "Barolo", and "Grignolino", each with a close button "x". To the right is a bar chart titled "Distribution for Category:" showing the distribution of these three categories. At the bottom right is a green button with a checkmark and the text "Confirm Classification".

Figure 6.3: Confirm Classification step

Tip

You can also remove all samples belonging to an entire class by clicking the **x** next to that class in the section highlighted in Figure 6.3.

6.3 Step 3 - Confirm Regression variables

From the dropdown menu, select each numeric column that you may want to use as a response variable (Y) in subsequent regression analyses, or as a continuous numeric variable for colouring plots. Then click **Confirm Regression**. If your dataset does not contain any variables of this type, simply click **Confirm Regression** without selecting any columns.

The screenshot shows a panel titled "Define Regression / Numeric Variables" with a red header. It contains a text input field for "Regression variables:" with a dropdown menu showing "Proline" and a close button "x". Below this is a yellow box labeled "Range for Proline:" containing a range slider with values "278", "478.285714285714", and "1536.9387755102". To the right is a histogram titled "Distribution for Proline:" showing the distribution of the "Proline" variable. At the bottom right is a red button with a checkmark and the text "Confirm Regression".

Figure 6.4: Confirm Regression step

Tip

If you want to restrict the dataset to a predefined range of response values, use the slider to set the acceptable range for each selected response variable (y). Samples with values below the lower limit or above the upper limit will be removed from the subsequent analyses. This step is optional; leave the full range selected if you do not want to exclude any samples. Figure 6.4.

6.4 Step 4 - Confirm Date variables

If your data contains date/time columns, select them and define a valid range. If no date/time columns are present, click **Skip - Continue**.

Note

Cactus recognises a date/time column if it matches one of the following formats:

- **Dates:** YYYY-MM-DD, DD/MM/YYYY, MM/DD/YYYY, YYYY/MM/DD, DD-MM-YYYY, or MM-DD-YYYY.
- **Dates with time:** any of the date formats above followed by HH:MM or HH:MM:SS (e.g. 2024-03-15 14:30:00).
- **Time only:** HH:MM or HH:MM:SS.
- **Excel date numbers** and **Unix timestamps** expressed in seconds or milliseconds are also detected automatically, as are columns already stored using a proper date/time type.

A column is recognised only if more than half of its values match one of these formats. If a date column is not detected, check that its values are consistent and that different formats are not mixed within the same column, for example 15/03/2024 and 2024-03-15.

Figure 6.5: Date variables step

Tip

To remove samples according to their date, click and drag along the date-range bar to select the interval that you want to retain. You can define multiple separate date ranges by clicking the + button highlighted in yellow. Use the - button to remove a previously added range. Samples whose dates fall outside the selected ranges will be excluded from the subsequent analyses. Figure 6.5.

6.5 Step 5 - Confirm X (predictor) variables

Review which columns will be used as predictors. By default, all remaining columns are selected. Once the predictor variables have been checked, click **Confirm X**.

Tip

The **Reverse order** switch is particularly useful for spectral plots, as it allows the spectrum to be displayed using the conventional orientation for the selected spectral axis. For example, spectra expressed in wavenumbers (cm^{-1}) are often displayed from higher to lower wavenumber values.

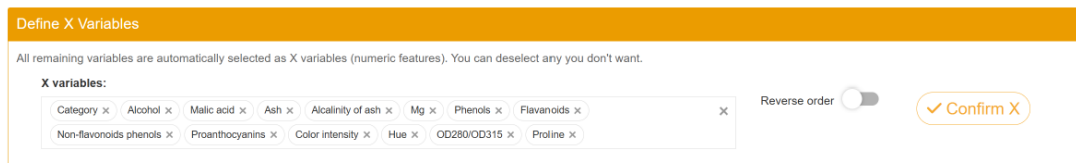


Figure 6.6: Confirm X (predictor variables) step

6.6 Step 6 - Approve

If all variables have been assigned correctly, the message **All variables have been assigned!** will appear. If one or more variables remain unassigned, Cactus will ask whether you want to delete them. You can always edit the configuration again later by clicking **Restart** and repeating the editing process from the beginning. Once everything is correct, click **Approve** to confirm your choices and proceed to the **Split Train/Test** tab.

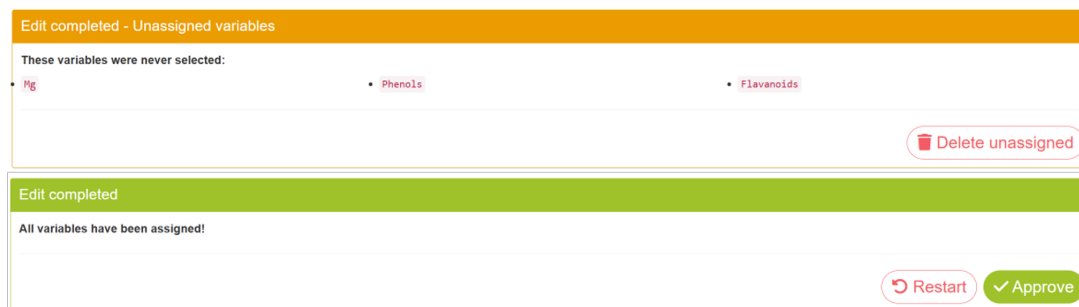


Figure 6.7: Approve pipeline step

6.7 Step 7 - Split into Train and Test (optional)

First, decide whether you want to split the dataset into separate **Training** and **Test** sets. If you plan to build and evaluate a predictive model, or simply want to divide your data, click the green **Yes, Enable split** button. To continue using the entire dataset without creating a separate test set, select **No, keep full dataset**, and continue in the Missing Values section.

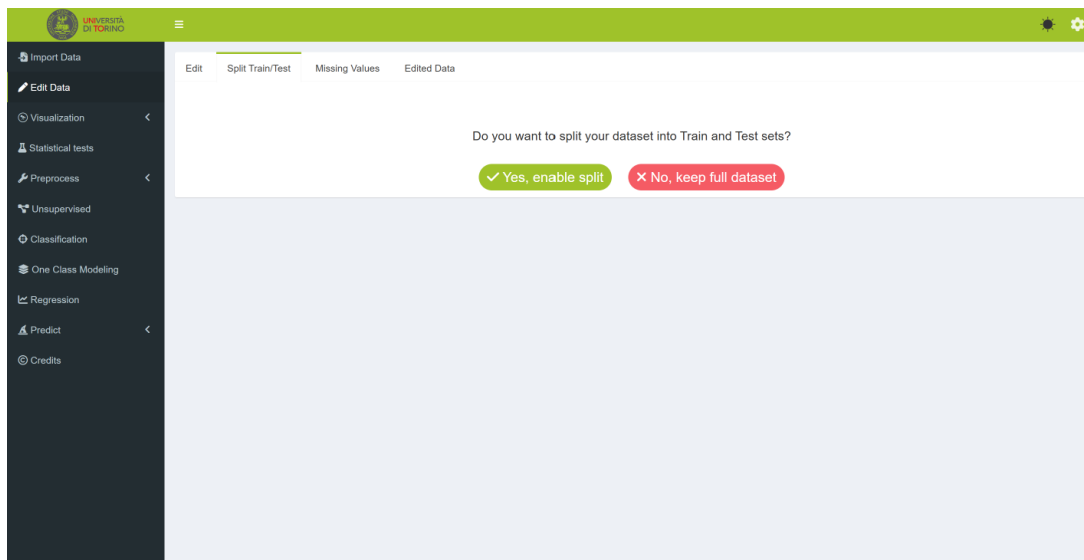


Figure 6.8: Split Train/Test configuration

If splitting is enabled, the **Method** dropdown menu will display the available splitting methods. After selecting a method, use the **Settings** section to configure its specific parameters and define the desired training-set percentage. The information box provides a brief description of the selected method. For a complete explanation of its principles, advantages, and limitations, refer to the bibliography reported in the same box. Once the configuration is complete, click **Run split**.

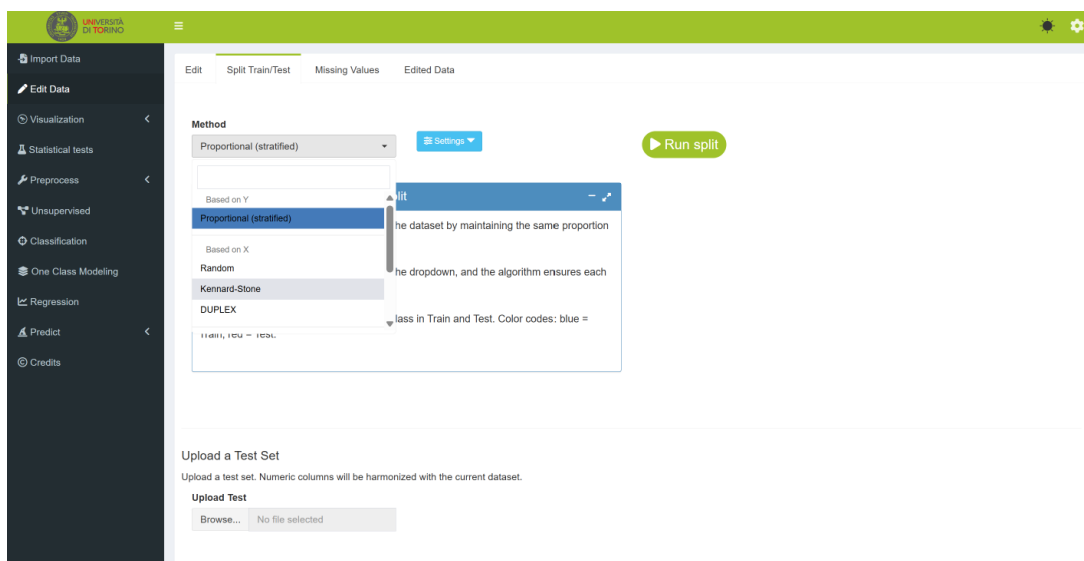


Figure 6.9: Split Train/Test Page

Note

Cactus offers six split methods, grouped by what they use to decide the split:

- **Proportional (stratified)** – based on your Y variable.
- **Random** – based on X, fully
- **Kennard-Stone** – based on X, picks a representative Training set.
- **DUPLEX** – based on X, picks a representative Test set.

- **Sample Order/Time** – based on a date column or import order.
- **Manual** – you choose the Te

Every method except Manual and Sample Order/Time us the same split can be reproduced later.

6.7.1 Proportional (stratified)

This method performs the split according to a selected response variable (Y), preserving its class or value distribution as closely as possible between the **Training** and **Test** sets. It is particularly useful when some classes contain relatively few samples, as it reduces the risk of assigning most or all samples from a rare class to only one of the two sets.

Use the slider to select the percentage of samples assigned to the Training set; the remaining samples will be assigned to the Test set. The default Training percentage is (70%). The **Seed** controls the random component of the split and ensures reproducibility: using the same dataset, settings, and seed value will produce the same Training/Test split.

The method can also be applied to a continuous numeric (Y) variable. In this case, its values are automatically divided into a predefined number of bins (**Y bins**, default: 10), and the split is then stratified according to these intervals.

The optional **Grouping factor (Replicates)** can be used to keep related observations together. Replicate information must already be included as a column in the original imported dataset. All measurements belonging to the same sample or replicate group must have exactly the same label, as shown in Figure 6.10b. Cactus will then assign all rows sharing that label entirely to either the Training or Test set, preventing repeated measurements of the same sample from being divided between the two sets.

(a) Settings available for the Proportional (stratified) splitting method.

ID	Category	Replicates
OLO1	Barolo	Wine1
OLO2	Barolo	Wine1
OLO3	Barolo	Wine1
OLO4	Barolo	Wine1
OLO5	Barolo	Wine1
OLO6	Barolo	Wine2
OLO7	Barolo	Wine2
OLO8	Barolo	Wine2
OLO9	Barolo	Wine2
OLO10	Barolo	Wine2

(b) Example of replicate labels included in the original dataset. Rows belonging to the same replicate group must share the same label.

Figure 6.10: Proportional stratification settings and example of replicate-group labels.

6.7.2 Random

Splits your data completely at random, according to the **Train percentage**. The **Seed** controls the random component of the split and ensures reproducibility: using the same dataset, settings, and seed value will produce the same Training/Test split. An optional **Grouping factor (Replicates)** keeps them, exactly as in Proportional.

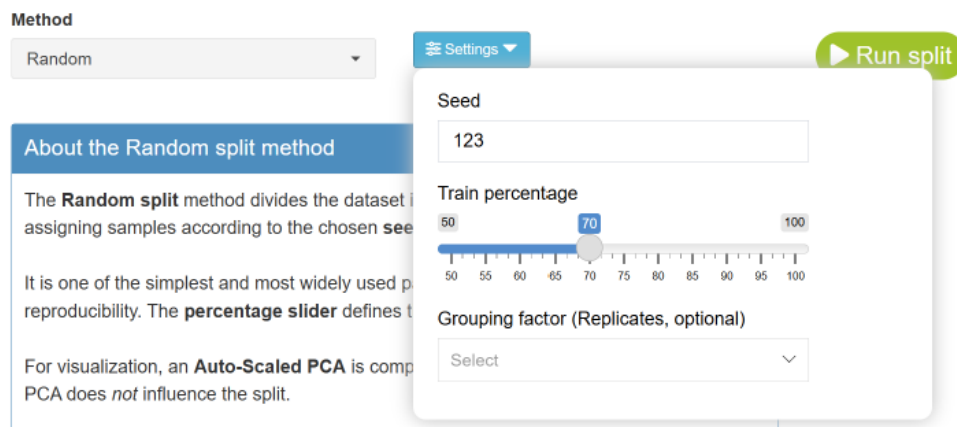


Figure 6.11: Random split settings

6.7.3 Kennard–Stone

Unlike random splitting methods, Kennard–Stone selects samples according to their position in the predictor space (X). The algorithm builds a Training set whose samples are distributed as evenly as possible across the available data space. This generally produces a representative Training set and reduces the risk of obtaining an unbalanced split due to random selection.

The available settings are:

- **Distance metric** - Defines how the distance between samples is calculated, Mahalanobis or Eclidean.
- **Preprocessing** - Applies **None**, **Center**, or **Autoscale** to the predictor variables before calculating the distances.
- **Use PCA** - Performs sample selection in PCA score space instead of using the original predictor variables. This option is enabled by default and can be useful when the dataset contains many variables or highly correlated predictors. The number of components can be specified directly or as a fraction of explained variance, for example 0.95 to retain 95% of the variance.
- **Grouping factor (Replicates)** - same that Proportional case.
- **Initial samples** - Optionally specifies samples that must be included in the Training set before the automatic selection begins.

Method

Kennard-Stone

About the Kennard-Stone split

The **Kennard-Stone algorithm** selects training samples uniformly as possible. It iteratively chooses samples based on Euclidean or Mahalanobis distance.

This deterministic method ensures the training samples are representative while remaining samples form the test set.

Reference: Kennard, R. W., & Stone, L. A. (1968). [Experiments.](#)

Upload a Test Set

Upload a test set. Numeric columns will be harmonized.

Upload Test

Browse... No file selected

Settings

Seed

123

Train percentage

50 55 60 65 70 75 80 85 90 95 100

Distance metric

☒ Mahalanobis ☐ Euclidean

Preprocessing

☐ None ☒ Center ☐ Autoscale

☒ Use PCA

PC components (count or <1 fraction)

0.95

Selected: 1 PCs (approx 95.4% cumulative variance)

Grouping factor (Replicates)

Select

Initial samples (optional, ordered)

Select

Run split

Figure 6.12: Kennard-Stone split settings

Warning

Kennard-Stone cannot be used when (X) contains missing values. Handle them first, save the imputed dataset, and then restart the app and return to the split step. Be careful: imputing before splitting may introduce **data leakage**.

6.7.4 DUPLEX

DUPLEX is a variant of Kennard-Stone that distributes both the Training and Test samples evenly across the predictor space (X). Unlike Kennard-Stone, it also aims to make the Test set representative rather than simply assigning the remaining samples to it.

It uses the same settings as Kennard-Stone, including distance metric, preprocessing, PCA, grouping factor, and initial samples. The same restriction regarding missing values also applies.

Method

DUPLEX

About the DUPLEX algorithm

The **DUPLEX** method extends Kennard-Stone to create training and test sets that are both representative of the data.

It ensures diversity in both subsets and avoids correlation in the partition.

Reference: Snee, R. D. (1977). *Validation of regression models*.

Settings

Seed: 123

Train percentage: 70

Distance metric: ☒ Mahalanobis ☐ Euclidean

Preprocessing: ☐ None ☒ Center ☐ Autoscale

☒ Use PCA

PC components (count or <1 fraction): 0,95

Selected: 1 PCs (approx 95.4% cumulative variance)

Grouping factor (Replicates): Select

Run split

Figure 6.13: DUPLEX split settings

6.7.5 Sample Order/Time

Splits the dataset according to a selected date/time variable or, if none is selected, according to the original sample order. The first percentage of samples is assigned to the Training set, while the remaining samples are assigned to the Test set.

This method does not use a random seed.

Method

Sample Order/Time

About the Sample Order/Time split

The **Sample Order/Time split** divides the dataset into two sets based on a selected **date/time variable**. The first $p\%$ of samples go to **Test**.

This approach preserves the original sequence of samples to predict earlier ones, reducing the risk of data leakage.

If a **date/time variable** is selected, the dataset is sorted by that column before splitting. Otherwise, the current **sample order** is preserved.

The visualization shows the sample sequence on the X-axis and the corresponding ordered rank on the Y-axis. A dashed vertical line marks the Train/Test boundary.

Settings

Train percentage: 70

Order variable: Sample order

The dataset is sorted by this variable and split at the chosen proportion. No randomness involved.

Run split

Figure 6.14: Sample Order/Time split settings

6.7.6 Manual

Allows you to search for and manually select the rows assigned to the Test set. All remaining rows are assigned to the Training set.

This method is useful when you already know which samples should be held out.

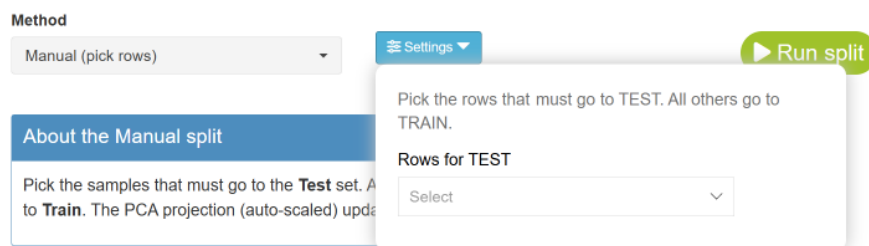


Figure 6.15: Manual split row selector

6.8 Run Split and Graphical Overview

After configuring the selected splitting method, click **Run Split** to generate the Training and Test sets.

Cactus then displays a graphical summary of the split. Depending on the selected method, the plot shows either the proportions of samples assigned to the Training and Test sets or a scatter plot illustrating how the samples were selected. Training samples are shown in **blue**, while Test samples are shown in **red**.

The visualization can be used to verify that the split is consistent with the chosen settings before continuing with the analysis.

Tip

If you entered the Split module by mistake and do not want to create a Test set, select either the **Random** or **Proportional** method and set the Training percentage to **100%**. All samples will remain in the Training set, allowing you to continue as if you had selected **No, keep full dataset** before entering the module.

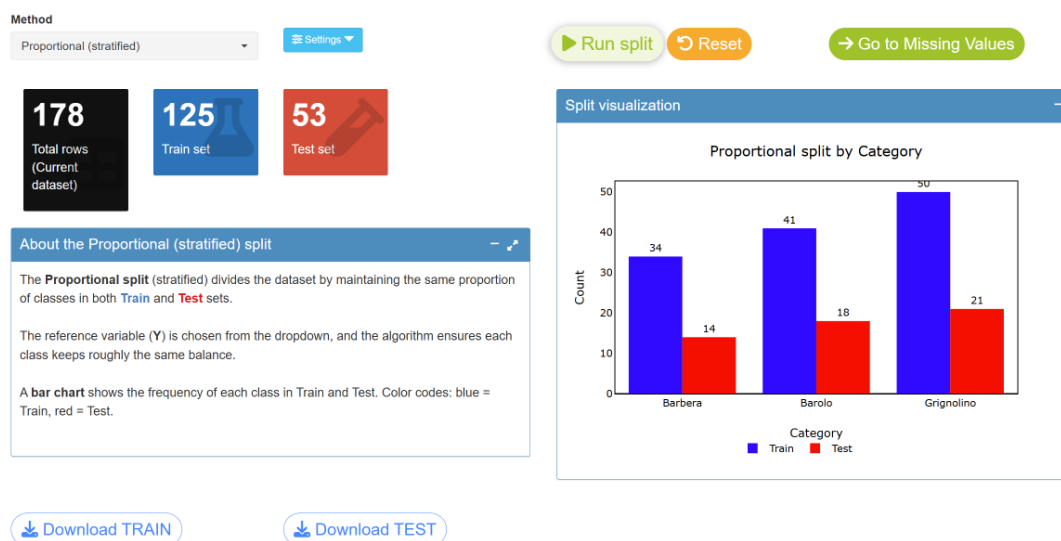


Figure 6.16: Split results and graphical overview

6.8.1 Uploading a Test Set

Instead of, or in addition to, splitting the main dataset, you can upload a separate file to use as an uploaded Test set from the **Upload a Test Set** tab.

The same file formats supported by the Import module are accepted (.csv, .txt, .xls, and .xlsx), together with the same advanced options, including delimiter, decimal symbol, row labels, header, and empty-value settings.

Before loading the file, Cactus compares it with the main dataset and performs the following checks:

- **Missing columns** - If the uploaded file does not contain a column present in the main dataset, the file is rejected and Cactus lists the missing columns.
- **Extra columns** - Columns present in the uploaded file but not in the main dataset are automatically removed, and a warning is displayed.
- **Type mismatches** - If a column has a different data type from the corresponding column in the main dataset, Cactus attempts to convert it. Values that cannot be converted are replaced with missing values, and a warning reports how many values were affected.
- **Duplicate row names** - If a sample name already exists in the main dataset, it is automatically modified to avoid duplicated row names.

Click **Load Test Set** to confirm the upload. A summary reports the number of loaded samples and columns and displays any warnings related to automatic corrections. Click **Delete Test Set** to remove the uploaded Test set.



Figure 6.17: Upload a Test Set panel

Tip

An uploaded Test set is stored separately from the internal Train/Test split. To use only the uploaded Test set, set the internal Training percentage to **100%**. Otherwise, the main dataset will also be split.

6.9 References

Random, Proportional, Sample Order/Time and Manual are general-purpose techniques with no single paper attached. Kennard-Stone and DUPLEX cite a specific reference, shown in their info box:

- **Kennard-Stone** – R. W. Kennard and L. A. Stone (1969). “Computer Aided Design of Experiments”. In: *Technometrics*. DOI: [10.1080/00401706.1969.10490666](https://doi.org/10.1080/00401706.1969.10490666)
- **DUPLEX** – R. D. Snee (1977). “Validation of regression models: Methods and examples”. In: *Technometrics*. DOI: [10.1080/00401706.1977.10489581](https://doi.org/10.1080/00401706.1977.10489581)

6.10 Step 8 - Handle missing values

Once your data is split, Cactus scans it for missing values (NAs). If none are found, you will see a green "No missing values detected" message – just click **Continue to Edited Data** and skip the rest of this step.

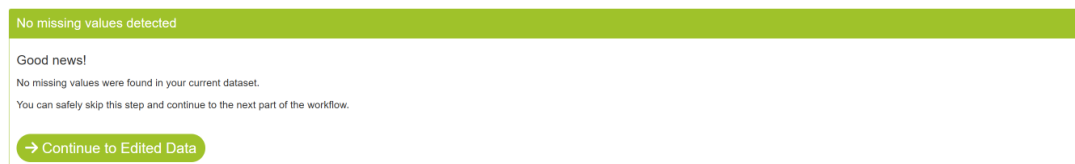


Figure 6.18: No missing values detected

Warning

Imputed values are **temporary** and only affect Visualization, Statistics and Unsupervised exploration. When you later train a supervised model (Classification, One Class Modeling, Regression) with cross-validation, your chosen imputation method is re-applied independently within each fold, so it never "leaks" information from one fold into another.

6.10.1 Missing values overview

At the top of the tab, a summary line shows the total number of unresolved missing values and how many columns they affect. Below it, two tables help you decide what to do:

- **Summary table** - one row per column, with the count and percentage of missing values overall, and separately for Train and Test (if a split exists). Cells are colour-coded: white = 0%, green < 30%, gold 30–50%, red > 50%. If a column is mostly missing, it is usually better to go back to the **Edit** tab and delete it instead of imputing it.
- **Positions table** - lists every individual missing cell (row name, column, type), and, once imputed, the value that was filled in and which method was used. Cells that have already been imputed are highlighted in gold.

Missing values overview									
Total missing values on active columns: 0 across 0 columns.									
Color Legend: 0% < 30% 30-50% > 50%									
If the missing percentage is high, it is usually better to delete the column. Go back to the Edit section and remove it there instead of imputing.									
Column	Type	Total N	Total %	row_index	row_name	column	type	imputed_value	meth
Alcohol	numeric	1	0.56	35	OLO35	Alcohol	numeric	13	Mean
Malic acid	numeric	1	0.56	19	OLO19	Malic acid	numeric	2.34	Mean
Category	factor	0	0						
Data	Date	0	0						
Ash	numeric	0	0						
Alcalinity of ash	numeric	0	0						
Mg	numeric	0	0						
Phenols	numeric	0	0						
Flavanoids	numeric	0	0						
Non-flavanoids phenols	numeric	0	0						

Figure 6.19: Missing values overview: summary and positions tables

6.10.2 Choosing an imputation method

Cactus offers six imputation methods, selected from the **Imputation method** dropdown:

Note

- **Mean (numeric)** – fills gaps with the column average.
- **Median (numeric)** – fills gaps with the column median (more robust to outliers).
- **KNN (numeric)** – estimates each value from similar samples.
- **Random (LOD–LOQ range)** – fills gaps with a random value in a chosen range.
- **Interpolation (spectral/adjacent)** – fills gaps using neighbouring columns.
- **Manual (single cell override)** – you type the replacement value yourself.

For every method except Manual, select the **Numeric columns to impute** and click **Run imputation**; a description of the selected method is also shown live on the tab.

Figure 6.20: Imputation method dropdown and column picker

6.10.3 Mean

Replaces missing values in the selected numeric columns with the column’s average, calculated only from the **Train** rows. The same value is then used for both Train and Test rows.

6.10.4 Median

Same idea as Mean, but uses the median instead of the average - calculated on Train rows only and applied to both Train and Test.

6.10.5 KNN

Estimates each missing value from the **K** most similar samples (nearest neighbours), based on the other numeric columns. Set **Number of neighbors (K)** (default 5). The neighbour search is fitted on Train rows and then applied to all rows.

Warning

KNN cannot use your classification, regression (Y), or date columns as imputation targets - Cactus removes them automatically from the selection and warns you if you tried to include one.

6.10.6 Random (LOD–LOQ range)

Fills gaps with a random value drawn from a uniform distribution between a **Random min value** and **Random max value** that you set – typically the instrument’s limit of detection/quantification (LOD–LOQ). A **Random seed** keeps the result reproducible.

Imputation missing values

Imputation method
Random (LOD-LOQ range)

Numeric columns to impute
Alcohol, Malic acid

Random min value
0

Random max value
1

Random seed
123

Run imputation

Method description

Random imputation (LOD-LOQ range)
Missing values in the selected numeric columns are replaced by random values drawn from a uniform distribution between the user-specified **min** and **max** (typically the LOD-LOQ range), using a reproducible random seed. *Imputed values will match the decimal precision of existing data.*

Figure 6.21: Random (LOD-LOQ) imputation settings

6.10.7 Interpolation (spectral/adjacent)

Fills a missing value by looking at the nearest non-missing *columns* to its left and right (by column position) and interpolating linearly between them; if only one side has data, that value is used directly. This is the recommended method for **spectroscopic data** (NIR, Raman, UV-Vis), where the signal varies smoothly across wavelengths. It is applied directly, with no training statistics involved, so there is no risk of data leakage during cross-validation.

6.10.8 Manual (single cell override)

Pick a **Column** and a **Row** (by name) that has a missing value, type the **New value**, then click **Save imputation (preview)**. Nothing is applied immediately – it is only saved as a preview, so you can review every manual change before committing them all at once in the next step.

6.11 Step 9 - Confirm and continue

Once every missing value has a method assigned, the **Confirm imputation & proceed** button becomes active (it stays disabled as long as any value is still unresolved). Use **Reset all imputations** at any point to discard everything and start over.

Actions

Reset all imputations

Confirm imputation & proceed

Figure 6.22: Missing Values overview and imputation controls

6.12 Step 9 - Approve the final dataset

On the **Edited Data** tab, use the radio buttons to switch between the available datasets: **Full**, **Training**, and **Test**. Check the preview table, and use the **Settings** menu, identified by the gear icon, to download the selected dataset. Once everything has been verified, click **Approve Dataset** to unlock the remaining sections of the application.

Dataset: ☒ Full ☐ Train ☐ Test

✓ Approve Dataset

Edited Data

Show number of rows: 20

Show number of columns: 10

☐ Show entire dataset

Download CSV

		Malic acid	Ash	Alkalinity of ash	Mg	Phenols	Flavanoids	Non-flavonoids phenols	Proanthocyanins
		2.36	2.67	18.6	101	2.8	3.24	0.3	2.81
		1.95	2.5	16.8	113	3.85	3.49	0.24	2.18
		2.59	2.87	21	118	2.8	2.69	0.39	1.82
		1.87	2.45	14.6	96	2.5	2.52	0.3	1.98
OLO8	Barolo	14.06	2.15	2.61	17.6	121	2.6	2.51	0.31
OLO9	Barolo	14.83	1.64	2.17	14	97	2.8	2.98	0.29
OLO10	Barolo	13.86	1.35	2.27	16	98	2.98	3.15	0.22
OLO11	Barolo	14.1	2.16	2.3	18	105	2.95	3.32	0.22
OLO12	Barolo	14.12	1.48	2.32	16.8	95	2.2	2.43	0.26
OLO13	Barolo	13.75	1.73	2.41	16	89	2.6	2.76	0.29

Showing 1 to 10 of 20 entries

Previous 1 2 Next

Figure 6.23: Edited Data preview with Approve Dataset button

Tip

Approving the dataset again later (after making more changes) resets any models you have already trained - this keeps your results consistent with the data.

6.13 Common mistakes

- Leaving a column unassigned - Cactus will ask you to delete it before you can Approve.
- Choosing **Kennard-Stone** or **DUPLEX** as a split method while missing values still exist - resolve missing values first, or pick a different method.
- Uploading an external test set that is missing a column from the main dataset - rejected outright.
- Trying to impute classification/date columns with **KNN** - not available for those columns.

Chapter 7

Visualization

The Visualization module lets you look at your data as charts. It has three sidebar sub-items: **One Variable**, **Two Variables** and **Multiple Variables**.

Note

Every chart is interactive: hover for details, zoom, pan, and export it as an image. This works the same way everywhere in Cactus - see Chapter 8 for the full guide.

7.1 Step 1 - Pick a tab

Choose **One Variable**, **Two Variables**, or **Multiple Variables** from the sidebar, depending on how many columns you want to compare at once.

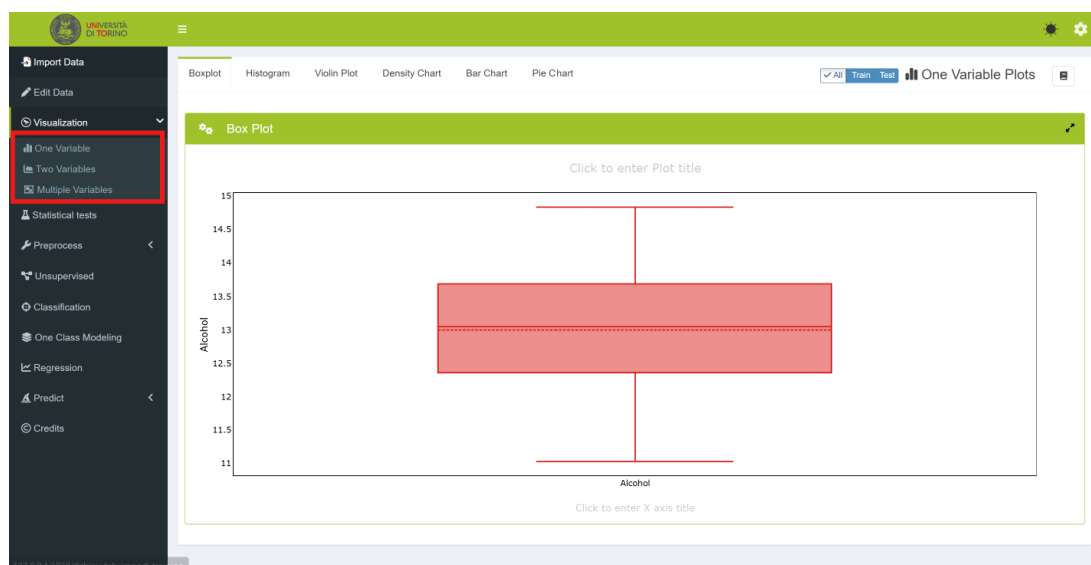


Figure 7.1: Visualization sidebar and tab selection

7.2 Step 2 - Pick a chart type

Each tab has its own set of chart types along the top. Click one to open it.

Tab	Chart types
One Variable	Boxplot, Histogram, Violin Plot, Density Chart, Bar Chart, Pie Chart
Two Variables	Scatter Plot, Bubble Plot, 2D Histogram, 2D Density Chart, Scatter Matrix, Time Series
Multiple Variables	3D Scatter Plot, Boxplots (multi), Heatmap, Spectra / Parallel Coordinates

7.3 Step 3 - Choose your variables or Adjust the chart

Open the **Settings** dropdown (gear-icon) to customize the appearance of the selected chart. Depending on the chart type, you can adjust options such as point size, number of bins, line appearance, transparency, labels, and other display parameters. The chart is updated automatically whenever a setting is changed.

Most charts also allow observations to be coloured according to either a categorical or continuous variable. Categorical variables can be used to distinguish classes or groups, with the option to customise the colour assigned to each category. Continuous numeric variables are instead represented using a colour gradient, making it easier to identify changes, trends, or patterns across their range of values.



Figure 7.2: Chart settings dropdown

7.4 Step 4 - Read the chart

Hover over any point, bar or line to see its exact value. Use the toolbar on the right to zoom, reset, or save the chart as an image (Chapter 8).

7.4.1 One Variable

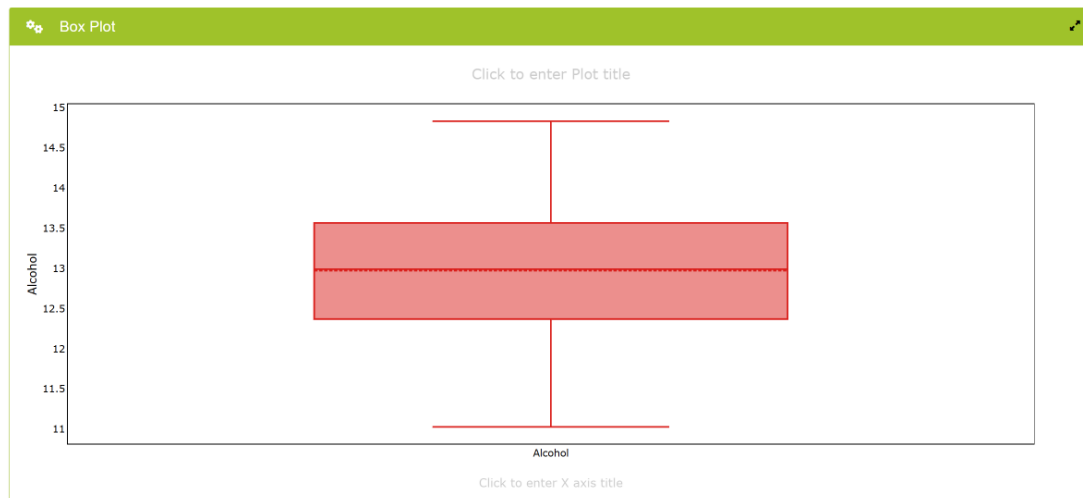


Figure 7.3: Boxplot



Figure 7.4: Histogram

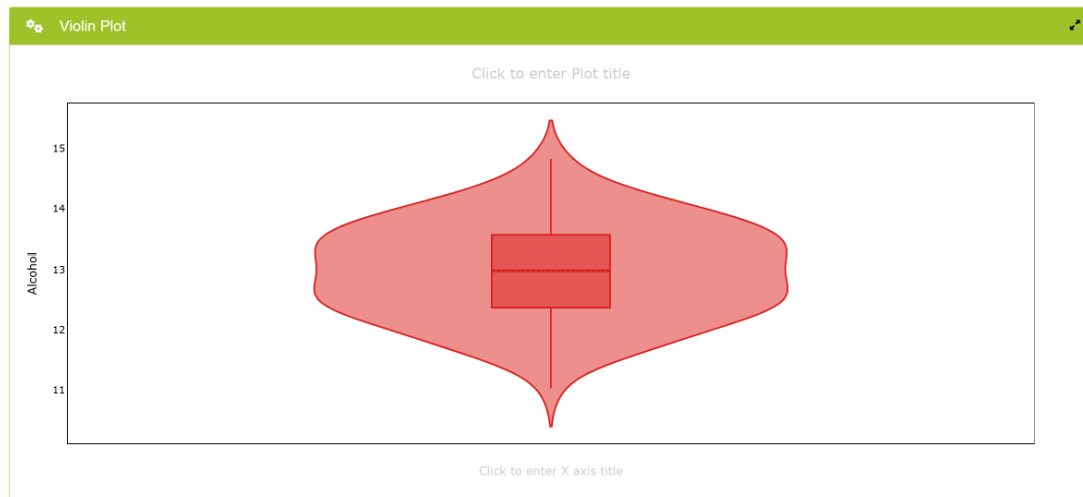


Figure 7.5: Violin Plot

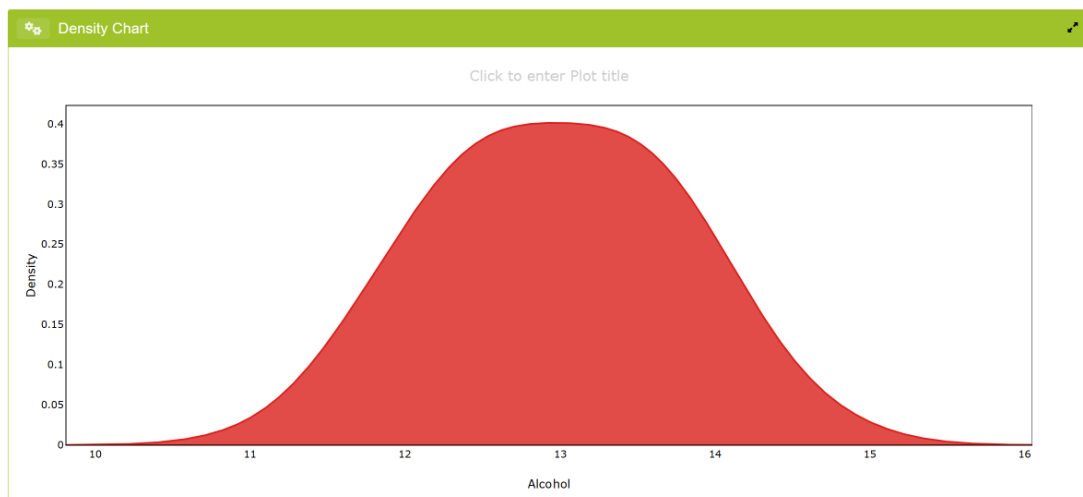


Figure 7.6: Density Chart

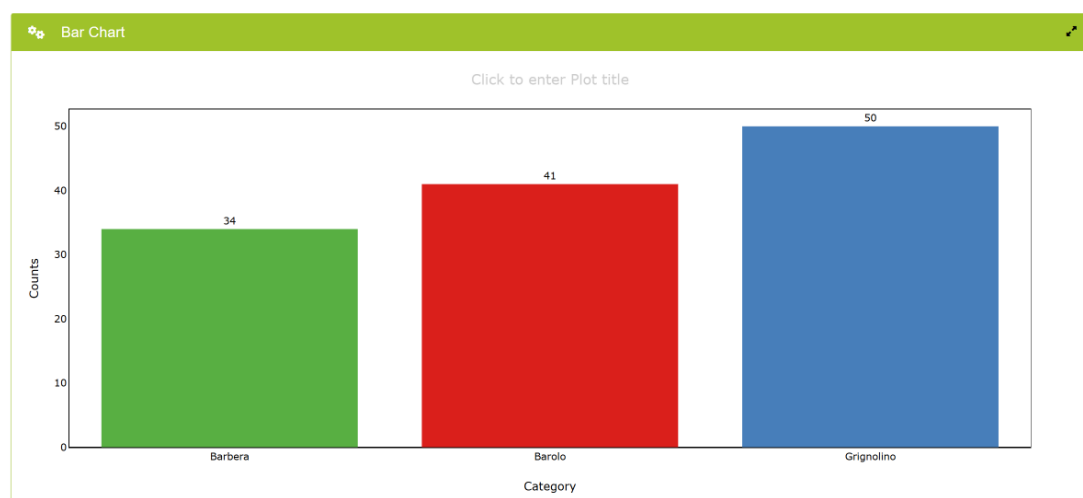


Figure 7.7: Bar Chart

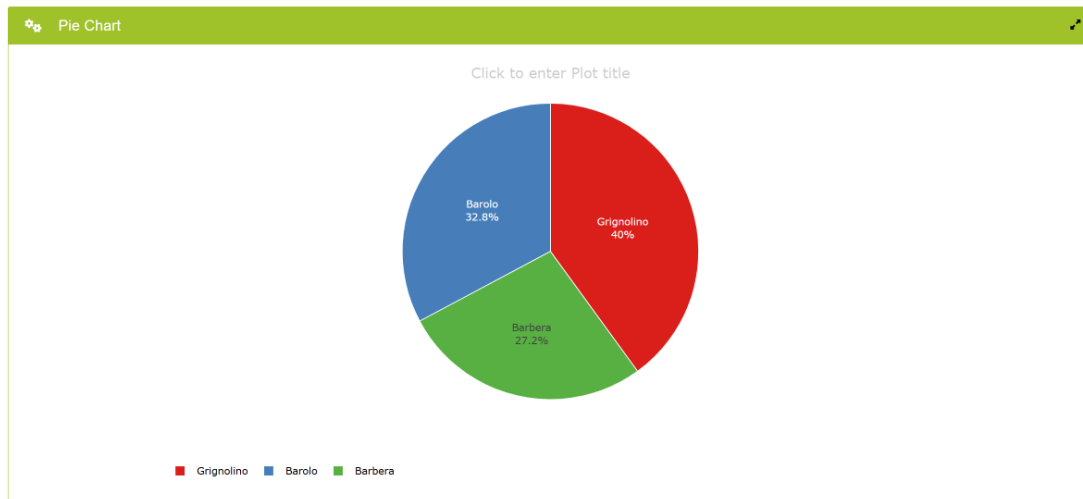
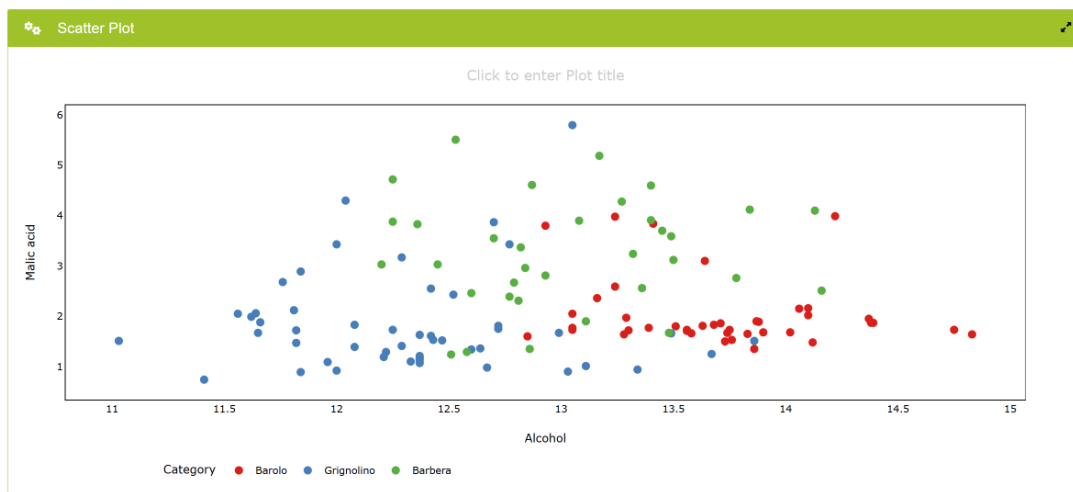


Figure 7.8: Pie Chart

7.4.2 Two Variables



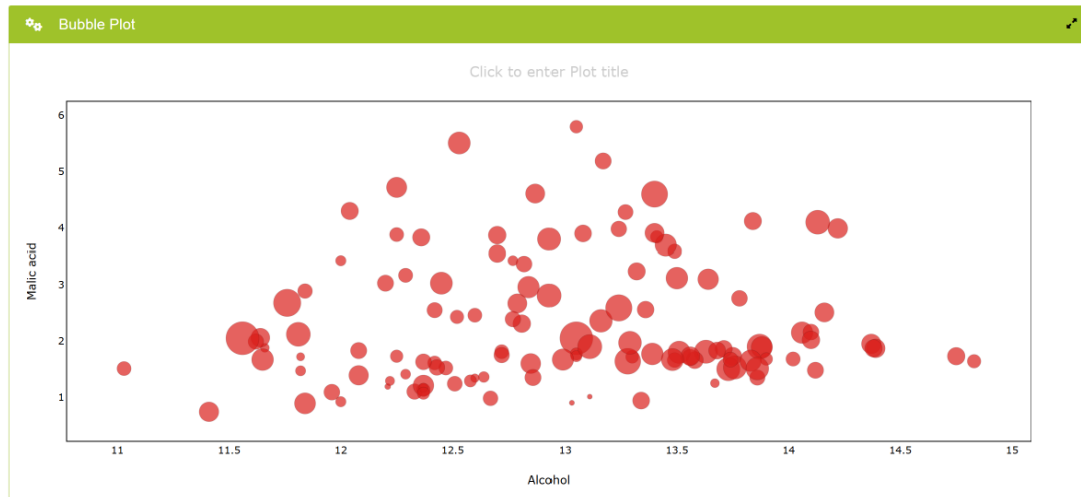


Figure 7.10: Bubble Plot

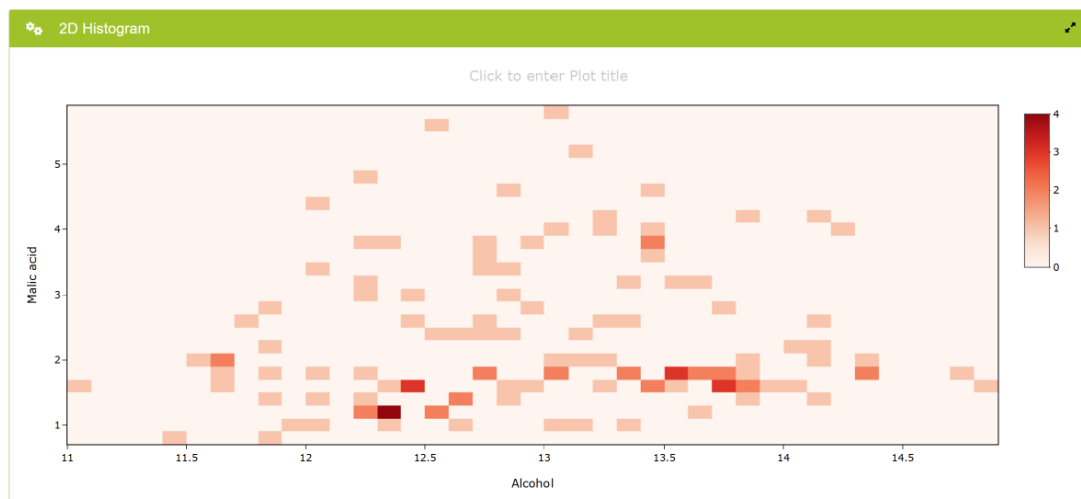


Figure 7.11: 2D Histogram

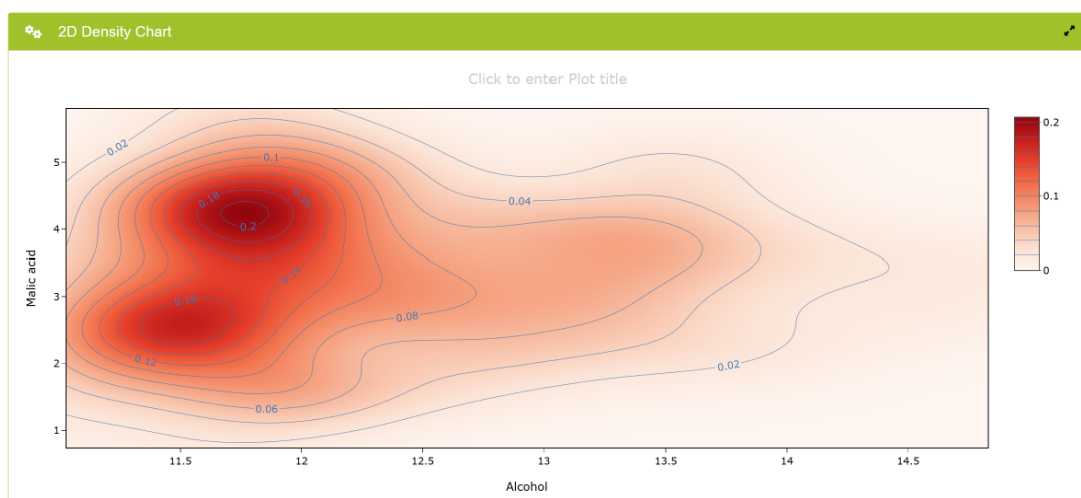


Figure 7.12: 2D Density Chart

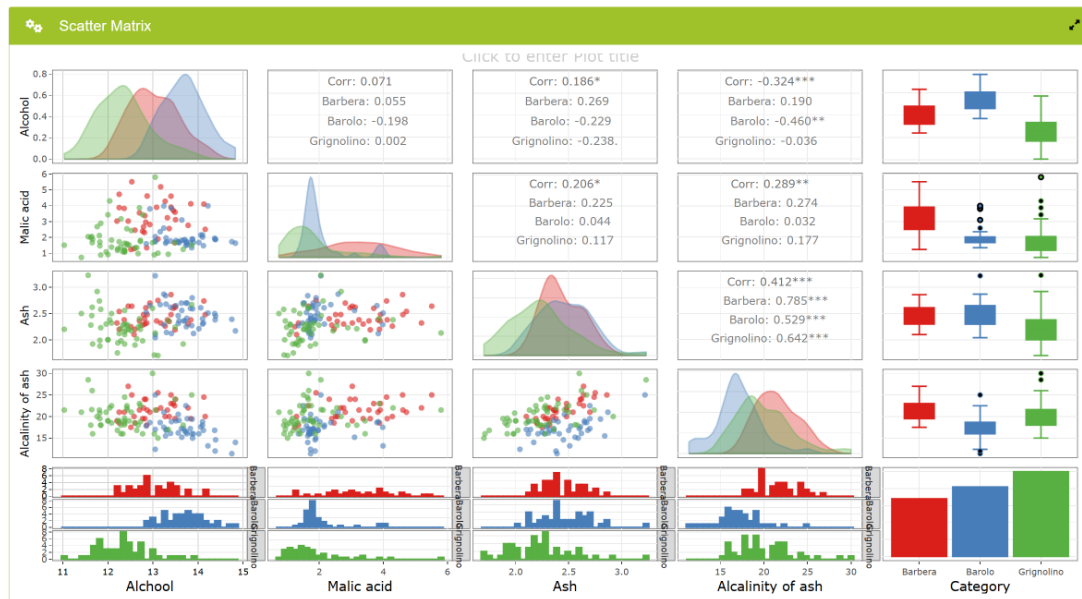


Figure 7.13: Scatter Matrix

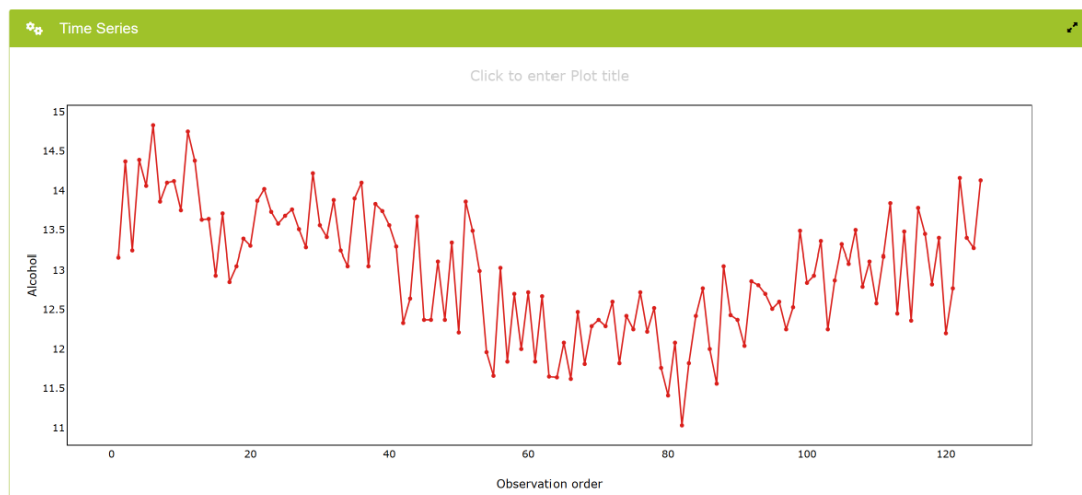


Figure 7.14: Time Series

7.4.3 Multiple Variables

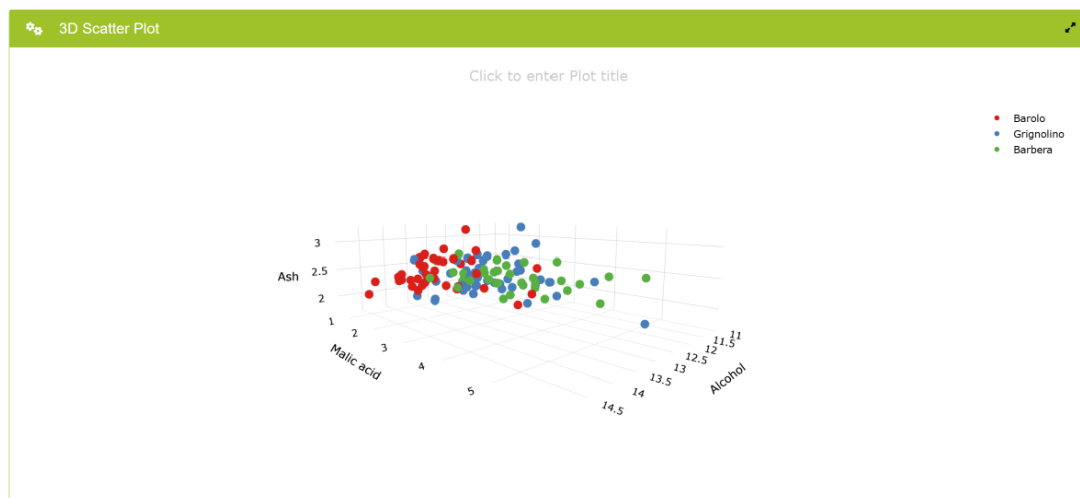


Figure 7.15: 3D Scatter Plot

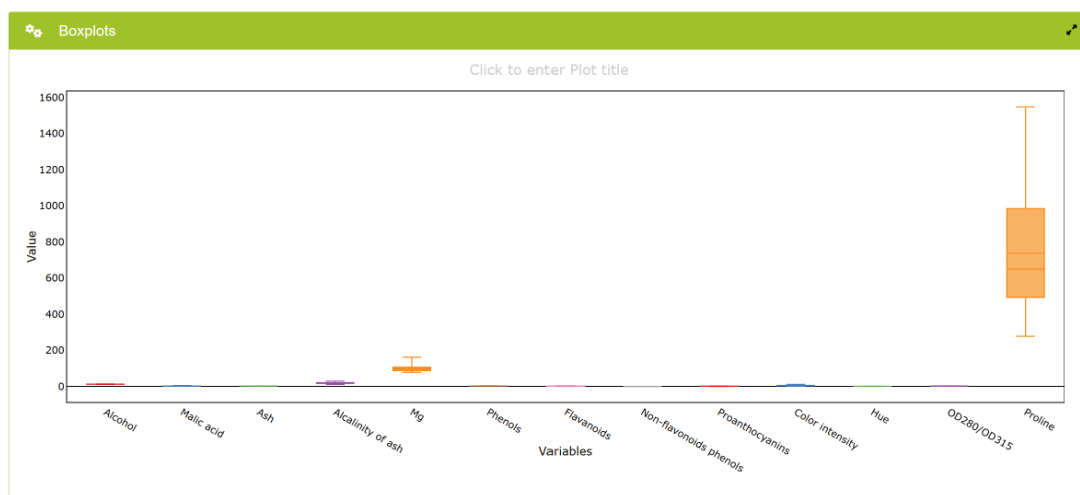


Figure 7.16: Boxplots (multi)

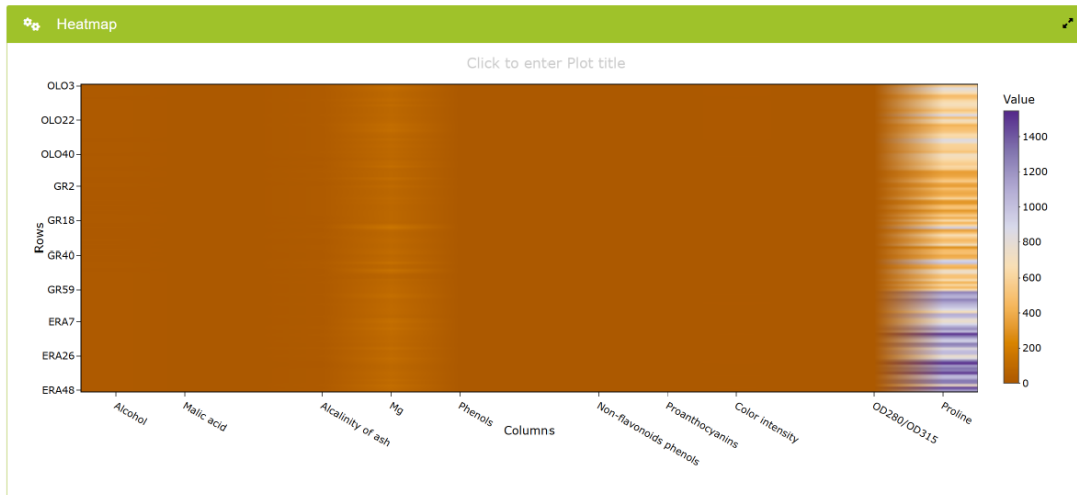


Figure 7.17: Heatmap

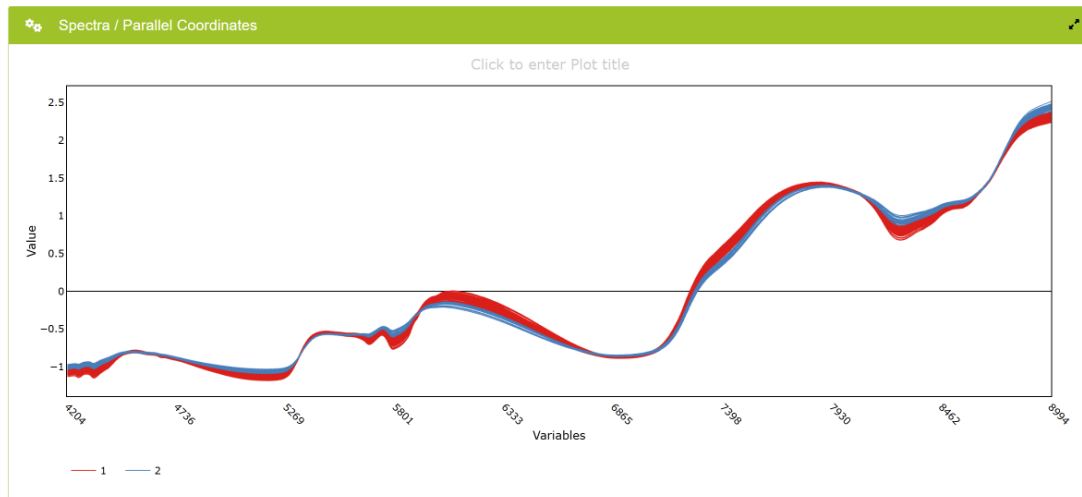


Figure 7.18: Spectra / Parallel Coordinates

7.5 Visualization Module After Train/Test Splitting or Preprocessing

When a Training/Test split has been created or preprocessing has been applied, additional buttons appear in the green bar at the top of the visualization panel. These buttons allow you to switch directly between the available datasets, such as the **Full**, **Training**, **Test**, and **Preprocessed** datasets.

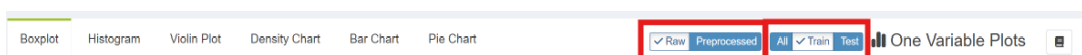
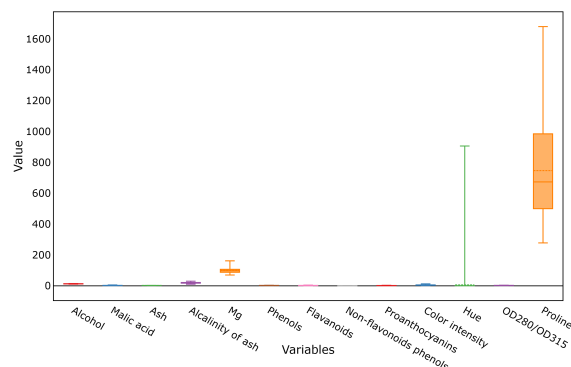
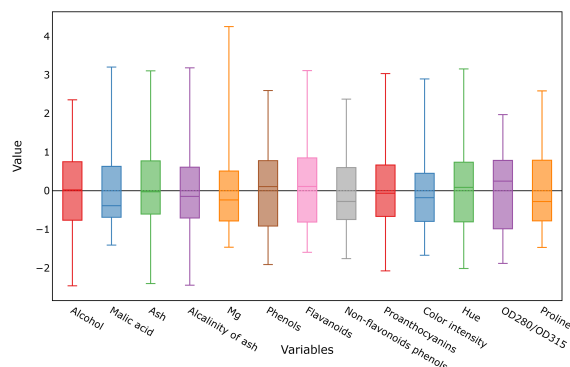


Figure 7.19: Different plot based on preprocessing and splitting

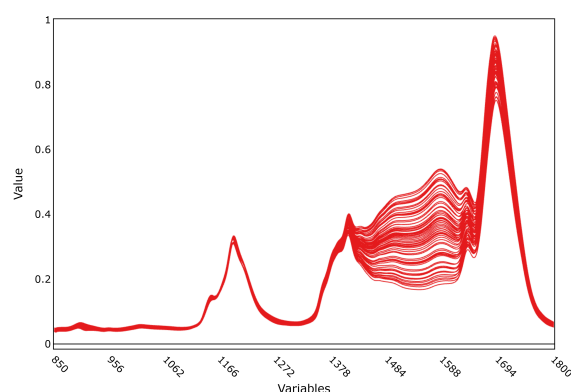
This makes it possible to compare the Training and Test groups visually, check whether the split produced representative subsets, and evaluate how preprocessing has modified the structure, distribution, or overall appearance of the data.



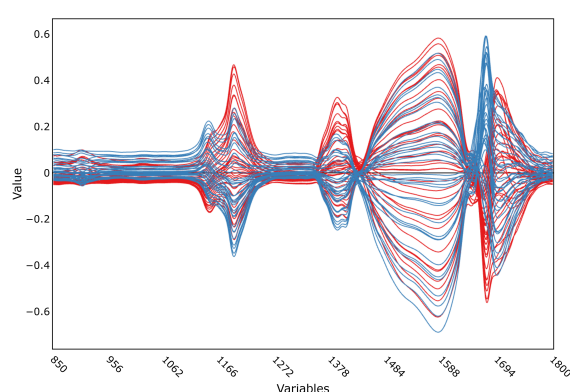
(a) Wines dataset raw data



(b) Wines dataset after autoscaling



(c) NIR Raw spectra



(d) NIR after SNV + Mean Centering

Figure 7.20: Comparison of the available dataset views in the Visualization module.

7.6 Common mistakes

- Comparing groups of very different sizes without checking how many samples are in each.
- Forgetting to click **Plot** again on the Scatter Matrix after changing your variable selection - it does not update automatically.
- Using a Pie Chart for a variable with many categories, which makes slices hard to compare.
- Choosing "Original order" on a Time Series chart and mistaking it for a real time axis - only an actual date column gives a true time axis.

7.7 References

The following citations are shown via the **book icon**, separately on each of the three Visualization tabs.

7.7.1 One Variable

- **General** – J. W. Tukey (1977). *Exploratory Data Analysis*. Vol. 2. Addison-Wesley. URL: https://openlibrary.org/works/OL9572429W/Exploratory_data_analysis
- M. L. Vigni, C. Durante, and M. Cocchi (2013). "Exploratory Data Analysis". In: *Comprehensive Analytical Chemistry*. Vol. 28. Elsevier. DOI: [10.1016/B978-0-444-59528-](https://doi.org/10.1016/B978-0-444-59528-)

7.00003-X

D. W. Scott (2015). *Multivariate Density Estimation: Theory, Practice, and Visualization*. Wiley. DOI: [10.1002/9781118575574](https://doi.org/10.1002/9781118575574)

- **Boxplot** – R. McGill, J. W. Tukey, and W. A. Larsen (1978). “Variations of box plots”. In: *The American Statistician* 32.1, pp. 12–16. DOI: [10.2307/2683468](https://doi.org/10.2307/2683468)
- D. L. Massart, J. Smeyers-Verbeke, et al. (2005). “Visual presentation of data by means of box plots”. In: *LC-GC Europe* 18.4. URL: <https://researchportal.vub.be/en/publications/visual-presentation-of-data-by-means-of-box-plots>
- **Histogram** – D. Freedman, R. Pisani, and R. Purves (1978). *Statistics*. W. W. Norton & Co. DOI: [10.2307/1268533](https://doi.org/10.2307/1268533)
- **Violin Plot** – J. L. Hintze and R. D. Nelson (1998). “Violin plots: a box plot-density trace synergism”. In: *The American Statistician* 52.2, pp. 181–184. DOI: [10.1080/00031305.1998.10480559](https://doi.org/10.1080/00031305.1998.10480559)
- **Density Chart** – S. Weglarczyk (2018). “Kernel density estimation and its application”. In: *ITM Web of Conferences* 23. DOI: [10.1051/itmconf/20182300037](https://doi.org/10.1051/itmconf/20182300037)

7.7.2 Two Variables

- **Scatter Plot** – K. L. Sainani (2016). “The value of scatter plots”. In: *PM&R* 8.12, pp. 1213–1217. DOI: [10.1016/j.pmrj.2016.10.018](https://doi.org/10.1016/j.pmrj.2016.10.018)
- **2D Histogram** – D. W. Scott (2010). “Histogram”. In: *WIREs Computational Statistics* 2.1, pp. 44–48. DOI: [10.1002/wics.59](https://doi.org/10.1002/wics.59)
- **2D Density Chart** – S. Weglarczyk (2018). “Kernel density estimation and its application”. In: *ITM Web of Conferences* 23. DOI: [10.1051/itmconf/20182300037](https://doi.org/10.1051/itmconf/20182300037)

7.7.3 Multiple Variables

- **3D Scatter Plot** – D. B. Wright and S. E. Williams (2005). “Three Dimensional (3D) Scatterplots”. In: *Encyclopedia of Statistics in Behavioral Science*. DOI: [10.1002/0470013192.bsa679](https://doi.org/10.1002/0470013192.bsa679)
- **Bubble Plot** – L. Bessler (2023). “Bubble Plots”. In: *Visual Data Insights Using SAS ODS Graphics*. Apress, pp. 263–281. DOI: [10.1007/978-1-4842-8609-8_7](https://doi.org/10.1007/978-1-4842-8609-8_7)
- **Scatter-plot Matrix** – J. W. Emerson et al. (2013). “The generalized pairs plot”. In: *Journal of Computational and Graphical Statistics* 22.1, pp. 79–91. DOI: [10.1080/10618600.2012.694762](https://doi.org/10.1080/10618600.2012.694762)
- **Boxplots** – R. McGill, J. W. Tukey, and W. A. Larsen (1978). “Variations of box plots”. In: *The American Statistician* 32.1, pp. 12–16. DOI: [10.2307/2683468](https://doi.org/10.2307/2683468)
- **Heat Maps** – N. Gehlenborg and B. Wong (2012). “Heat maps”. In: *Nature Methods* 9.3, p. 213. DOI: [10.1038/nmeth.1902](https://doi.org/10.1038/nmeth.1902)
- **Parallel Coordinates** – R. E. Moustafa (2011). “Parallel coordinate and parallel coordinate density plots”. In: *WIREs Computational Statistics* 3.2, pp. 134–148. DOI: [10.1002/wics.145](https://doi.org/10.1002/wics.145)

Chapter 8

Plots and Visual Outputs

Every chart in Cactus - from a simple boxplot to a PCA scores plot - is built with the same interactive plotting engine (Plotly) and shares one common toolbar and set of gestures. This chapter documents that shared behaviour once, and then indexes every plot type produced by the application.

8.1 Interacting with any plot

Every plot uses the same customised toolbar (right-hand side, appears on hover) on top of Plotly's native interactions - pan, zoom, box/lasso select, and hover tooltips all work as usual.

8.1.1 Toolbar buttons

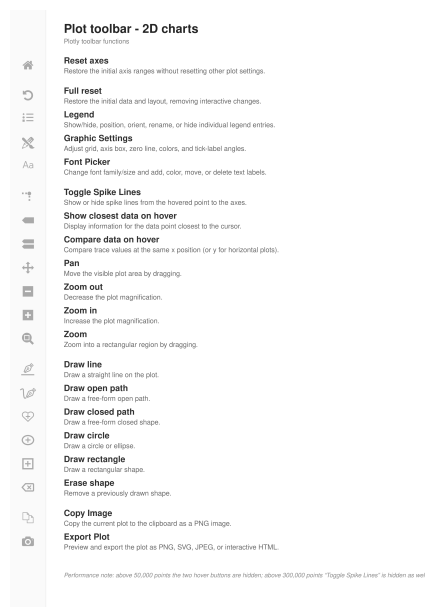


Figure 8.1: Custom toolbar shown on every plot

From top to bottom, the toolbar groups are:

- **Reset axes** (home icon) - restore the initial axis ranges without resetting any other plot setting (2D plots only).

- **Full reset** (undo icon) - restore the initial data and layout, undoing any zoom, style, legend or label change made through this toolbar.
- **Font Picker** - change font family/size, and add, colour, move or delete text labels; **Graphic Settings** (brush icon) - adjust grid, axis box, zero line, colours and tick-label angles; **Legend** - show/hide, position, orient, rename, or hide individual legend entries. Each opens its own window, described in the sections below. On 3D plots only Graphic Settings (background colour only) and Font Picker remain; Legend is not available.
- **Pan / zoom / hover controls** - Plotly's native navigation: zoom (drag a region), zoom in/out, pan, compare data on hover, show closest data on hover, and toggle spike lines. On very large plots (roughly 50 000+ points) spike lines and hover comparison are dropped automatically to keep the plot responsive; above about 300 000 points hovering is disabled entirely and only pan/zoom remain. 3D plots use rotate/pan/zoom controls instead.
- **Drawing tools** (erase shape, rectangle, circle, closed path, open path, line) - Plotly's native annotation shapes, drawn directly on the plot (2D plots only).
- **Copy Image** - copies the plot as a PNG directly to your clipboard, ready to paste into a document or email. The same action is available as a keyboard shortcut, **Ctrl+C** (**Cmd+C** on macOS), while the plot is focused (hover over it first).
- **Export Plot** (camera icon) - opens a dialog to preview and save the plot as PNG, SVG, JPEG, or as a self-contained interactive HTML file. Choose the format, width, height and scale, click **Preview** to check the result, then **Save**.

Tip

Titles, axis titles, legend text and annotation text can also be edited directly by clicking on them in the plot itself (Plotly's "editable" text mode) - you do not need to open a settings window just to fix a typo in a title.

8.1.2 Legend Settings

Click the **Legend** button to open **Legend Settings**: show or hide the legend, choose its orientation (vertical/horizontal) and placement (inside or outside the plot area), fine tune its anchor and X/Y position, and, for each trace, rename it or toggle it on/off in the list at the bottom. Click **Apply** to confirm.

Legend Settings

☐ Show legend

Orientation: Vertical Placement: Inside plot

X anchor: right Y anchor: top

X position: 1 Y position: 1

Legend labels

☒ malic_acid

Close Apply

Figure 8.2: Legend Settings window

8.1.3 Graphic Settings

Click the **Graphic Settings** button (brush icon) to open a window controlling the plot's background colour, and - for 2D plots - the grid, axis box and zero-line visibility, axis/text colour, and the tick-label angle on both axes (via sliders). On 3D plots only the background colour is available, since 3D axes are not exposed the same way. Click **Apply** to confirm.

Graphic Settings

☐ Show grid
☒ Show axis box
☒ Show zeroline

Background color: [color picker]
Axis & text color: [color picker]

X axis tick angle: 0° Y axis tick angle: 0°

Close Apply

Figure 8.3: Graphic Settings window

8.1.4 Font Picker

Click the **Toggle Font Picker** button (its window is titled **Select Font**) to change the font family and size used for the title, axis titles, tick labels and legend all at once, or to add free-standing, movable text labels to the plot. You can pick from a list of common system fonts, or import any [Google Fonts](#) name directly (requires an internet connection). Each added label keeps its own colour and can be deleted individually, or all at once with **Delete all labels**.

Figure 8.4: Select Font window, with the Add label section

Note

Copying or exporting a plot always produces a white background with black text, unless you explicitly chose different colours yourself in Graphic Settings - this keeps a plot made in dark mode (Section 3.3) readable once pasted into a document or printed, instead of exporting with its dark background baked in.

8.2 Plot catalog

The table below lists every plot type produced by Cactus, grouped by module, with a pointer to the chapter that explains its required data, options and interpretation in full.

Module	Plot types	Full details
Visualization - One Variable	Boxplot, Histogram, Violin Plot, Density Chart, Bar Chart, Pie Chart	Chapter 7
Visualization - Two Variables	Scatter Plot, Bubble Plot, 2D Histogram, 2D Density Chart, Scatter Matrix, Time Series	Chapter 7
Visualization - Multiple Variables	3D Scatter Plot, Boxplots (multi), Heatmap, Spectra / Parallel Coordinates	Chapter 7
Statistics	Q-Q plot (Normality), Box plot with significance brackets (t-test, ANOVA), Correlation Heatmap, Clustered Heatmap, Volcano Plot	Chapter 9
Preprocessing	Original vs. Pre-processed comparison plots (Predictors and Outcome tabs)	Chapter 10

Module	Plot types	Full details
Unsupervised PCA	- Scree Plot, Scores Plot (2D/3D/BiPlot), Loadings Plot, Hotelling's T^2 vs. Q Residuals Plot, Contribution Plots	Chapter 11
Unsupervised HCA	- Dendrogram, Silhouette Plot, CA/Original data plots	Chapter 11
Unsupervised - K-means	- Silhouette-score-vs-K curve, Cluster Space plot, Original Variables plot	Chapter 11
Unsupervised - t-SNE	- t-SNE Scores Plot, Feature Coloring plot	Chapter 11
One Class Modeling (SIMCA)	- Acceptance Plot (Score Distance vs. Orthogonal Distance), Contribution Plots, Scores/Loadings/Scree plots, confusion-matrix heatmap	Chapter 14
Classification KNN	- Decision-region scatter plot	Chapter 13
Classification LDA / QDA	- Scores plot, Loadings plot, Class Probability plot, ROC curves	Chapter 13
Classification PLS-DA	- Scores/Loadings/Weights plots, VIP plot, Scree plot (CV accuracy)	Chapter 13
Regression - MLR	- Residuals plot, Influence (Cook's-distance) plot	Chapter 15
Regression - PCR	- PC Scores plot, Loadings plot, Scree plot (CV metric)	Chapter 15
Regression - PLS	- Scores/Loadings/Weights plots, VIP plot, Leverage plot, Scree plot, Measured-vs-Predicted plot, residuals histogram	Chapter 15
Predict	- Same plot families as the originating model type (PCA, SIMCA, KNN, LDA/QDA, PLS-DA, MLR/PCR/PLS), recomputed for the new data	Chapter 16

Chapter 9

Statistics Module

The Statistics module (sidebar entry **Statistical tests**) runs standard statistical tests on your data. Nothing is calculated until you click the Calculate/Test button on a tab.

9.1 Step 1 - Pick a test

Choose a tab: **Normality**, **t-test**, **ANOVA**, **Correlation**, or **Volcano Plot**.

Test	What it tells you
Normality	Whether a variable follows a "normal" bell-curve distribution.
t-test	Whether two groups differ, for one variable.
ANOVA	Whether three or more groups differ, for one variable.
Correlation	How strongly pairs of variables (or samples) move together.
Volcano Plot	Which variables differ the most between two groups, screening many at once.

9.2 Step 2 - Choose your variable(s) and run the test

On each tab, select the numeric variable(s) to test (and a grouping/category column where the test needs one), then click **Calculate** (or **Test**). Results appear below: a table with a p-value, and usually a chart.

Tip

Significant results are marked with stars: *** for $p < 0.001$, ** for $p < 0.01$, * for $p < 0.05$, and **ns** ("not significant") for $p \geq 0.05$.

9.2.1 Normality

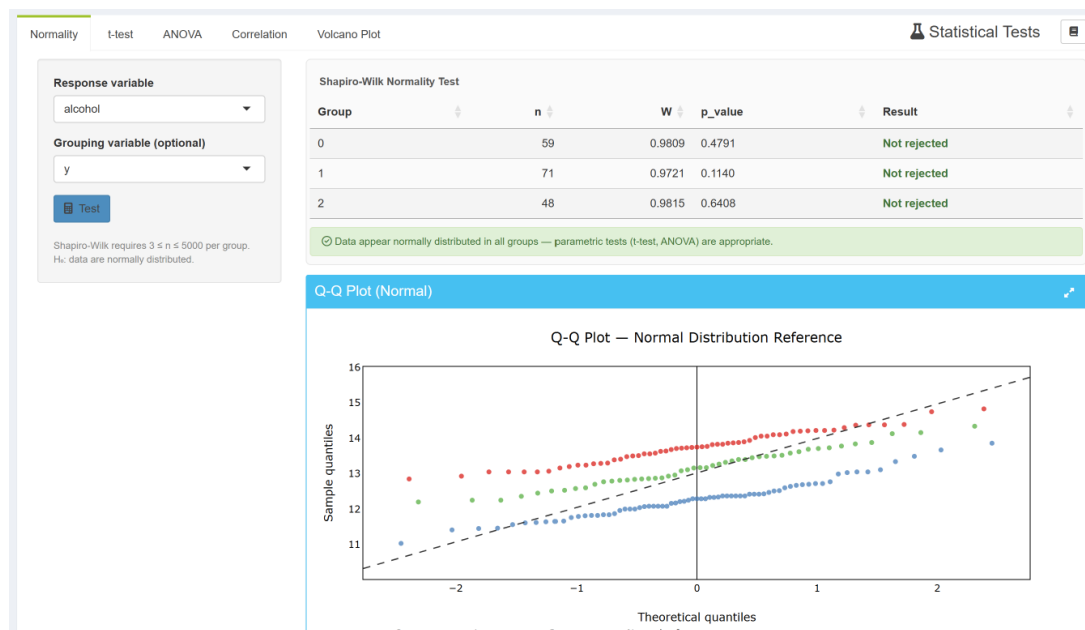


Figure 9.1: Normality tab with Q-Q plot

9.2.2 t-test

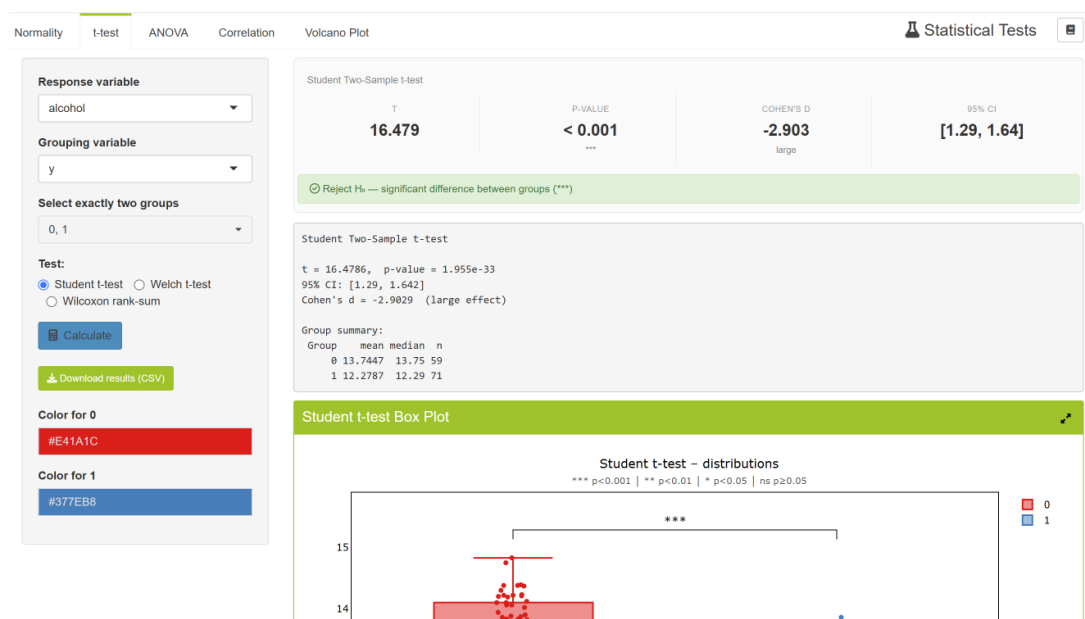


Figure 9.2: t-test tab with box plot

9.2.3 ANOVA

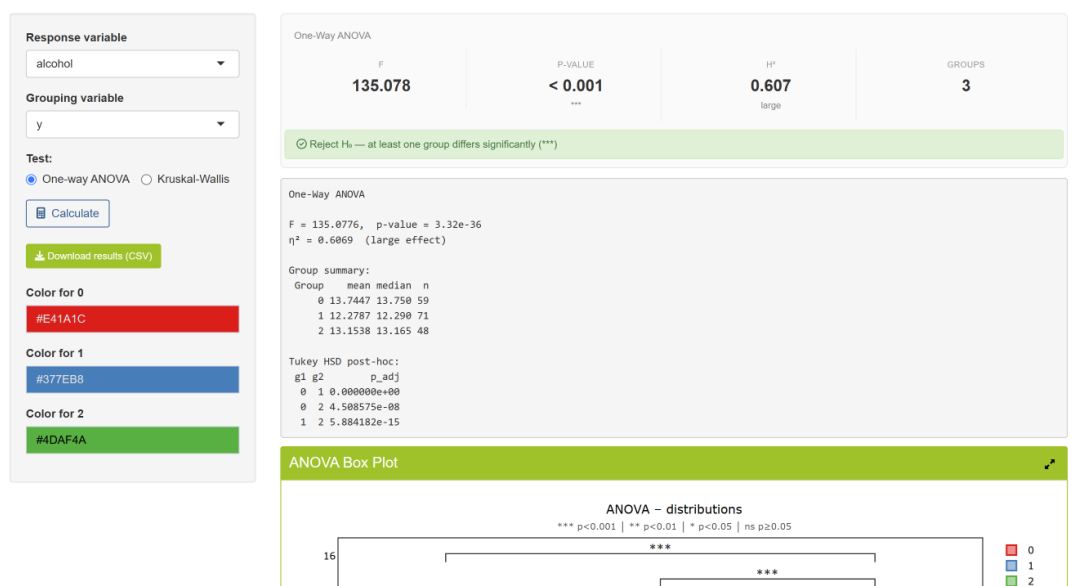


Figure 9.3: ANOVA tab with post-hoc brackets

9.2.4 Correlation

The Correlation tab has two views, **Heatmap** and **Groups/Clustering**, selectable at the top of the tab.

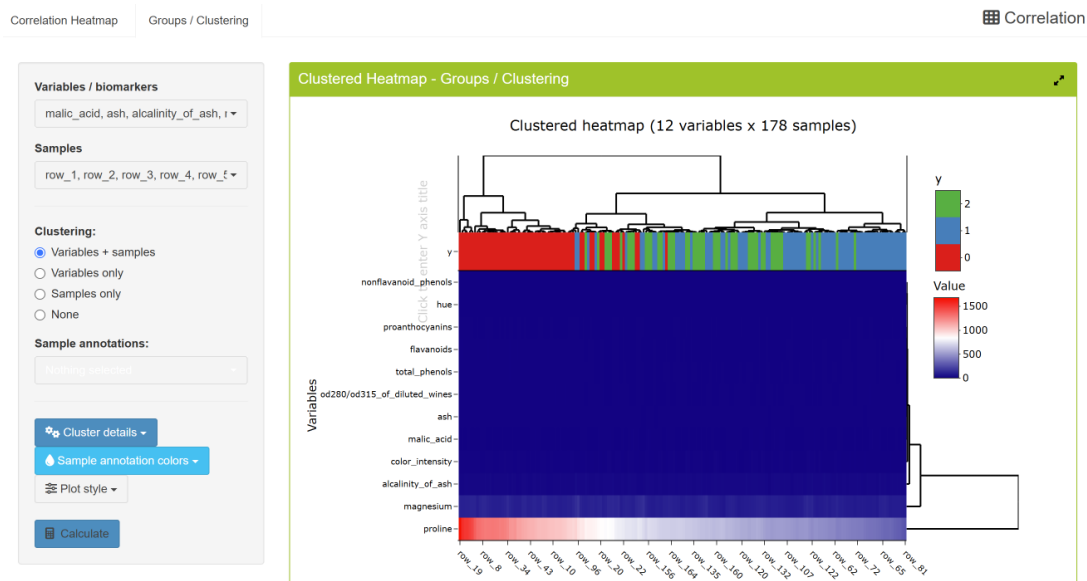


Figure 9.4: Correlation tab - Heatmap view

9.2.5 Volcano Plot

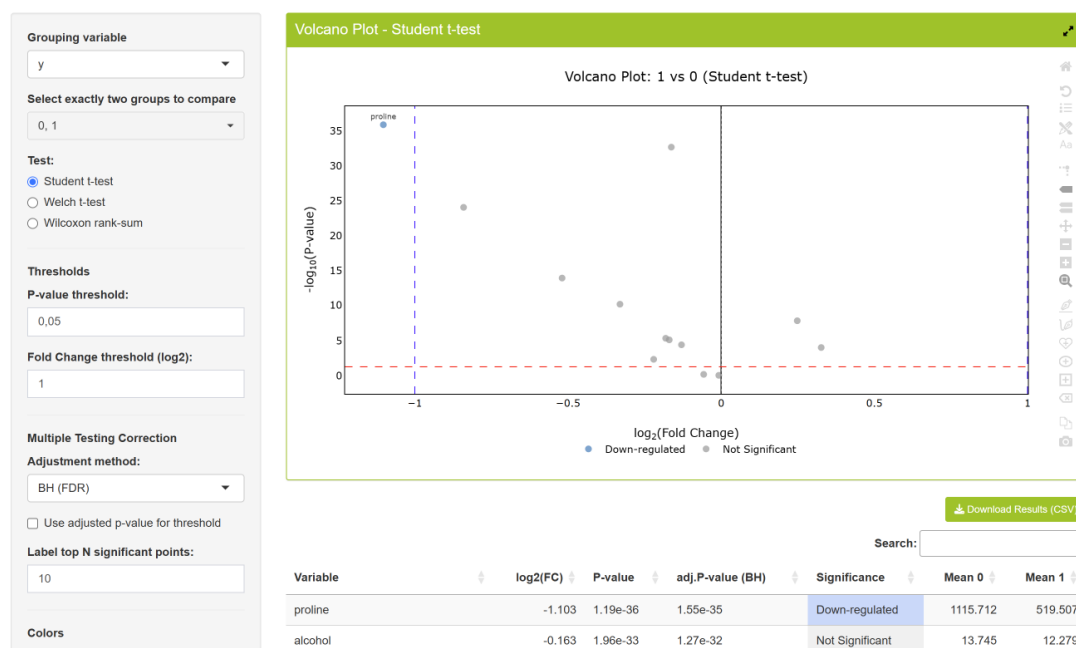


Figure 9.5: Volcano Plot tab

9.3 Step 3 - Download your results

Most tests have a **Download CSV** button next to the results table.

9.4 References

The following citations are shown via the **book icon** at the top of the Statistics module.

- **Normality (Shapiro-Wilk)** - S. S. Shapiro and M. B. Wilk (1965). “An analysis of variance test for normality”. In: *Biometrika* 52.3-4, pp. 591–611. DOI: [10.1093/biomet/52.3-4.591](https://doi.org/10.1093/biomet/52.3-4.591)
- **t-test / Wilcoxon** - Student (1908). “The probable error of a mean”. In: *Biometrika*, pp. 1–25. DOI: [10.2307/2331554](https://doi.org/10.2307/2331554)
F. Wilcoxon (1945). “Individual comparisons by ranking methods”. In: *Biometrics Bulletin* 1.6, pp. 80–83. URL: <https://www.jstor.org/stable/3001968>
- **ANOVA / Kruskal-Wallis** - R. A. Fisher (1934). *Statistical Methods for Research Workers*. Edinburgh: Oliver and Boyd. URL: https://openlibrary.org/works/OL1153861W/Statistical_methods_for_research_workers
W. H. Kruskal and W. A. Wallis (1952). “Use of ranks in one-criterion variance analysis”. In: *Journal of the American Statistical Association* 47.260, pp. 583–621. DOI: [10.1080/01621459.1952.10483441](https://doi.org/10.1080/01621459.1952.10483441)
- **Correlation** - K. Pearson (1920). “Notes on the history of correlation”. In: *Biometrika* 13.1, pp. 25–45. DOI: [10.1093/biomet/13.1.25](https://doi.org/10.1093/biomet/13.1.25)
R. G. Brereton (2007). *Applied Chemometrics for Scientists*. Wiley. DOI: [10.1002/9780470057780](https://doi.org/10.1002/9780470057780)

- **Volcano Plot** - S. H. Holmes and W. Huber (2018). *Modern Statistics for Modern Biology*. Cambridge University Press. URL: <https://www.huber.embl.de/msmb/>

Chapter 10

Preprocessing Module

The Preprocessing module cleans up and transforms your data before modelling. It appears in the sidebar only after you click **Approve Dataset** in the Edit module, under three tabs: **Predictors**, **Outcome**, and **Data Preprocessed**.

10.1 Step 1 - Open the Predictors tab

This is where you preprocess your X (predictor) variables.

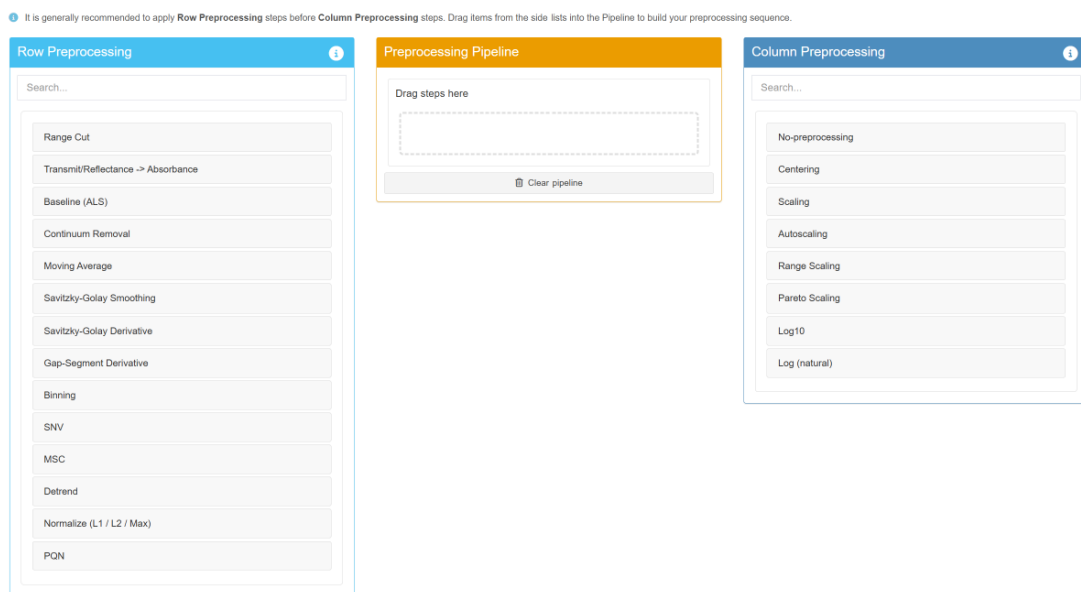


Figure 10.1: Predictors tab overview

10.2 Step 2 - Build your pipeline

Drag steps from the left and right lists into the middle **Pipeline** box. Reorder them by dragging, or click **Clear pipeline** to start over.

Tip

For spectral data is generally recommended to apply **Row Preprocessing** steps before **Column Preprocessing** steps. You can always come back and change it.

Step	What it does
Savitzky-Golay	Smooths noisy spectra (can also compute a derivative).
Moving Average	Simple smoothing.
Baseline (ALS)	Removes a slowly drifting background.
SNV / MSC	Corrects light-scattering effects between samples.
Detrend	Removes a curved baseline trend.
Normalize / PQN	Corrects for overall differences in signal strength between samples.
Range Cut	Keeps only a chosen range of columns, discards the rest.
Centering / Scaling / Autoscaling	Puts all variables on a comparable scale.
Log10 / Log (natural)	Compresses data that spans very different orders of magnitude.

A full description of every step, with formulas, is available from the info (book) icon on the tab.

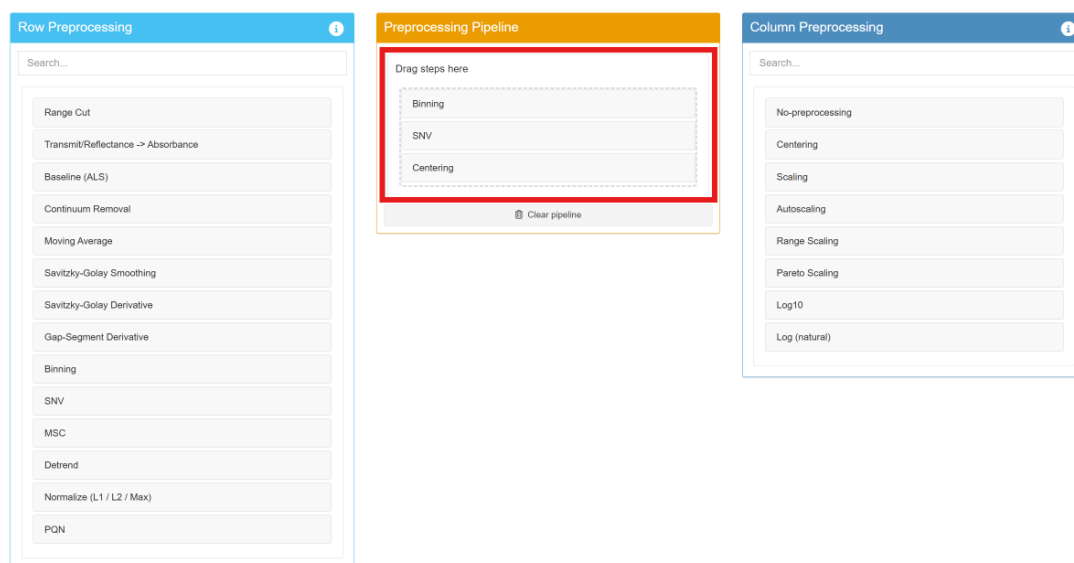


Figure 10.2: Dragging a step into the pipeline

10.3 Step 3 - Set each step's options (optional)

Once a step is in the pipeline, its settings panel appears below - most steps work fine with the default values.

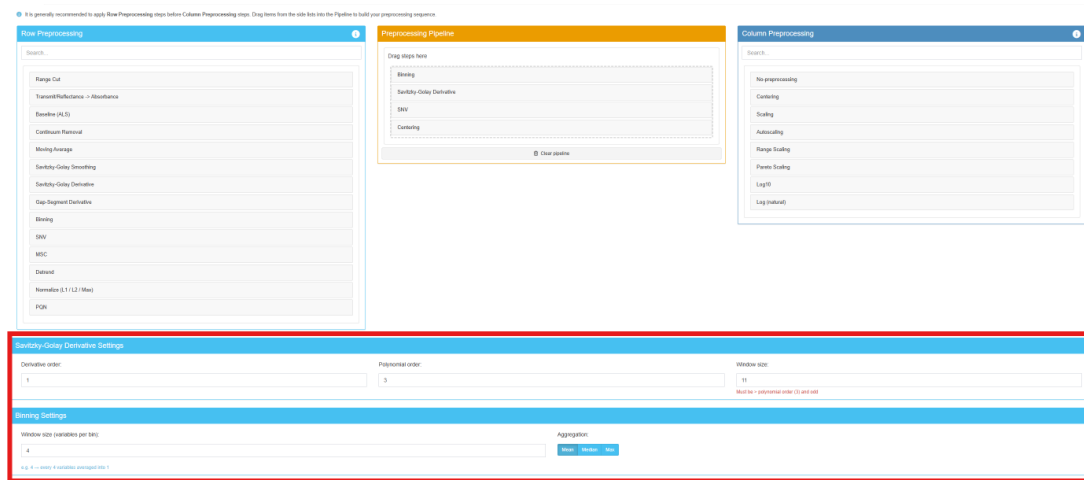


Figure 10.3: Step settings panel (e.g. Savitzky-Golay)

10.4 Step 4 - Preview the result

Click **Preprocess** to apply the selected preprocessing methods. The methods are applied sequentially, following the exact order in which they were added to the preprocessing pipeline. Each method therefore operates on the output produced by the previous one.

Once the preprocessing is complete, two charts are displayed side by side: **Original Data** and **Preprocessed Data**, allowing you to directly compare the dataset before and after the complete preprocessing sequence. The individual preprocessing steps are also listed below the charts in their order of application.

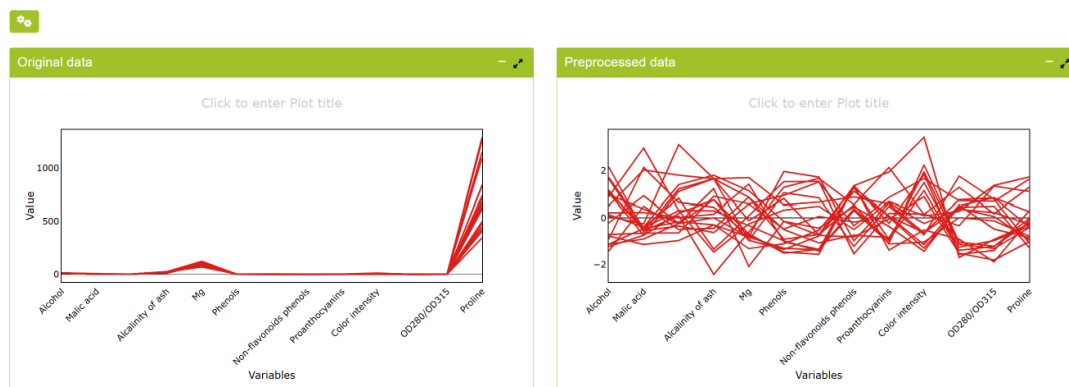


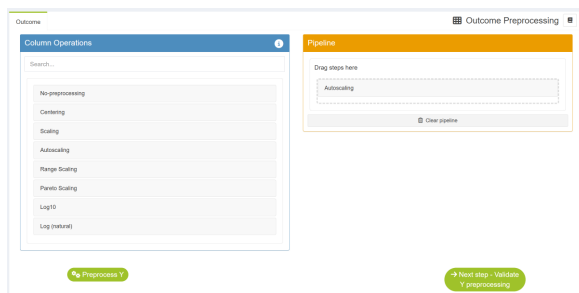
Figure 10.4: Original vs. Pre-processed comparison plots

10.5 Step 5 - Confirm and move on

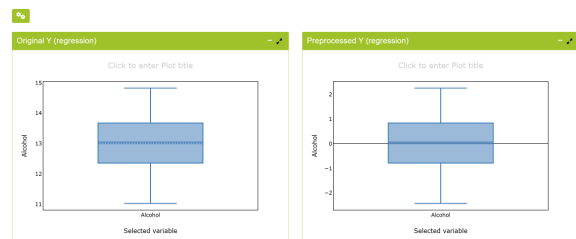
Happy with the result? Click **Next step - Validate preprocessing**.

10.6 Step 6 - Preprocess the Outcome (if you have one)

If you defined a regression target earlier, go to the **Outcome** tab and repeat the same drag-and-drop process, then click **Preprocess Y** and **Next step - Validate Y preprocessing**. If you did not define one, this tab is skipped automatically.



(a) Box preprocessing Y



(b) Outcome preprocessed after autocasting

10.7 Step 7 - Check the final result

The **Data Preprocessed** tab shows a read-only summary: the final data table and a count of samples/variables. Nothing can be changed here - go back to Predictors/Outcome to make edits.

Preprocessed Data

10

entries

Search:

	ID	Alcohol	Category	Data	Malic acid	Ash	Alcalinity of ash	Mg	Phenols	Flavanoids	Non-flavanoids phenols
1	OLO1	1.520	Barolo	1995-01-07	-0.565	0.231	-1.166	1.909	0.807	1.032	-0.658
2	OLO2	0.249	Barolo	1995-01-08	-0.502	-0.826	-2.484	0.018	0.567	0.732	-0.818
3	OLO3	0.200	Barolo	1995-01-09	0.017	1.106	-0.268	0.088	0.807	1.212	-0.497
4	OLO4	1.692	Barolo	1995-01-10	-0.350	0.487	-0.807	0.928	2.484	1.462	-0.979
5	OLO5	0.299	Barolo	1995-01-11	0.224	1.835	0.451	1.278	0.807	0.661	0.226
6	OLO6	1.483	Barolo	1995-01-12	-0.520	0.304	-1.286	0.858	1.558	1.362	-0.176
7	OLO7	1.717	Barolo	1995-01-13	-0.422	0.304	-1.466	-0.262	0.327	0.491	-0.497
8	OLO8	1.310	Barolo	1995-01-14	-0.171	0.888	-0.567	1.488	0.487	0.481	-0.417
9	OLO9	2.259	Barolo	1995-01-15	-0.628	-0.716	-1.645	-0.192	0.807	0.952	-0.577
10	OLO10	1.063	Barolo	1995-01-16	-0.888	-0.352	-1.047	-0.122	1.094	1.122	-1.140

Showing 1 to 10 of 20 entries

Previous

1

2

Next

178

Samples

2

Factor/Classification variables

1

Regression variables

12

X variables

Figure 10.6: Data preprocessed

10.8 Common mistakes

- Applying scaling steps before smoothing/correction steps - add Row steps first, then Column steps.
- Stacking several scaling steps together (e.g. Centering + Scaling + Autoscaling) - pick just one.
- Skipping the Outcome tab by accident when you actually have a regression target - double check it was detected in the Edit module.

Warning

Important. All preprocessing parameters are estimated exclusively from the **Training set**. The resulting values are then used to transform the **Test set**; no information from the Test set is used during the estimation step.

The preprocessing calculations performed in this section are used only for the **Visualization** and **Statistics** modules, where the preprocessed data must be generated in advance for plotting and statistical analysis.

For the **Unsupervised**, **Classification**, and **Regression** modules, the selected preprocessing sequence is not applied beforehand. Instead, it is fitted separately within each cross-validation iteration using only the corresponding training fold, and then applied to the validation fold. This procedure prevents data leakage and ensures an unbiased evaluation of model performance.

Warning

One-Class Modeling. In the **One-Class Modeling** module, preprocessing is handled directly within the module because the underlying procedure is conceptually different from standard classification and regression.

Preprocessing parameters are estimated exclusively from the samples belonging to the selected target class and are not calculated using the entire dataset. Further details about this procedure are provided in Section 14.

10.9 References

The following citations are shown via the **book icon** at the top of the Preprocessing tabs.

- P. Oliveri et al. (2019). “The impact of signal pre-processing on the final interpretation of analytical outcomes – A tutorial”. In: *Analytica Chimica Acta* 1058, pp. 9–17. DOI: [10.1016/j.aca.2018.10.055](https://doi.org/10.1016/j.aca.2018.10.055)
- S. Brown, R. Tauler, and B. Walczak, eds. (2020). *Comprehensive Chemometrics: Chemical and Biochemical Data Analysis*. Elsevier. DOI: [10.1016/C2017-1-02312-7](https://doi.org/10.1016/C2017-1-02312-7)
- J. M. Andrade and R. Leardi, eds. (2025). *Basic Chemometrics for Analytical Chemists*. World Scientific. DOI: [10.1142/q0495](https://doi.org/10.1142/q0495)

Chapter 11

Unsupervised

The Unsupervised module explores the natural structure of your data without using any class or target column. It provides four techniques as separate sub-tabs: **PCA**, **HCA**, **K-means**, and **t-SNE**. The module becomes available once the preprocessing workflow has been defined.

A feature shared by all methods allows samples or variables to be removed directly after the exploratory evaluation, without restarting the entire analysis workflow. When a reduced dataset is created, the selected preprocessing steps are recalculated and applied to the updated data, rather than reusing the values previously computed on the original dataset. This ensures that the removed samples or variables do not influence the new preprocessing results and avoids introducing data leakage.

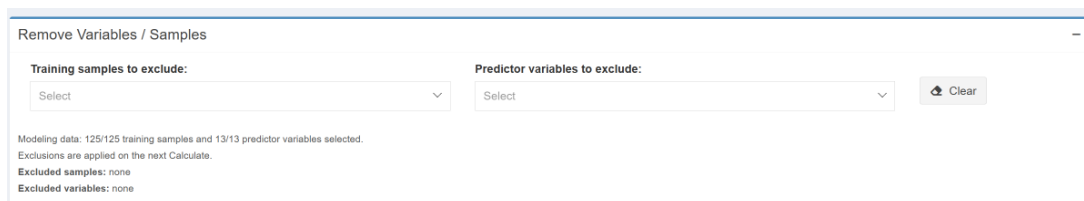


Figure 11.1: Remove samples or variables

11.1 PCA (Principal Component Analysis)

Use PCA to see your samples on a simple 2D/3D map and spot groups or outliers.

1. Set the **number of components**. The field is automatically pre-filled with **10 components**, or with a lower value when the dataset contains fewer than 10 predictor variables.
2. Select the PCA calculation algorithm from the dropdown menu: **SVD** or **NIPALS**.
3. Click **Calculate** to fit the PCA model using the selected settings.
4. Check the **Scree Plot** to inspect the variance explained by each component and the cumulative variance retained by the model.
5. Explore the **Scores Plot** and the corresponding **Loadings Plot**. The Scores Plot displays the samples in the selected component space, while the Loadings Plot shows the contribution of the predictor variables to those components. Use the available controls to change the components displayed in both plots.

- Open the **Hotelling's T^2 vs. Q Residuals** plot to inspect the samples according to their score and residual distances and to identify potentially unusual observations. When a sample lies outside the T^2 or Q acceptance limits, click its point to display the corresponding **Contribution Plot** and inspect which variables contribute most to its unusual behaviour.
- Click **Save PCA model** to store the fitted model and reuse it later in the **Predict** module.

Tip

As throughout the application, clicking the **gear-icon button** opens the plot settings. These options allow you to customise the graph, including its colours and point shapes, and to select the variables or samples to display.

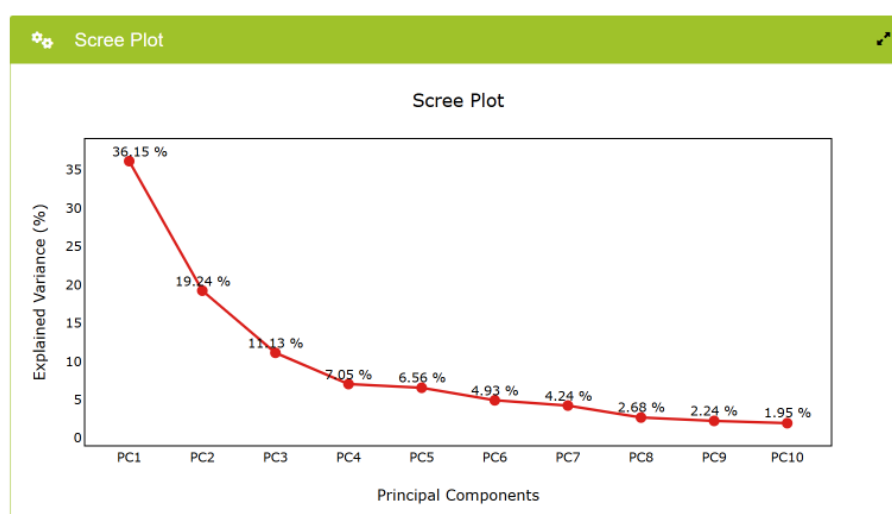


Figure 11.2: PCA Scree Plot

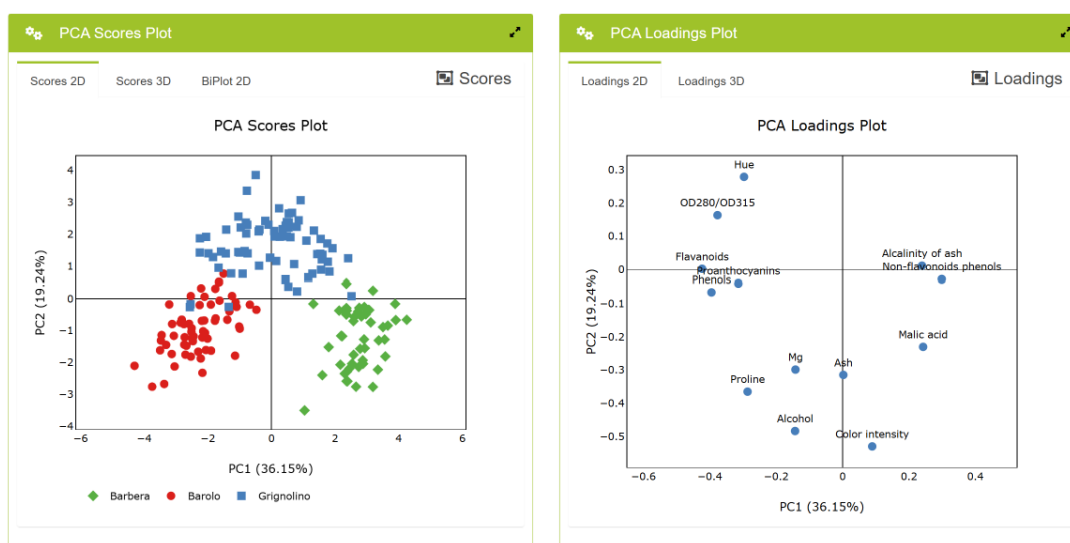


Figure 11.3: PCA Scores and Loading Plots coloured by a categorical variable

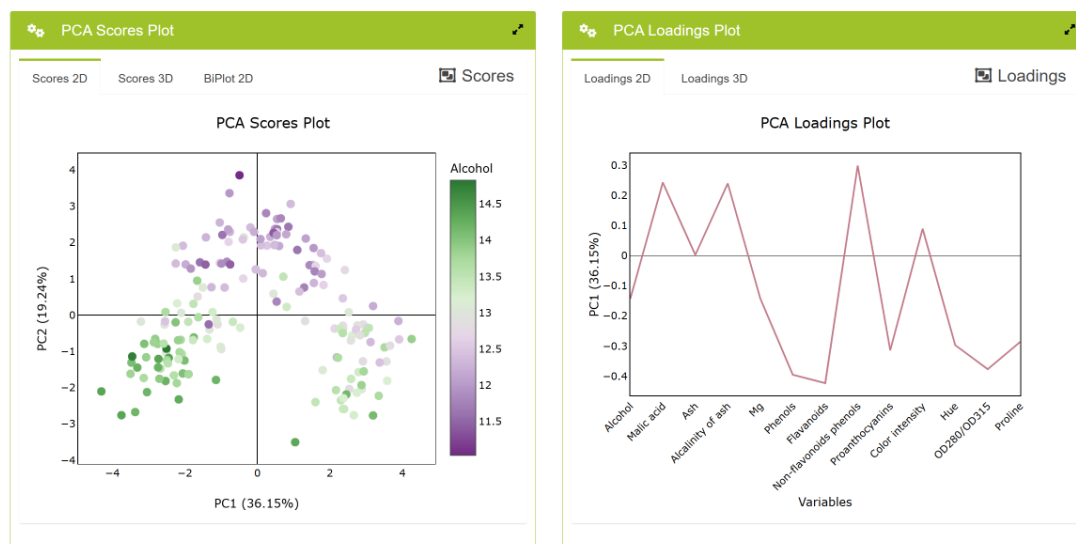


Figure 11.4: PCA Scores and Loading Plots coloured by a continuous variable and loadings plot in "Spectra mode"

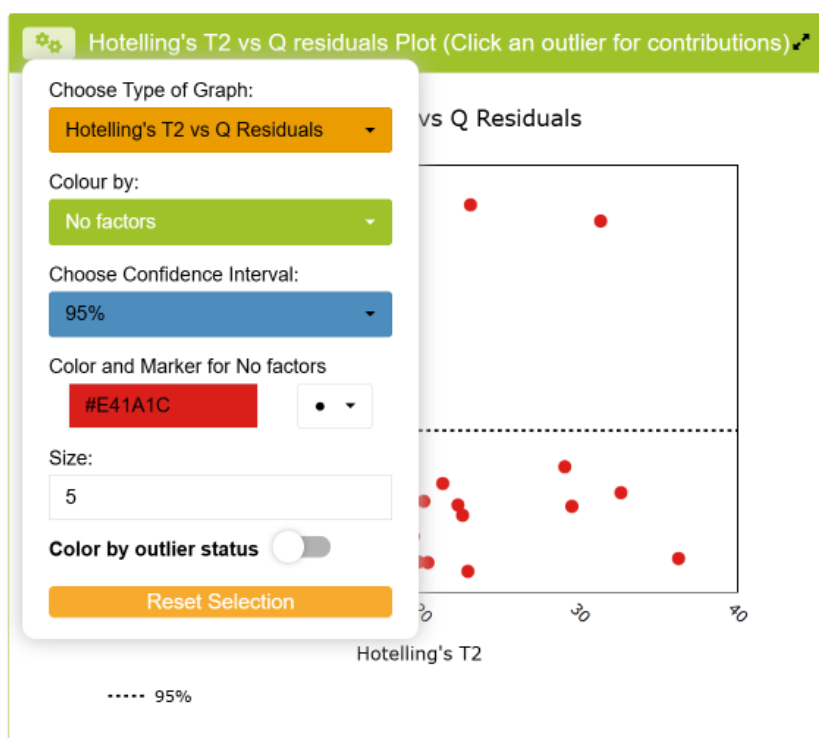


Figure 11.5: PCA Hotelling's T^2 vs. Q plot

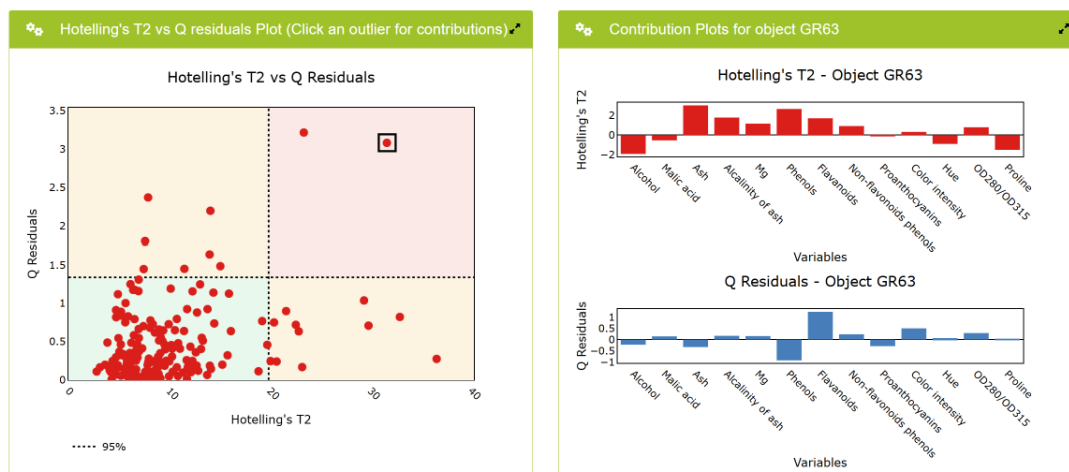


Figure 11.6: Contribution Plot

Tip

The **Contribution Plot** is available only for samples located outside the Hotelling's T^2 limit, the Q residual limit, or both limits. Samples that fall within both acceptance limits cannot be selected and no contribution results or plots are generated for them.

If you have a Test set, a **Test** tab appears once a model is trained: click **Run PCA projection** to see where the new samples fall.

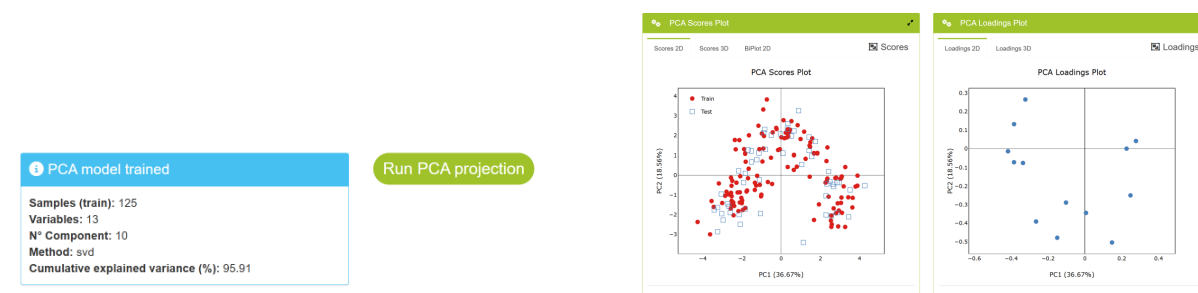


Figure 11.7: PCA test tab interface

11.2 HCA (Hierarchical Cluster Analysis)

Use HCA to group similar samples into clusters and display their relationships as a tree diagram.

1. Choose the **distance** and **linkage** methods.
2. Optionally select **Perform PCA first?** to run HCA on PCA scores instead of the original variables.
3. Set the **number of clusters**.
4. Click **Calculate**.
5. Inspect the **Dendrogram**: samples that merge at lower levels are more similar, while the selected clusters are highlighted in the final tree.

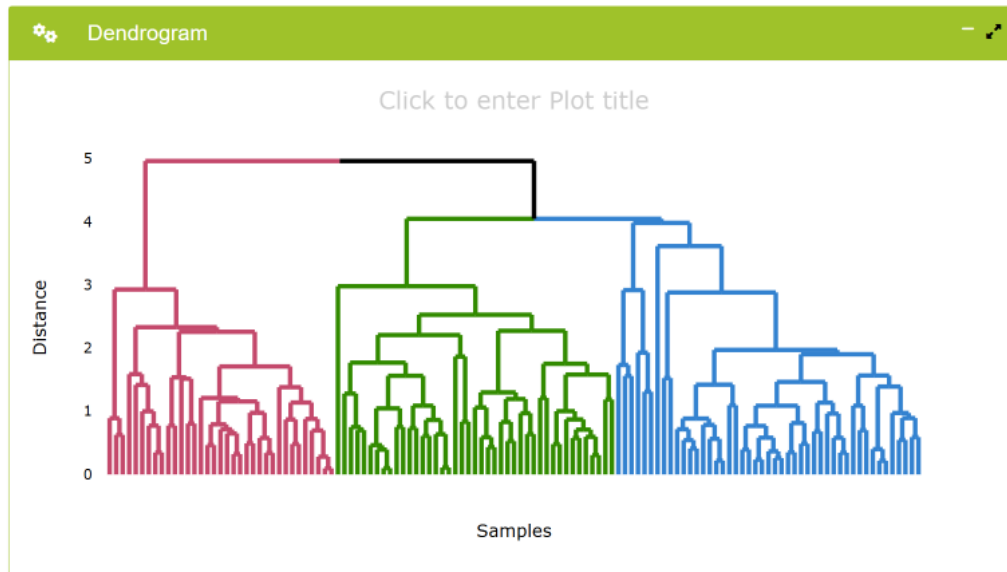


Figure 11.8: HCA Dendrogram



Figure 11.9: HCA Silhouette Plot



Figure 11.10: Comparison Cluster Analysis vs Original Data after PCA decomposition

11.3 K-means

Use K-means to group samples into a fixed number of clusters.

1. Optionally select **Perform PCA first?** to run K-means on PCA scores instead of the original variables.
2. Click **Calculate** to test several cluster numbers and obtain a suggested solution.
3. Adjust **Nr. of clusters** if a different grouping is preferred.
4. Click **Validate** to fit and display the final clustering result.

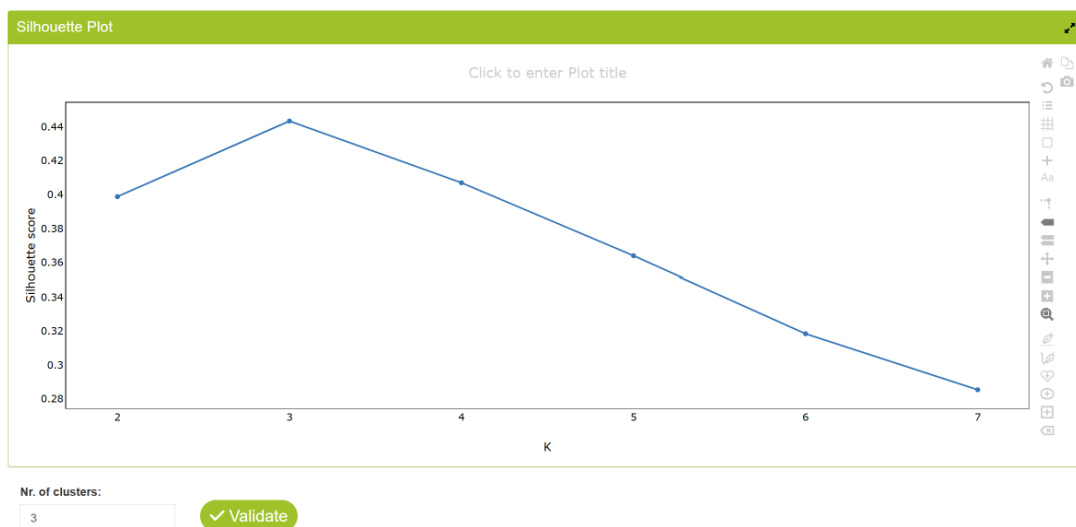


Figure 11.11: K-means Silhouette plot

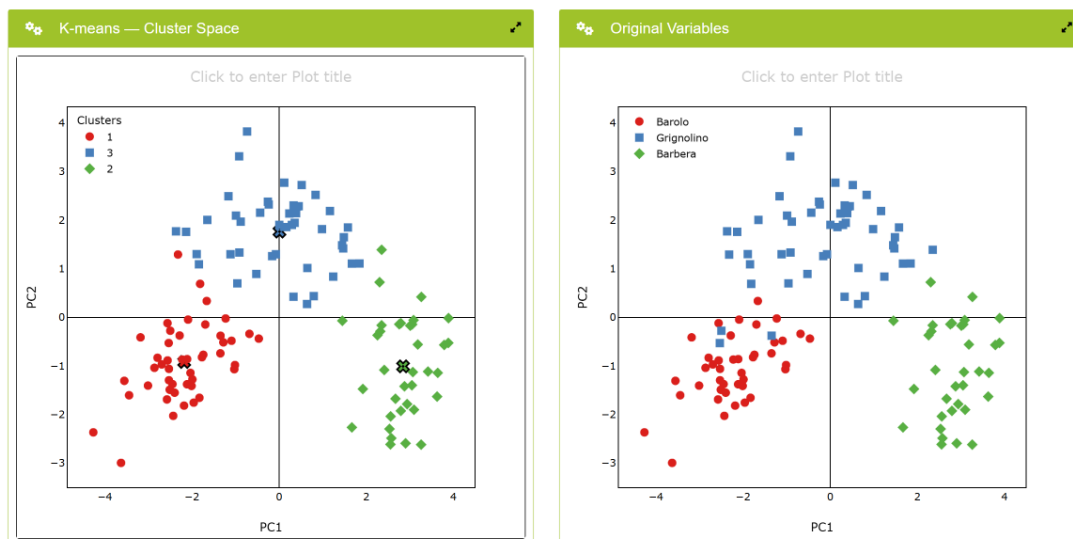


Figure 11.12: K-means Cluster Space plot

11.4 t-SNE (t-distributed stochastic neighbor embedding)

Use t-SNE for a more flexible 2D/3D map when PCA does not separate your samples well.

1. Open the **t-SNE settings** dropdown menu to configure the model parameters.
2. Set the number of **Dimensions** to 2 or 3 and adjust the remaining settings if needed.
3. Click **Calculate** to run t-SNE and display the resulting sample map.



Figure 11.13: t-SNE embedding plot

Warning

Only the distance between *neighbouring* points in a t-SNE plot is meaningful. Do not read too much into the distance between separate clusters.

11.5 References

The following citations are shown via the **book icon** at the top of the Unsupervised module.

- **PCA** – R. Bro and A. K. Smilde (2014). “Principal component analysis”. In: *Analytical Methods* 6.9, pp. 2812–2831. DOI: [10.1039/C3AY41907J](https://doi.org/10.1039/C3AY41907J)
S. Wold, K. Esbensen, and P. Geladi (1987). “Principal component analysis”. In: *Chemometrics and Intelligent Laboratory Systems* 2.1-3, pp. 37–52. DOI: [10.1016/0169-7439\(87\)80084-9](https://doi.org/10.1016/0169-7439(87)80084-9)
- **HCA** – D. L. Massart and L. Kaufman (1983). *The Interpretation of Analytical Chemical Data by the Use of Cluster Analysis*. URL: <https://catalogue.nla.gov.au/Record/2170828>
F. Murtagh and P. Contreras (2012). “Algorithms for hierarchical clustering: an overview”. In: *WIREs Data Mining and Knowledge Discovery* 2.1, pp. 86–97. DOI: [10.1002/widm.53](https://doi.org/10.1002/widm.53)
F. Murtagh and P. Contreras (2017). “Algorithms for hierarchical clustering: an overview, II”. in: *WIREs Data Mining and Knowledge Discovery* 7.6, e1219. DOI: [10.1002/widm.1219](https://doi.org/10.1002/widm.1219)
- **K-means** – J. MacQueen et al. (1967). “Some methods for classification and analysis of multivariate observations”. In: *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability*. Vol. 1, pp. 281–297. URL: <https://projecteuclid.org/euclid.bsmmsp/1200512992>
P. J. Rousseeuw (1987). “Silhouettes: a graphical aid to the interpretation and validation of cluster analysis”. In: *Journal of Computational and Applied Mathematics* 20, pp. 53–65. DOI: [10.1016/0377-0427\(87\)90125-7](https://doi.org/10.1016/0377-0427(87)90125-7)
A. M. Ikotun et al. (2023). “K-means clustering algorithms: A comprehensive review”. In: *Information Sciences* 622, pp. 178–210. DOI: [10.1016/j.ins.2022.11.139](https://doi.org/10.1016/j.ins.2022.11.139)
E. Schubert (2023). “Stop using the elbow criterion for k-means and how to choose the number of clusters instead”. In: *ACM SIGKDD Explorations* 25.1, pp. 36–42. DOI: [10.1145/3606274.3606278](https://doi.org/10.1145/3606274.3606278)
- **t-SNE** – G. E. Hinton and S. Roweis (2002). “Stochastic neighbor embedding”. In: *Advances in Neural Information Processing Systems*. Vol. 15. URL: <https://papers.nips.cc/paper/2276-stochastic-neighbor-embedding>
L. van der Maaten and G. Hinton (2008). “Visualizing data using t-SNE”. in: *Journal of Machine Learning Research* 9, pp. 2579–2605. URL: <https://jmlr.org/papers/v9/vandermaaten08a.html>

Chapter 12

Cross-Validation

Cross-validation checks how well a model generalises to data it has not seen, instead of only fitting the data it was trained on. It is available, with the same configuration window, in **Classification** (Chapter 13), **One Class Modeling** (SIMCA, Chapter 14), and **Regression** (Chapter 15) - this chapter documents it once for all three.

12.1 Strategies

Click **Cross-validation** on any supervised method's tab to open the configuration window and pick a strategy:

Strategy	What it does
Random k-fold	Data split randomly into k equal folds.
Stratified k-fold	Maintains class proportions in each fold (Regression: the numeric response is binned first).
Grouped k-fold	Keeps replicates together - the same group is never split across train/test.
Stratified + Grouped k-fold	Both of the above at once.
Leave-one-out	Each observation becomes a test set once. Computationally intensive for large datasets.
Venetian blinds	Interleaved folds (fold i = samples $i, i+k, i+2k, \dots$). Requires samples in a meaningful order (e.g. by concentration or time).
Contiguous blocks	Sequential folds (fold i = an adjacent block of samples). Requires ordered data - standard practice in multivariate chemometrics.

Cross-validation settings ?

CV strategy:

☐ No cross-validation
☐ Leave-one-out
☒ Random k-fold
☐ Venetian blinds
☐ Contiguous blocks
☐ Stratified k-fold
☐ Grouped k-fold (replicates together)
☐ Stratified + Grouped k-fold

Number of folds (k):

☐ Use repeated CV

Seed:

Click 'Confirm cross-validation' to generate folds and view row assignments.

Confirm cross-validation
Apply

Fold assignments preview

Excel
Search:

Fold	Rows.Train	Rows.Test
Fold1	row_1, row_2, row_3, row_4, row_5, row_6, row_7, row_8, row_9, row_10, row_12, row_13, row_14, row_15, row_16, row_18, row_19, row_20, row_22, row_23, row_24, row_26, row_27, row_28, row_30, row_31, row_32, row_33, row_34, row_36, row_37, row_38, row_39, row_40, row_41, row_42, row_43, row_46, row_47, row_50, row_52, row_55, row_56, row_58, row_59, row_61, row_62, row_64, row_65, row_66, row_67, row_68, row_69, row_70, row_71, row_73, row_74, row_75, row_76, row_77, row_78, row_80, row_81, row_82, row_85, row_86, row_87, row_88, row_89, row_90, row_92, row_93, row_95, row_97, row_98, row_99, row_100, row_101, row_102, row_103,	row_11, row_17, row_21, row_25, row_29, row_35, row_44, row_45, row_48, row_49, row_51, row_53, row_54, row_57, row_60, row_63, row_72, row_79, row_83, row_84, row_91.

Figure 12.1: Cross-validation configuration window

Note

One Class Modeling (SIMCA) offers only **Random k-fold**, **Grouped k-fold**, **Leave-one-out**, **Venetian blinds**, and **Contiguous blocks** - the two **Stratified** variants are not offered there, since a one-class model has no other classes to stratify against.

12.2 Folds, repeats and seed

For every strategy except Leave-one-out, set the **number of folds (k)**. For the randomised strategies (not Venetian blinds/Contiguous blocks, which are deterministic) you can also turn on **repeated CV** (repeat the whole k-fold split several times with different random splits) and set a **seed** for reproducibility. Stratified and/or Grouped strategies additionally ask for the stratification and/or grouping variable.

Click **Confirm cross-validation** to generate the folds and preview which rows fall in each one, then **Apply** to use that configuration when you click **Calculate**.

Warning

In One Class Modeling, folds are built from the **target class only**. The non-target class is only used for scoring if you choose a *compliant* approach; with a *rigorous* approach it is not used in model construction at all, and non-target rows never move between Train and Test.

Chapter 13

Classification Module

The Classification module builds a model that predicts a **category** (e.g. product type, authentic vs. counterfeit) from your data. It offers four methods: **KNN**, **LDA**, **PLS-DA** and **QDA**. Available once Preprocessing is complete.

13.1 Workflow (same for every method)

1. Open a method tab and select your classification (Y) variable.
2. Set the method's own parameters - each method has different ones, see Section 13.3 below.
3. *Optional* - click **Cross-validation** to check how the model generalises to new data instead of just fitting the training data. See Chapter 12 for the available strategies.
4. Click **Calculate**. For KNN and PLS-DA (methods with a tunable parameter - k and the number of latent variables respectively), a tuning plot appears first: pick the value that looks best (or keep the automatically-selected one) and click **Fit Best Model**. LDA and QDA fit directly, with no tuning step.
5. Check the results, shown as two panels side by side: **Model** (calibration, i.e. on the training data itself) and **Cross-Validation**. Both have several tabs: Metrics (**Accuracy**, **Sensitivity** and **Specificity** are always shown - Section 13.2 below lets you add more), Confusion Matrix, ROC, Misclassified samples, and Metrics per Class; Cross-Validation also adds Predictions and a Permutation Test tab.
6. If you have a Test set, a **Test** tab appears once trained - click **Run ... Projection** to apply the model to your Test rows.
7. Use the download option on the Train tab to save the model for later reuse in Predict (Chapter 16).

Tip

Cross-validation is optional, but recommended - a model's calibration score alone tends to look better than it will actually perform on new data.

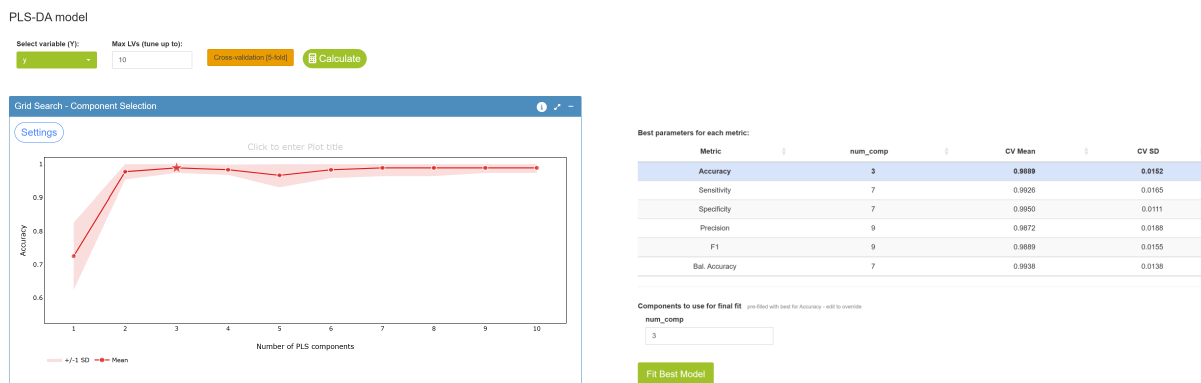


Figure 13.1: Tuning plot and Fit Best Model (KNN or PLS-DA)

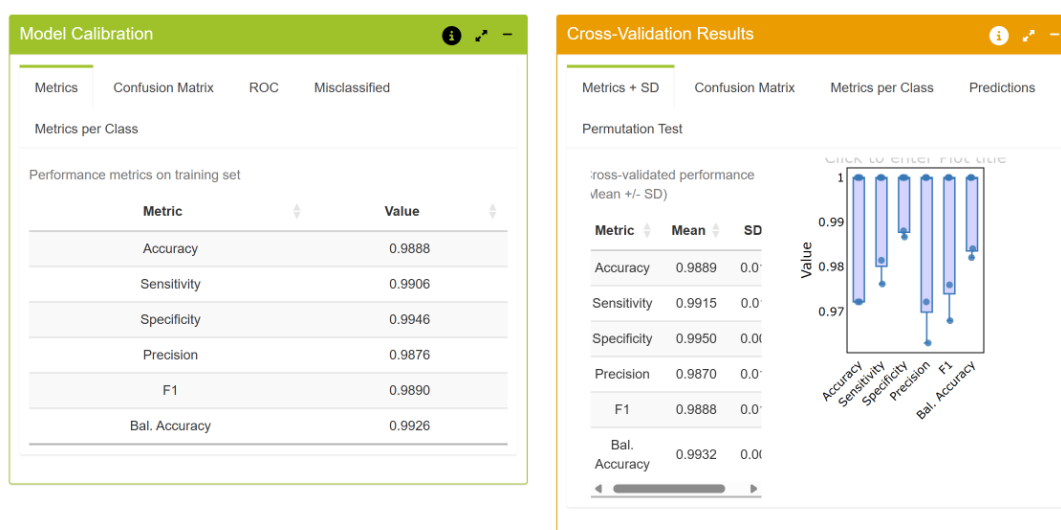


Figure 13.2: Model and Cross-Validation result panels, side by side

13.2 Customizing metrics and CV strategies

The right-hand settings panel (**Configure App**, Section 3.3) has a **Classification** entry with two sub-panels useful for a teaching setting, where you may want to keep the results simpler:

- **Metrics** - show or hide **Precision**, **F1**, and **Balanced Accuracy** in the Metrics tabs above (Accuracy, Sensitivity and Specificity cannot be hidden).
- **CV Methods** - show or hide individual cross-validation strategies (Chapter 12) from the Cross-validation configuration window, to limit which ones students can pick. **No cross-validation** and **Random k-fold** are always available.

This applies to every Classification method tab, not just the one you have open, and is remembered the next time you open Cactus.

13.3 What you can configure, per method

Beyond the Y variable and cross-validation (shared by all four methods above), each method exposes its own settings.

13.3.1 KNN

Perform PCA first? (+ number of components); k range to tune (min/max/step); weight function; distance metric.



KNN model

Select variable (Y): y

☒ Perform PCA first?

Number of components: 3

⚙️ KNN Settings Cross-validation [None] Calculate

Figure 13.3: KNN settings panel

13.3.2 LDA

Perform PCA first? (+ number of components); *Use equal priors*.



LDA model

Select variable (Y): y

☒ Perform PCA first?

Max PCs (tune up to): 3

☐ Use equal priors:

Cross-validation [None] Calculate

Figure 13.4: LDA settings panel

Tip

Perform PCA first? is useful for LDA when you have more predictor columns than samples, which LDA cannot otherwise handle.

13.3.3 QDA

Perform PCA first? (+ number of components); *Use equal priors*.



QDA model

Select variable (Y): y

☒ Perform PCA first?

Max PCs (tune up to): 3

☐ Use equal priors:

Cross-validation [None] Calculate

Figure 13.5: QDA settings panel

13.3.4 PLS-DA

Max number of latent variables to tune.



PLS-DA model

Select variable (Y): y

Max LVs (tune up to): 10

Cross-validation [5-fold] Calculate

Figure 13.6: PLS-DA settings panel

13.4 Typical outputs

Once fitted, each method also produces its own plots (scores, loadings, VIP, decision regions, etc., depending on the method) - see the plot catalog in Chapter 8 for the full list. As an example, PLS-DA's Scores/Loadings/VIP view:

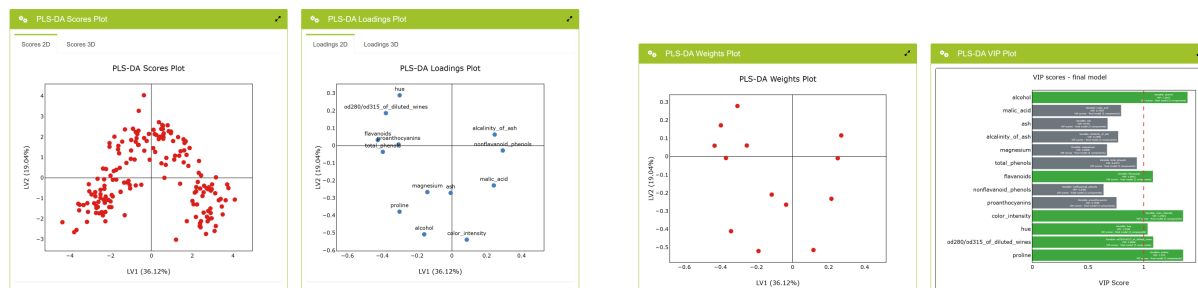


Figure 13.7: PLS-DA scores, loadings and VIP plots

13.5 References

The following citations are shown via the **book icon** at the top of the Classification module.

- **LDA** - R. A. Fisher (1936). “The use of multiple measurements in taxonomic problems”. In: *Annals of Eugenics* 7, pp. 179–188. DOI: [10.1111/j.1469-1809.1936.tb02137.x](https://doi.org/10.1111/j.1469-1809.1936.tb02137.x)
P. Xanthopoulos, P. M. Pardalos, and T. B. Trafalis (2012). “Linear discriminant analysis”. In: *Robust Data Mining*, pp. 27–33. DOI: [10.1007/978-1-4419-9878-1_4](https://doi.org/10.1007/978-1-4419-9878-1_4)
- **QDA** - A. Tharwat (2016). “Linear vs. quadratic discriminant analysis classifier: a tutorial”. In: *International Journal of Applied Pattern Recognition* 3.2, pp. 145–180. DOI: [10.1504/ijapr.2016.079050](https://doi.org/10.1504/ijapr.2016.079050)
P.-N. Tan et al. (2016). *Introduction to Data Mining*. Pearson Education India. URL: https://openlibrary.org/works/OL20999884W/Introduction_to_Data_Mining
- **PLS-DA** - M. Barker and W. Rayens (2003). “Partial least squares for discrimination”. In: *Journal of Chemometrics* 17.3, pp. 166–173. DOI: [10.1002/cem.785](https://doi.org/10.1002/cem.785)
M. Cocchi, A. Biancolillo, and F. Marini (2018). “Chemometric methods for classification and feature selection”. In: *Comprehensive Analytical Chemistry*. Vol. 82, pp. 265–299. DOI: [10.1016/bs.coac.2018.08.006](https://doi.org/10.1016/bs.coac.2018.08.006)
- **KNN** - E. Fix and J. L. Hodges (1951). “Discriminatory analysis, nonparametric discrimination”. In: *USAF School of Aviation Medicine, Report 4*. DOI: [10.1080/01621459.1951.10500769](https://doi.org/10.1080/01621459.1951.10500769)
B. D. Ripley (1996). *Pattern Recognition and Neural Networks*. Cambridge University Press. DOI: [10.1007/978-1-4612-0257-1](https://doi.org/10.1007/978-1-4612-0257-1)
- **Validation** - F. Westad and F. Marini (2015). “Validation of chemometric models – a tutorial”. In: *Analytica Chimica Acta* 893, pp. 14–24. DOI: [10.1016/j.aca.2015.06.056](https://doi.org/10.1016/j.aca.2015.06.056)
K. H. Esbensen and P. Geladi (2010). “Principles of proper validation: use and abuse of re-sampling for validation”. In: *Journal of Chemometrics* 24.3-4, pp. 168–187. DOI: [10.1002/cem.1310](https://doi.org/10.1002/cem.1310)
D. Ballabio, V. Consonni, and A. Manganaro (2018). “Classification tools in chemistry. Part 1: Linear models. PLS-DA”. in: *Chemometrics and Intelligent Laboratory Systems* 174, pp. 33–44. DOI: [10.1016/j.chemolab.2017.12.004](https://doi.org/10.1016/j.chemolab.2017.12.004)

Chapter 14

One Class Modeling Module (SIMCA)

One Class Modeling answers a different question than Classification: *"does this sample belong to class X?"* (accept/reject) rather than *"which class does it belong to?"*. It uses the SIMCA method.

Warning

SIMCA never compares classes to each other - a "Rejected" result only means the sample does not fit the target class, not that it belongs to another specific one.

14.1 Step 1 - Set Up the Target Class

Select the classification (**Y**) variable and the **target class** to be modelled.

Warning

SIMCA models only one class at a time. It does not automatically build one model for each available class; only a single model is created for the selected target class.

The screenshot shows the 'Setup' tab of the SIMCA module. At the top, there are two dropdown menus: 'Select Y variable:' with 'Category' selected, and 'Select target class:' with 'Barbera' selected. To the right of these is a green button labeled 'Accept & Continue →'. Below these is a section titled 'Predictors' with a sub-header 'Predictors Preprocessing'. A blue information icon is followed by the text: 'It is generally recommended to apply Row Preprocessing steps before Column Preprocessing steps. Drag items from the side lists into the Pipeline to build your preprocessing sequence.' The main area is divided into three panels: 'Row Preprocessing' (blue header), 'Preprocessing Pipeline' (orange header), and 'Column Preprocessing' (blue header). The 'Row Preprocessing' panel contains a search bar and a list of steps: 'Range Cut', 'Transmit/Reflectance -> Absorbance', 'Baseline (ALS)', 'Continuum Removal', 'Moving Average', 'Savitzky-Golay Smoothing', and 'Savitzky-Golay Derivative'. The 'Preprocessing Pipeline' panel contains a dashed box labeled 'Drag steps here' and a 'Clear pipeline' button. The 'Column Preprocessing' panel contains a search bar and a list of steps: 'No-preprocessing', 'Centering', 'Scaling', 'Autoscaling', 'Range Scaling', 'Pareto Scaling', and 'Log10'.

Figure 14.1: One-Class Modeling - Setup tab

Note

The preprocessing settings used in this module are independent of those configured in the main **Preprocessing** module. In one-class modelling, preprocessing must be estimated exclusively from the training samples belonging to the target class. Configure the required preprocessing steps again in this tab, then click **Accept & Continue**.

14.2 Step 2 - (Optional) Set up cross-validation

Click **Cross-validation** to check how well the model generalises. See Chapter 12 for the available strategies.

14.3 Step 3 - Train the model

Set **Max components** and **alpha** (a smaller alpha is stricter), then click **Calculate**. If you set up cross-validation, review the suggested number of components and click **Fit Best Model**.



Figure 14.2: One Class Modeling - Grid Search results

14.4 Step 4 - Check the results

The **Acceptance Plot** shows which samples fall inside (accepted) or outside (rejected) the target class boundary. Check **Efficiency**, **Sensitivity** and **Specificity** below it.

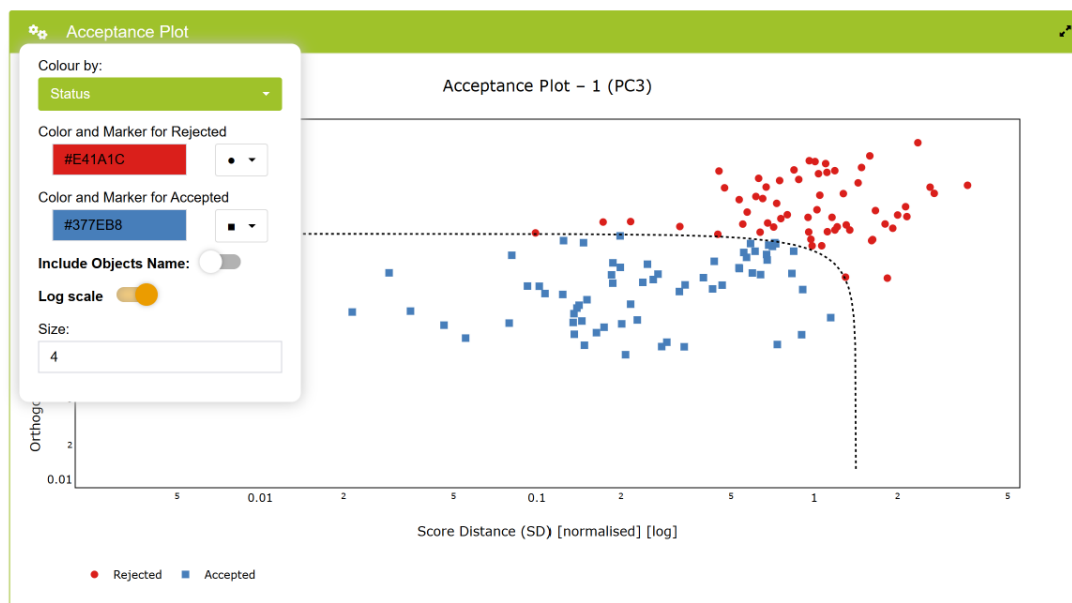


Figure 14.3: One Class Modeling - Acceptance Plot

14.5 Step 5 - Test on new data (if you have a Test set)

Once trained, a **Test** tab appears. Click **Run SIMCA Projection** to apply the model to your Test rows.

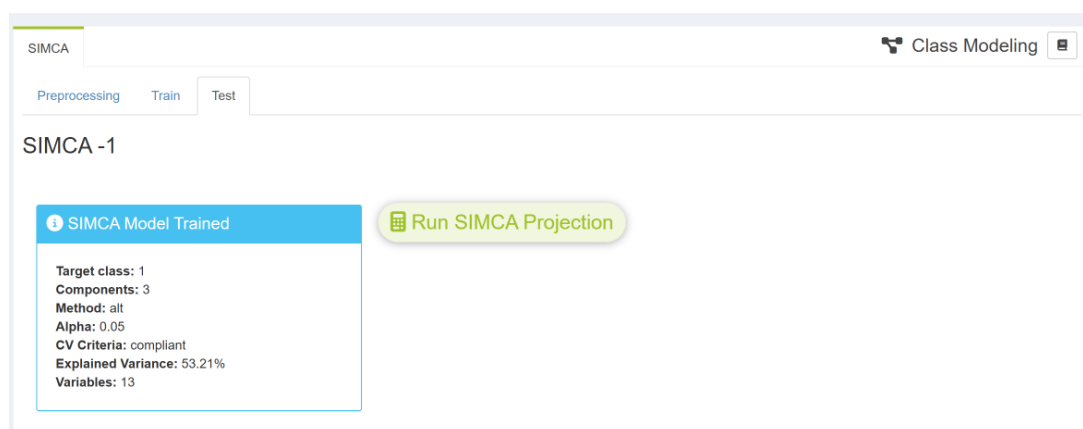


Figure 14.4: One Class Modeling - Test tab

14.6 Customizing decision methods, limits and CV strategies

The right-hand settings panel (**Configure App**, Section 3.3) has a **One Class Modeling** entry with four sub-panels:

- **CV Methods** - show or hide individual cross-validation strategies (Chapter 12) from the Cross-validation configuration window. **No cross-validation** and **Random k-fold** are always available.
- **Decision Method** - show or hide the **sim** (Simple), **ci** (Combined index) and **dd** (Data-Driven) decision methods from the **Decision method** picker on the Train tab. **alt** (Alternative) is always available.

- **Limit Methods** - each decision method shown above gets its own pair of radio buttons here to choose how its **OD limit (Q)** and **SD limit (T²)** are computed (Jackson-Mudholkar, Chi²-Box, Chi² (moments), Chi²-Pom or Percentile for Q; F-distribution, F-dist rigorous, Chi² (df=A), Chi²-Pom or Percentile for T²). **dd** only offers Chi² (moments)/Chi²-Pom for Q and Chi² (df=A)/Chi²-Pom for T², plus an optional **gamma limit** with its own value. Changing a decision method's limit choice here also changes how that method's results are computed on the Train tab, not just what is displayed.
- **Metrics** - show or hide **FAR** (False Acceptance Rate) and **FRR** (False Rejection Rate) next to Efficiency, Sensitivity and Specificity (Step 4 above), which cannot be hidden.

This applies to the whole One Class Modeling module and is remembered the next time you open Cactus.

14.7 References

The following citations are shown via the **book icon** at the top of the One Class Modeling module.

- **SIMCA** - Vitale, R., Cocchi, M., Biancolillo, A., Ruckebusch, C. & Marini, F. (2023). Class modelling by SIMCA: why, when, how? A tutorial. *Analytica Chimica Acta*, 1270, 341304.
Pomerantsev, A.L. & Rodionova, O.Y. (2014). Concept and role of extreme objects in PCA/SIMCA. *Journal of Chemometrics*, 28(5), 429-438.
Kucheryavskiy, S., Rodionova, O.Y. & Pomerantsev, A.L. (2024). A comprehensive tutorial on Data-Driven SIMCA. *Journal of Chemometrics*, 38(7).
Strani, L., Cocchi, M., Tanzilli, D., Biancolillo, A., Marini, F. & Vitale, R. (2025). One class classification (class modelling): State of the art and perspectives. *TrAC Trends in Analytical Chemistry*, 183, 118117.
- **Validation** - Camacho, J. (2025). A Set of Rules for Model Validation. arXiv:2511.20711.
Westad, F. & Marini, F. (2015). Validation of chemometric models - a tutorial. *Analytica Chimica Acta*, 893, 14-24.
Esbensen, K.H. & Geladi, P. (2010). Principles of proper validation: use and abuse of re-sampling for validation. *Journal of Chemometrics*, 24(3-4), 168-187.
Bro, R. et al. (2008). Cross-validation of component models: a critical look at current methods. *Analytical and Bioanalytical Chemistry*, 390(5), 1241-1251.
Ezenarro, J. & Schorn-Garcia, D. (2025). How are chemometric models validated? *Journal of Chemometrics*, 39(6), e70036.
Ballabio, D. et al. (2018). Classification tools in chemistry. Part 1: Linear models. PLS-DA. *Chemometrics and Intelligent Laboratory Systems*, 174, 33-44.
Lopez, A. et al. (2025). Recipes for (avoiding) overfitting in predictive modelling. *Analytica Chimica Acta*.

Chapter 15

Regression Module

The Regression module builds a model that predicts a **number** (e.g. a concentration) from your data. It offers three methods: **MLR** (Multiple Linear Regression), **PCR** (Principal Component Regression) and **PLS** (Partial Least Squares). Available once Preprocessing is complete.

15.1 Workflow (same for every method)

1. Open a method tab and select your numeric (Y) variable to predict.
2. Set the method's own parameters - each method has different ones, see Section 15.3 below.
3. *Optional* - click **Cross-validation** to check how the model generalises to new data instead of just fitting the training data. See Chapter 12 for the available strategies.
4. Click **Calculate**. For PCR and PLS (methods with a tunable number of components/latent variables), a tuning plot appears first: pick the value that looks best (or keep the automatically-selected one) and click **Fit Best Model**. MLR fits directly, with no tuning step.
5. Check the results, shown as two panels side by side: **Model** (calibration, i.e. on the training data itself) and **Cross-Validation**. Both have several tabs: Metrics (**RMSE** and R^2 are always shown - Section 15.2 below lets you add more), Measured vs Predicted, Residuals, and Residuals vs Fitted; Cross-Validation also adds RMSEC vs RMSE-CV and a Permutation Test tab.
6. If you have a Test set, a **Test** tab appears once trained - click **Run Predictions ...** to apply the model to your Test rows.
7. Use the download option on the Train tab to save the model for later reuse in Predict (Chapter 16).

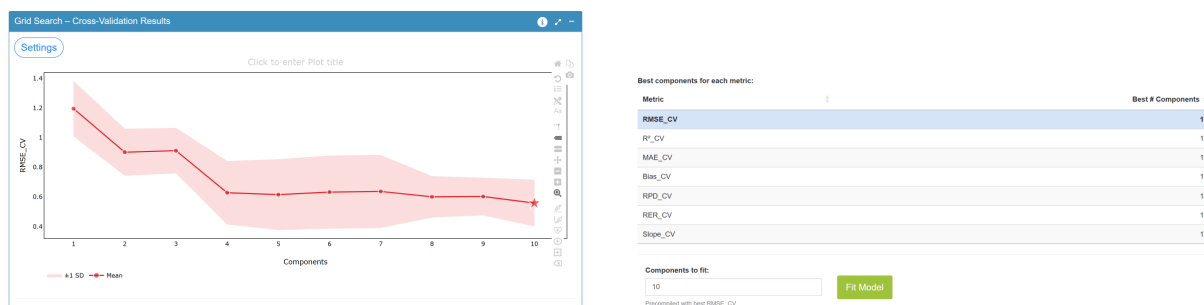


Figure 15.1: Tuning plot and Fit Best Model (PCR or PLS)

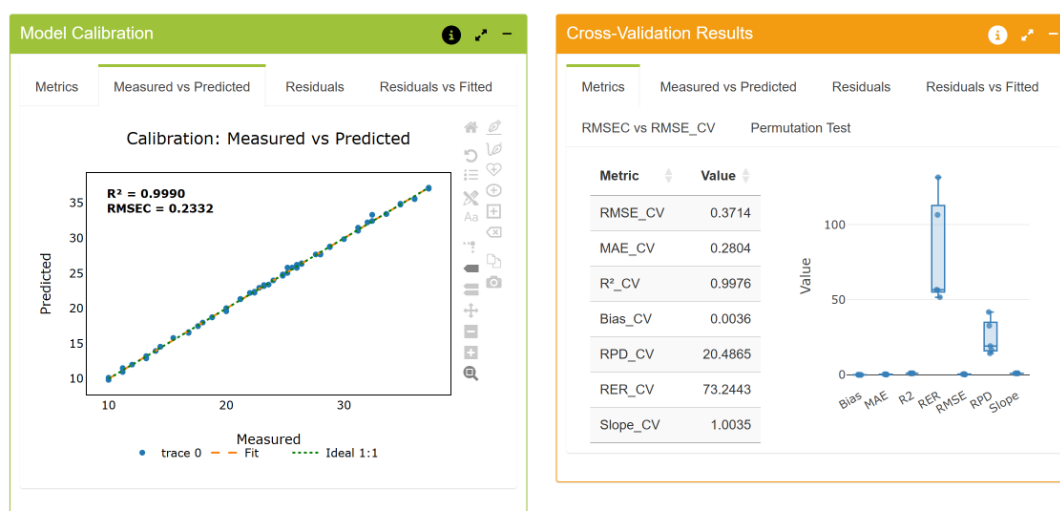


Figure 15.2: Model and Cross-Validation result panels, side by side

15.2 Customizing metrics and CV strategies

The right-hand settings panel (**Configure App**, Section 3.3) has a **Regression** entry with two sub-panels useful for a teaching setting, where you may want to keep the results simpler:

- **Metrics** - show or hide **MAE**, **Bias**, **RPD**, **RER**, and **Slope** in the Metrics tabs above (RMSE and R^2 cannot be hidden).
- **CV Methods** - show or hide individual cross-validation strategies (Chapter 12) from the Cross-validation configuration window, to limit which ones students can pick. **No cross-validation** and **Random k-fold** are always available.

This applies to every Regression method tab, not just the one you have open, and is remembered the next time you open Cactus.

15.3 What you can configure, per method

Beyond the Y variable and cross-validation (shared by all three methods above), each method exposes its own settings.

15.3.1 MLR

No tunable parameter - MLR fits directly on all available predictors. Its results include two extra views not shared with PCR/PLS: **VIF** (variance inflation factors, flags redundant predictors) and **Assumptions** (linear-model diagnostic tests).

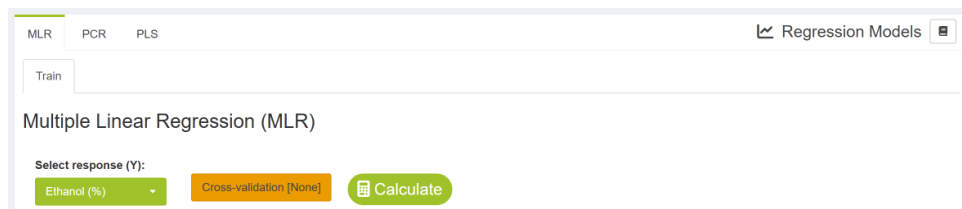


Figure 15.3: MLR settings panel

Warning

MLR needs at least as many samples as predictor columns to fit - with more predictors than samples (common with spectral data), the fit becomes unstable or fails outright. Check the **VIF** tab for warning signs, or use **PCR**/**PLS** instead, which handle many correlated predictors via component reduction.

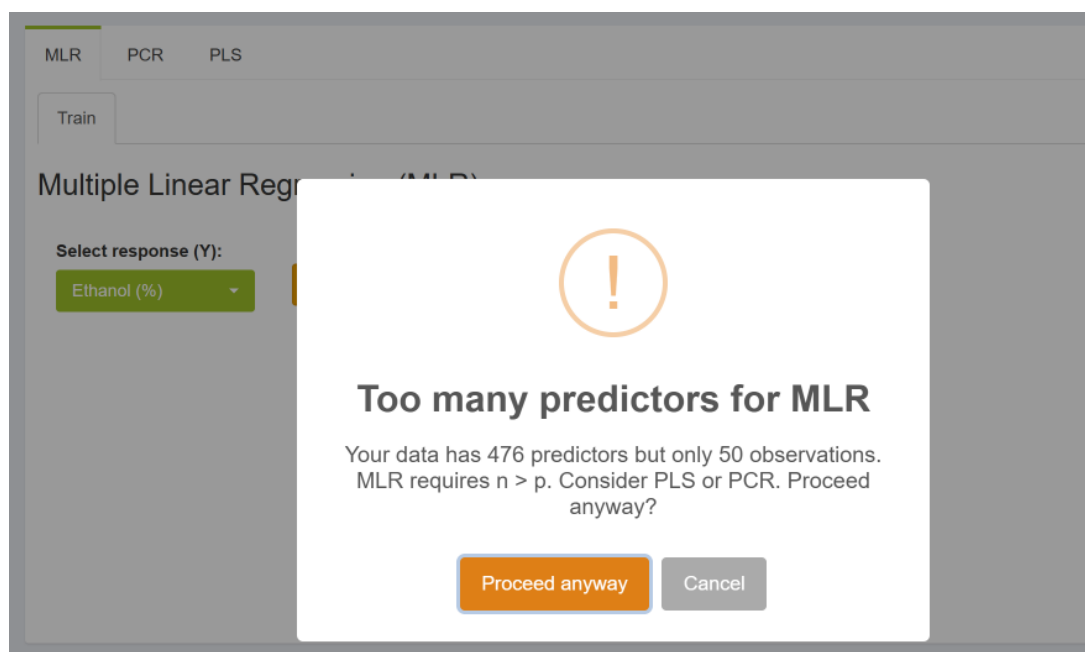


Figure 15.4: MLR settings panel

15.3.2 PCR

Max number of principal components to tune.

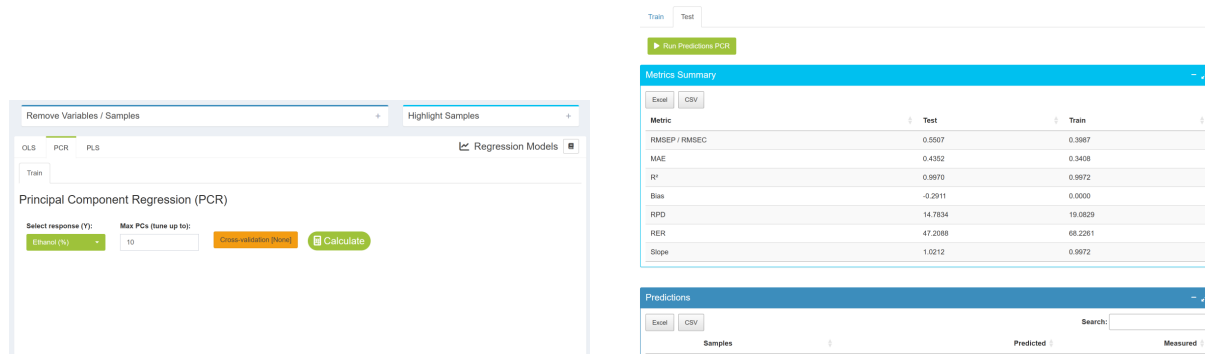


Figure 15.5: PCR settings panel and example of outputs of test set

15.3.3 PLS

Max number of latent variables to tune.



Figure 15.6: PLS settings panel

15.4 Typical outputs

Once fitted, each method also produces its own plots (scores, loadings, VIP, coefficients, etc., depending on the method) - see the plot catalog in Chapter 8 for the full list. As an example, PLS's Scores/Loadings/VIP view:

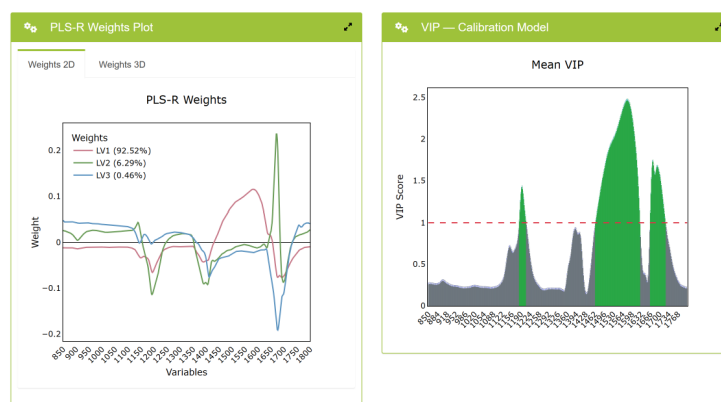


Figure 15.7: PLS scores, loadings and VIP plots

15.5 References

The following citations are shown via the **book icon** at the top of the Regression module.

- **PLS** - H. Wold (1966). “Estimation of principal components and related models by iterative least squares”. In: URL: <https://scholar.google.com/scholar?q=Estimation+of+principal+components+Wold+1966>
P. Geladi and B. R. Kowalski (1986). “Partial least-squares regression: a tutorial”. In: *Analytica Chimica Acta* 185, pp. 1–17. DOI: [10.1016/0003-2670\(86\)80028-9](https://doi.org/10.1016/0003-2670(86)80028-9)
S. Wold, H. Martens, and H. Wold (1983). “The multivariate calibration problem in chemistry solved by the PLS method”. In: DOI: [10.1007/BFb0062108](https://doi.org/10.1007/BFb0062108)
S. Wold, M. Sjostrom, and L. Eriksson (2001). “PLS-regression: a basic tool of chemometrics”. In: *Chemometrics and Intelligent Laboratory Systems* 58.2, pp. 109–130. DOI: [10.1016/s0169-7439\(01\)00155-1](https://doi.org/10.1016/s0169-7439(01)00155-1)
- **MLR** - K. R. Beebe and B. R. Kowalski (1987). “An introduction to multivariate calibration and analysis”. In: *Analytical Chemistry* 59.17, 1007A–1017A. DOI: [10.1021/ac00144a725](https://doi.org/10.1021/ac00144a725)
H. Martens and T. Naes (1992). *Multivariate Calibration*. John Wiley & Sons. URL: <https://books.google.com/books?id=6lVcUeVDg9IC>
D. A. Freedman (2009). *Statistical Models: Theory and Practice*. Cambridge University Press. DOI: [10.1142/9789814280860_0002](https://doi.org/10.1142/9789814280860_0002)
- **PCR** - H. Martens and T. Naes (1992). *Multivariate Calibration*. John Wiley & Sons. URL: <https://books.google.com/books?id=6lVcUeVDg9IC>
M. Maeder and Y.-M. Neuhold (2007). “Practical Data Analysis in Chemistry”. In: vol. 26. Elsevier. DOI: [10.1016/s0922-3487\(07\)x8002-x](https://doi.org/10.1016/s0922-3487(07)x8002-x)
M. Otto (2023). *Chemometrics: Statistics and Computer Application in Analytical Chemistry*. John Wiley & Sons. DOI: [10.1002/9783527843800](https://doi.org/10.1002/9783527843800)
- **Validation** - J. Camacho (2025). *A Set of Rules for Model Validation*. arXiv:2511.20711. URL: <https://arxiv.org/abs/2511.20711>
F. Westad and F. Marini (2015). “Validation of chemometric models – a tutorial”. In: *Analytica Chimica Acta* 893, pp. 14–24. DOI: [10.1016/j.aca.2015.06.056](https://doi.org/10.1016/j.aca.2015.06.056)
K. H. Esbensen and P. Geladi (2010). “Principles of proper validation: use and abuse of re-sampling for validation”. In: *Journal of Chemometrics* 24.3-4, pp. 168–187. DOI: [10.1002/cem.1310](https://doi.org/10.1002/cem.1310)
R. Bro, K. Kjeldahl, et al. (2008). “Cross-validation of component models: a critical look at current methods”. In: *Analytical and Bioanalytical Chemistry* 390.5, pp. 1241–1251. DOI: [10.1007/s00216-007-1790-1](https://doi.org/10.1007/s00216-007-1790-1)
J. Ezenarro and D. Schorn-Garcia (2025). “How are chemometric models validated?” In: *Journal of Chemometrics* 39.6, e70036. DOI: [10.1002/cem.70036](https://doi.org/10.1002/cem.70036)
A. Lopez et al. (2025). “Recipes for (avoiding) overfitting in predictive modelling”. In: *Analytica Chimica Acta*. DOI: [10.1016/j.aca.2025.344838](https://doi.org/10.1016/j.aca.2025.344838)

Chapter 16

Predict Module

The Predict module is designed to apply a model you already saved (Chapter 13, 14, 15 or 11) to new data, without repeating the whole Import/Edit/Preprocess/Train pipeline.

Note

This current free version of Cactus does not allow prediction of novel instances. Opening the Predict tab shows a message instead of the module itself: *"This current free version does not allow prediction of novel instances. If you need to download the model and perform predictions, you can visit the Cactus repository"*, linking to github.com/Lebolebo95/Cactus.

Chapter 17

Credits

17.1 Purpose

The Credits tab (sidebar entry **Credits**) is a static information page: who develops Cactus, where to find the documentation, how to report a bug or send feedback, how to cite Cactus, and a pointer to another tool from the same team. It has no inputs, parameters, or interactive analysis.

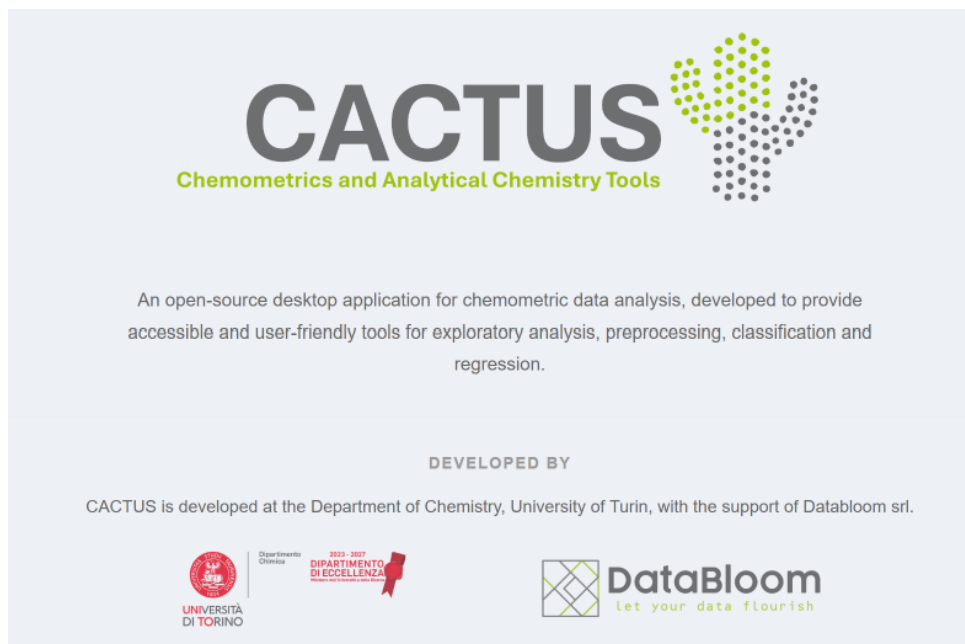


Figure 17.1: Credits tab

17.2 Developed by

Below the Cactus logo and a short description of the application, this section states that Cactus is developed at the Department of Chemistry, University of Turin, with the support of Databloom srl, and shows two clickable logos:

- Department of Chemistry, University of Turin - chimica.unito.it
- Databloom srl - databloom.it

Clicking either logo opens the corresponding website in your default browser.

17.3 Documentation and Feedback cards

Two cards let you reach the most common resources without leaving the application:

- **Documentation** - the **Open User Guide** button opens this user guide (as a PDF) in a new tab.
- **Feedback** - two buttons:
 - **Report a Bug** opens the [Cactus GitHub Issues](#) page, for technical problems.
 - **Feedback Form** opens a [Google Form](#) for suggestions, comments, or feature requests that are not necessarily bugs.

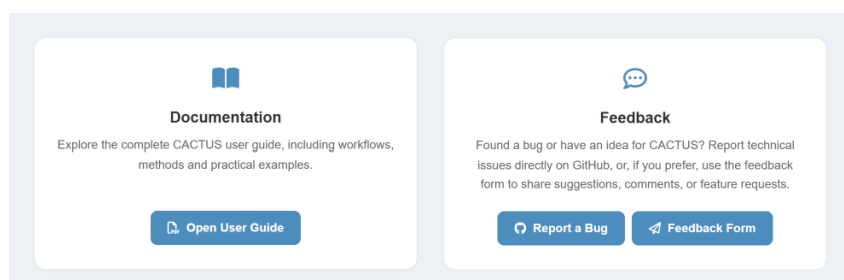


Figure 17.2: Documentation and Feedback cards

17.4 How to cite

If you use Cactus in scientific research, publications, presentations, or other academic work, this section asks you to cite the associated publication. The full citation and DOI will be added here once the associated publication is available.

17.5 Also from us

You can also explore the **Methods Validation App (MVA)**, another dashboard developed by **Databloom** for analytical method validation in regulated laboratories. Click **Visit MVA** to open mva.databloom.it in your default browser.

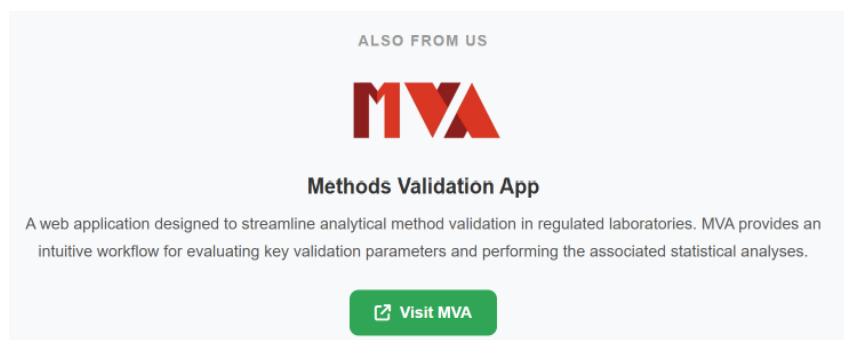


Figure 17.3: Methods Validation App

i Note

This page is informational only - there is nothing to configure or troubleshoot here. If you need to report a problem with the application, use the **Report a Bug** or **Feedback Form** buttons on this tab.

Chapter 18

Troubleshooting

This chapter lists problems users are likely to encounter, grouped by symptom, with practical fixes. If your problem is not listed here, check the chapter for the specific module you were using - each one lists its own common mistakes.

18.1 Installer problems

- **Windows: "Unknown publisher" warning** - expected, since the installer is not (yet) signed with a paid code-signing certificate; click **More info** then **Run anyway** (Chapter 2).
- **macOS: "Cactus is damaged and can't be opened"** - right-click (or Control-click) the app and choose **Open** instead of double-clicking, the first time only (Chapter 2). If the warning persists after that, the download may be corrupted - delete it and download the installer again.
- **Ubuntu/Debian: apt reports missing dependencies** - make sure you installed with `sudo apt install ./CactusSetup.deb` (not by double-clicking the file), so **apt** can resolve them automatically.
- **macOS: updates never seem to apply / "Download" does nothing useful** - Cactus is almost certainly still running from the installer's disk image instead of from **Applications** (Chapter 2); a read-only disk image cannot be updated. Move the app to Applications, eject the disk image, and relaunch from there.

18.2 Application will not launch

- **Stuck on the splash screen** - the first launch after installing (or after an update) can take longer than usual; wait a little longer before assuming it is frozen.
- **Nothing happens after that** - check the log file for the exact error:
 - Windows: %LOCALAPPDATA%\Cactus\cactus_log.txt
 - macOS: ~/Library/Application Support/Cactus/cactus_log.txt
 - Ubuntu/Debian: ~/.local/share/Cactus/cactus_log.txt

- If the log shows "Startup failed" or "Shiny startup timeout", or if it stops part-way through the `[CACTUS ...] ...loaded` package-loading messages instead of reaching "helpers + custom steps sourced - global.R COMPLETE", the installation is likely incomplete or corrupted - try reinstalling Cactus. If that does not fix it, contact us (Section 19.2) with the log file attached.

18.3 Signing up / logging in

- **"Please check your internet connection"** - creating an account, confirming the OTP code, and a normal login all require an internet connection (Cactus talks to its authentication service). Offline use is only available to previously-authorized accounts that have already logged in online once.
- **Confirmation code (OTP) never arrives** - check your spam/junk folder; the code is only valid for a limited time, so if it expired, go back to **Sign Up!** and register again with the same email.
- **Forgot your password** - use **Forgot Password?** on the login screen (Chapter 2).

18.4 Import problems

- **"File type mismatch"** - the selected **Type of file** does not match the uploaded file's extension; pick the matching type.
- **Wrong file format / garbled data in the preview** - check the **Delimiter** and **Decimal symbol** advanced options; files exported with European conventions (semicolon delimiter, comma decimal) need those settings adjusted.
- **Duplicate values in first column** - disable **Use first column as row labels**, or make the identifiers in that column unique before importing.
- **Column mismatch** (Split module's external test-set upload, or Predict module's new data) - the uploaded file is missing a column that the reference dataset/model expects; add the missing column or use a file that matches the original schema.

18.5 Non-numeric variables where numeric ones are expected

- A column intended to be numeric appears as text if the source file mixes numbers with units, symbols, or inconsistent decimal separators - clean the column in the source file and re-import, or adjust the **Decimal symbol** option.
- Modelling tabs (Classification, Regression, Unsupervised) only use numeric columns as predictors; if a variable you expect to see as a predictor is missing, confirm in the Edit Pipeline (Chapter 6) that it was actually assigned the **X (predictor)** role and is numeric.

18.6 Missing values

- If the **Confirm imputation & proceed** button in the Missing Values tab stays disabled, some missing values are still unresolved - check the positions table and continue imputing.
- If a column has a very high percentage of missing values (shown in red in the overview table), consider removing it entirely in the Edit Pipeline rather than imputing it.

- **KNN imputation** cannot be applied to classification, regression or date columns - these are excluded automatically.
- **Kennard-Stone / DUPLEX split** cannot run while predictor data still has missing values - impute first, or choose a different split method.

18.7 Too many variables / too few samples

- **LDA** is automatically disabled with a warning when the number of numeric predictors is greater than or equal to the number of samples - enable the **Perform PCA first?** option to reduce dimensionality before running LDA.
- **QDA** needs enough samples per class to estimate a separate covariance matrix per class; very small classes, or high dimensionality without a PCA pre-step, can fail or produce unstable fits.
- **MLR** with many correlated predictors and few samples is prone to multicollinearity (check the **VIF** tab) - consider PCR or PLS, which handle this better via component reduction.
- **One Class Modeling (SIMCA)** with too few target-class training samples relative to the number of components chosen will overfit - reduce the number of components or gather more target-class samples.

18.8 Plot not appearing / looks empty

- Some plots only render after an explicit action, not automatically: click **Calculate/Preprocess/Plot** (e.g. the Scatter Matrix in Visualization does not auto-update on setting changes - click **Plot** again).
- A sidebar tab showing "complete the previous step" instead of a plot means an earlier pipeline stage (Approve Dataset, Preprocessing) has not been completed yet - see Chapter 3.
- A chart type that used to be visible but has disappeared from a menu may have been hidden via **Configure App** in the right-hand settings panel (Section 3.3) - check there before assuming something is broken.
- Very large plots may drop some interactive features (hover tooltips, certain toolbar buttons) automatically to stay responsive - this is expected behaviour, not an error.

18.9 Model will not run

- **"Complete the Preprocessing step first"** - Unsupervised, Classification, One Class Modeling and Regression all require Preprocessing (Chapter 10) to be finished first.
- **Run ...Projection / Test tab missing** - the Test tab for any modelling method only appears once a model has been trained *and* Test-split rows exist; go back to Split Train/Test (Chapter 6) if you need a Test set.
- **"Run KNN Projection" blocked** - click **Show Plots** on the Train tab first.
- **Cross-validation error about a missing class in a fold** - Venetian Blinds/Contiguous Blocks folds require ordered/sufficiently represented class data; try a Stratified or Random k-fold strategy instead, or reduce the number of folds.

18.10 Slow / unresponsive interface

- Checking **Show entire dataset** on a very large table will prompt a confirmation warning first, since it can slow the app down - prefer paging through a limited number of rows/columns instead.
- Heatmaps and Spectra/Parallel Coordinates plots on very large datasets are rendered as a single rasterised image for performance; reduce the number of rows/columns/axis labels shown if the app feels sluggish.

Chapter 19

Appendix

19.1 Glossary

Term	Practical meaning
Predictor (X) variable	An input measurement used to explain or predict something else (e.g. a spectral intensity at one wavelength).
Classification (Y) variable	A categorical column used as a class label to discriminate between groups.
Regression (Y) variable	A numeric column used as the target a regression model tries to predict.
Train / Test split	Dividing samples into a set used to build a model (Train) and a separate set used to honestly evaluate it (Test), so performance is not judged on the same data the model was fitted to.
Cross-validation (CV)	Repeatedly holding out part of the Training data, fitting on the rest, and predicting the held-out part, to get a more realistic performance estimate than fitting and testing on the exact same data.
Leave-one-out (LOO)	A cross-validation strategy where each single sample is held out and predicted once.
Stratified k-fold	Cross-validation folds built so each fold preserves the overall class/response proportions.
PCA (Principal Component Analysis)	An unsupervised technique that summarises many correlated variables into a smaller number of uncorrelated "components" capturing most of the variation in the data.
Scores	The coordinates of each sample in a reduced-dimension space (e.g. PCA/PLS components).
Loadings	How strongly each original variable contributes to a given component/latent variable.
Explained variance	The percentage of the total information (variance) in the data captured by a given component.
Hotelling's T^2	A statistic measuring how extreme a sample is <i>within</i> a model (large values = unusual combination of scores).
Q residual (SPE)	A statistic measuring how poorly a sample is explained/reconstructed <i>by</i> a model (large values = the model does not fit this sample well).

Term	Practical meaning
HCA (Hierarchical Cluster Analysis)	An unsupervised technique that groups samples into a tree (dendrogram) based on similarity.
Dendrogram	A tree diagram showing the order and distance at which samples/clusters merge in HCA.
Silhouette score	A per-sample measure of how well it fits its assigned cluster compared to the next-closest cluster; higher/positive is better.
K-means	An unsupervised technique that partitions samples into a fixed number (k) of clusters around estimated centroids.
t-SNE	A non-linear technique for visualising high-dimensional data in 2D/3D, preserving local neighbourhood structure (not global distances).
SIMCA	Soft Independent Modelling of Class Analogies: a one-class modelling technique that accepts or rejects new samples based on whether they fit a single target class's own PCA-based envelope.
LDA / QDA (Linear / Quadratic Discriminant Analysis)	Classification methods that separate classes using straight (LDA) or curved (QDA) boundaries based on each class's estimated distribution.
PLS-DA	Partial Least Squares Discriminant Analysis: a classification method built on PLS latent variables chosen to separate classes.
KNN (K-Nearest Neighbours)	A classification method that assigns a sample to the majority class among its k closest training samples.
VIP (Variable Importance in Projection)	A score (PLS/PLS-DA) summarising how much a variable contributes across all retained latent variables; larger VIP = more influential variable.
MLR (Multiple Linear Regression / OLS)	A regression method fitting a straight-line relationship between all predictors and the response simultaneously.
PCR (Principal Component Regression)	Regression performed on PCA scores instead of the raw predictors, to handle many correlated variables.
PLS (Partial Least Squares)	Regression (or, as PLS-DA, classification) performed on latent variables chosen to correlate with the response, handling many correlated predictors.
Latent variable (LV) / Component	A constructed variable (a combination of the original predictors) used by PCA/PCR/PLS/PLS-DA instead of the raw variables.
RMSE (Root Mean Squared Error)	Average prediction error of a regression model, in the same units as the response; lower is better.
R^2 (Coefficient of determination)	The proportion of variation in the response explained by the model (0-1, or as a %); higher is better.
Accuracy / Sensitivity / Specificity	Classification performance measures: overall correct-rate, correct-rate within the positive class, and correct-rate within the negative class, respectively.
Confusion matrix	A table of actual vs. predicted classes, showing exactly which classes are being confused with which.
FAR / FRR	False Acceptance Rate / False Rejection Rate: in one-class modelling (SIMCA), the rate of wrongly accepting non-target samples / wrongly rejecting target samples.

Term	Practical meaning
p-value	The probability of seeing a result at least as extreme as observed if there were truly no effect; conventionally, $p < 0.05$ is treated as "statistically significant".
Effect size	A measure of <i>how large</i> a difference or association is, independent of sample size - always worth checking alongside a p-value.
Savitzky-Golay smoothing/derivative	A spectral preprocessing technique that locally fits a polynomial to smooth a spectrum or estimate its derivative.
SNV (Standard Normal Variate)	A row-wise spectral preprocessing step correcting scatter effects by centring/scaling each individual spectrum.
MSC (Multiplicative Scatter Correction)	A row-wise spectral preprocessing step correcting scatter effects relative to a reference spectrum.
Baseline correction (ALS)	A preprocessing step that estimates and removes a slowly varying background/drift from each spectrum.
Autoscaling	Centring and scaling a variable to zero mean and unit variance (a "z-score").
Pareto scaling	Centring a variable and dividing by the square root of its standard deviation - a compromise between no scaling and full autoscaling.

19.2 Where to get help

- Application-specific problems: Chapter 18.
- Found a bug? Report it on [Cactus GitHub Issues](#) - the same link as the **Report a Bug** button on the Credits tab (Chapter 17).
- Suggestions, comments, or feature requests: use the [Feedback Form](#) on the Credits tab.
- Anything else: contact us at info.cactus@databloom.it.

Chapter 20

Third-Party Libraries

Cactus is built in R and Shiny, and relies on a number of open-source packages to provide its interface, charts, and statistical/chemometric methods. This appendix lists every package loaded by the application (as declared in `global.R`), grouped by the part of the workflow it supports, with a link to its official documentation.

Note

These are development dependencies, not something you need to install or configure yourself - they are bundled with the application (Chapter 2). This page is provided for transparency and for anyone who wants to read more about the methods behind a given feature.

20.1 Interface (core UI)

Package	Used for
shiny	The web application framework Cactus is built on.
shinydashboard	The dashboard layout (sidebar, header, boxes).
shinydashboardPlus	Extra dashboard components (the right-hand settings panel).
shinyWidgets	Extra input controls (pickers, switches, action buttons) used throughout the app.
shinyjs	Show/hide interface elements and run small JavaScript actions from R.
DT	Interactive, searchable data tables.
plotly	All interactive charts (zoom, pan, hover, export) - see Chapter 8.
htmlwidgets	The underlying framework that lets R render interactive widgets like plotly and DT in the browser.
jsonlite	Reading/writing JSON data (e.g. saved user preferences, login status).
RColorBrewer	The colour palettes offered in the Qualitative Palette picker.
colourpicker	The colour-picker input used for categorical colours.
curl	Network requests (used for the optional external login service).
shinydisconnect	The "connection lost" screen shown if the app disconnects.

20.2 Data import

Package	Used for
readxl	Reading Excel files (<code>.xls/.xlsx</code>).
data.table	Fast reading of CSV/TXT files.

20.3 Edit module

Package	Used for
lubridate	Detecting and parsing date/time columns.
dplyr	General data manipulation (Edit pipeline, Missing Values).
tibble	Modern data-table utilities used internally.
readr	Writing CSV files (e.g. the Edited Data download).
rlang	Internal programming utilities.

20.4 Split module

Package	Used for
withr	Makes random splits reproducible (the Seed setting).
prospectr	The Kennard-Stone and DUPLEX split methods.

20.5 Missing values

Package	Used for
recipes	The preprocessing/imputation pipeline framework (also used by Preprocessing and every modelling module).

20.6 Visualization

Package	Used for
scales	Mapping values to visual scales (e.g. Bubble Plot size).
MASS	The 2D Density Chart calculation.

Package	Used for
ggplot2	Building certain charts before they are made interactive.

20.7 Unsupervised modelling

Package	Used for
mdatools	PCA, and the underlying engine behind PLS and SIMCA.
reshape2	Reshaping data for the Loadings bar chart.
writexl	Excel export (e.g. "Save PCA data").
dendextend	Colouring the HCA dendrogram.
cluster	The Silhouette Plot calculation (K-means, HCA).
Rtsne	The t-SNE technique.

20.8 Classification

Package	Used for
discrim	The LDA and QDA engines.
parsnip	A common interface for defining models (shared by Classification and Regression).
workflows	Bundles a preprocessing recipe together with a model.
tune	Hyperparameter tuning (e.g. choosing the best k for KNN).
rsample	Builds the cross-validation folds (Section 12.1).
yardstick	Calculates model metrics (Accuracy, Sensitivity, RMSE, R^2 , etc.).
dials	Generates the tuning grids used when searching for the best model settings.
plsmod	The PLS-DA engine.
butcher	Shrinks a trained model before saving it, for smaller downloaded files.
kknm	The KNN engine.

20.9 Preprocessing

Package	Used for
shinyFeedback	Inline validation warnings (e.g. invalid Savitzky-Golay settings).
shinycustomloader	Loading spinners while a preprocessing preview is calculated.

Package	Used for
sortable	The drag-and-drop pipeline builder.

20.10 Regression

Package	Used for
mixOmics	The PLS and PLS-DA engine (Bioconductor package).
broom	Tidies model output for the MLR Coefficients/Model Statistics tables.
gvlma	The MLR Assumptions diagnostic test.
tidyr	Reshaping data for scores/loadings charts.
colorspace	Generates the colour palette used for loadings colour pickers.